T2K 7.482×10^{20}-POT FHC and 7.471×10^{20}-POT RHC (1-7c) Joint 3-Flavour $\nu_\mu/\bar{\nu_\mu}$ Disappearance and ν_e / $\bar{\nu_e}$ Appearance Analysis with additional ν_e CC1π^+-like sample.

Costas Andreopoulos[a,b], Christopher Barry[b], Thomas Dealtry[d], Steve Dennis [a,b], Davide Sgalaberna[e], Raj Shah[a,c]

[a] *Science & Technology Facilities Council (STFC), Rutherford Appleton Laboratory, Harwell Oxford Campus, OX11 0QX, UK*
[b] *University of Liverpool, Physics Department, Liverpool, L69 7ZE, UK*
[c] *University of Oxford, Subdepartment of Particle Physics, Oxford, OX1 3RH, UK*
[d] *Lancaster University, Physics Department, Lancaster, LA1 4YB, UK*
[e] *University of Geneva, Departement de physique nucleaire et corpusculaire, 1211 Geneve, Switzerland*

Abstract

We report the results of the 3-flavour $\nu/\bar{\nu}$ joint analysis with the addition of the ν_e charged current single pion like sample, performed by the *VALOR* group, on the Run 1+7c dataset, Run using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for CP violation by setting limits on δ_{CP} and measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) (Δm^2_{32} for normal hierarchy and Δm^2_{13} for inverted hierarchy). Though T2K has limited sensitivity to the mass hierarchy given the relatively short baseline, fits under both hypotheses are performed to show which is preferred by data. Furthermore the search for $\bar{\nu}_e$ appearance and the measurement of $\bar{\nu}_\mu$ disappearance are shown as well.

This adapts the work presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] [10] and [11] to this dataset.

The Run 1-7c dataset corresponds to an integrated J-PARC neutrino beam exposure of 7.482×10^{20} POT in neutrino mode and 7.471×10^{20} POT in antineutrino mode.

This analysis predicts 135.8 ± 6.9 (syst) for neutrino μ-like, 64.2 ± 3.3 (syst) for antineutrino μ-like, 28.7 ± 1.8 (syst) for neutrino e-like, and 6.0 ± 0.4 (syst) for antineutrino e-like single-ring 0π events in Super-K and **3.1** ± **0.47** (syst) ν_e CC1π^+ events, given the oscillation parameters of the Asimov data set A in table 13, corresponding to $\sin^2\theta_{23} = 0.528$, $\delta_{CP} = -1.601$ and normal hierarchy, and observes respectively 135 for neutrino μ-like, 66 for antineutrino μ-like, 32 for neutrino e-like, 4 for antineutrino e-like and **5** for ν_e CC1π^+-like.

A joint analysis of all the five selected samples is performed in a framework of 3-flavour oscillations including matter effects in constant-density matter.

For μ-like samples the templates are built as a function of the reconstructed neutrino energy, while for e-like the analysis templates are built as a function of the reconstructed neutrino energy and lepton angle, in order to improve the separation of different flavour/reaction components and increase the sensitivity to δ_{CP}. The observed single-ring events spectra are compared to the expected spectra in order to infer all the 3-flavour oscillation parameters.

As in [11], all the systematic parameters are marginalised (see section 6) in the fit and confidence intervals are quoted for all the oscillation parameters. Since the most important result of this analysis release is the search for CP violation, the Feldman-Cousins method [64] is performed only for the simultaneous fit of δ_{CP} and mass hierarchy.

Contents

1 Introduction **4**

2 Additional ν_e $CC(1\pi^+)$ sample at Super-K **5**

3 Predictions of single ring event spectra in Super-K **6**
3.1 Methodology . . . 6
3.2 Construction of the nominal Super-K Monte Carlo templates . . . 6
3.2.1 Input Monte Carlo samples . . . 6
3.2.2 Normalisation of Monte Carlo samples . . . 7
3.2.3 Single μ-like ring event selection cuts . . . 7
3.2.4 Single e-like ring event selection cuts . . . 8
3.2.5 $CC1\pi^+$-like ring event selection cuts . . . 8
3.2.6 List of Monte Carlo templates . . . 9
3.2.7 Binning of Monte Carlo templates . . . 10
3.3 Applying flux tuning based on NA61/SHINE data . . . 13
3.4 Applying flux and cross section (BANFF) tuning based on ND280 data . . . 13
3.5 Applying neutrino oscillations . . . 13
3.6 Predictions of nominal and tuned-MC spectra . . . 14

4 Systematic effects in the prediction of the single μ-like ring event energy spectrum **17**
4.1 Input systematic parameters . . . 17

5 Validation of the $\bar{\nu}/\nu$ joint analysis with Run 1-7c statistics **25**
5.1 Prediction of event rate and effect of systematic uncertainties . . . 25
5.2 Evaluation of the effects of systematic parameters on the predicted reconstructed energy spectrum 26

6 Analysis procedure **50**
6.1 Fit method . . . 50
6.2 Marginalization . . . 50
6.3 Best-fit spectra with marginalization . . . 52
6.4 Confidence intervals . . . 52
6.5 Contours with profiled mass hierarchy . . . 54

7 Sensitivity studies **56**
7.1 Additional sensitivity studies . . . 60

8 Coverage study of the simultaneous fit of δ_{CP} and mass hierarchy **61**
8.1 Validation of the minimisation algorithm . . . 64

9 Results of the $\bar{\nu}/\nu$ joint analysis with the Run 1-7c dataset **65**
9.1 1-dimensional χ^2 distributions and best-fit . . . 65
9.2 Feldman-Cousins and significance for the simultaneous fit of δ_{CP} and mass hierarchy for Run 1-7c . . . 65
9.3 2-dimensional confidence intervals . . . 70
9.4 $CC1\pi^+$ vs 1 Ring comparison . . . 70

10 Comparison between ν and $\bar{\nu}$ only contours with the Run 1-7c dataset **70**
10.1 Best-fit spectra . . . 76
10.2 Goodness of fit . . . 77
10.3 Discussion of the Run 1-7c data fit results . . . 78
10.4 Comparison of δ_{CP} vs mass hierarchy Run 1-7c result with sensitivity . . . 79
10.4.1 Expected sensitivity method . . . 79
10.4.2 $\delta_{CP} = -\frac{\pi}{2}$ - normal hierarchy . . . 79
10.4.3 $\delta_{CP} = -1.728$ - normal hierarchy . . . 80

10.4.4 $\delta_{CP} = 0$ - normal hierarchy . . . 80
10.4.5 $\delta_{CP} = \pi$ - normal hierarchy . . . 81
10.4.6 $\delta_{CP} = -1.445$ - inverted hierarchy . . . 81
10.4.7 $\delta_{CP} = -\frac{\pi}{2}$ - inverted hierarchy . . . 82
10.5 Discussion of the Run 1-7c data fit results with sensitivity . . . 82

11 Summary **87**

Appendices **92**

A Predicted number of events **92**

B Effect of systematic uncertainties **97**

C Method for significance calculation **109**

D Statistical uncertainty on weighted p-value and critical values for confidence interval **111**

E FC results **113**

F VALOR, MaCH3 and p-θ comparison **113**

G Further coverage studies **125**
G.1 VALOR - pTheta Comparison . . . 125
G.2 Summer vs Current analysis . . . 125
G.2.1 Study of the new sample . . . 125
G.2.2 MC toy extraction method . . . 126

H Signal vs background **129**

1. Introduction

In this technical note, we report the results of a 3-flavour $\nu/\bar{\nu}$ joint analysis including the ν_e CC1π^+ sample, performed by the *VALOR* group, on the combined Run 1-7c dataset, using data taken in both neutrino mode (7.482×10^{20} POT), with the polarity set to forward horn current (FHC), and antineutrino (7.471×10^{20} POT) mode with polarity set to reverse horn current (RHC). The Run 7a-c data set was collected in the following periods: Run 7a ν mode (0.88×10^{19} POT, 6.42×10^{19} POT) (01-03 Feb. 2016, 03-17 Feb. 2016), Run 7b $\bar{\nu}$ mode (13.66×10^{19} POT, 14.52×10^{19} POT) (20 Feb.-29 Mar. 2016, 13 Apr.-18 May. 2016) Run 7c ν mode (3.92×10^{19} POT) (19-27 May. 2016)

This analysis builds on our other analyses presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], and [11]. In this analysis a hybrid frequentist-Bayesian method [65] is employed, which consists of marginalising instead of profiling the nuisance parameters, as was used in the analyses [10] and [11].

The consistency of the data with the expectation is given by comparing the predicted reconstructed neutrino energy and outgoing lepton angle spectra of 1 ring CCQE-like events and ν_e CC1π^+-like events at Super-K to the observed spectra. The method of calculation of these predicted spectra is presented in section 3. This analysis uses flux tuning based on NA61/SHINE 2009 data [19][55]. It also uses an indirect extrapolation from ND280 to Super-K, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ for different event topologies with the neutrino interaction vertex either in FGD1 or FGD2 (BANFF fit) [31].

As in the previous analysis [11], the e-like samples, which now includes the ν_e CC1π^+ sample, have analysis templates that are built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ and $\nu_\mu \rightarrow \nu_e$, fundamental in measuring δ_{CP}. Furthermore the ν_e intrinsic beam and NC components can also be better separated from signal. The addition of the variable θ increases the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to retain the previous sensitivity to oscillations. For μ-like samples the templates are built as a function of the reconstructed neutrino energy only as in the past analyses.

Nominal predictions of the reconstructed neutrino energy and lepton angle spectrum of CC 0π events and ν_e CC1π^+-like events in Super-K are presented for Run 1-7c in section 3.

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is performed using the likelihood-ratio method [61]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 110 parameters including energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state re-scattering systematics (nuisance parameters) plus the oscillation parameters. All the nuisance parameters are marginalised [65]. The nuisance parameters and their effects on the extrapolated single-ring event spectra are described in section 4. The analysis method is presented in detail in section 6.

Fit validations are presented in section 5. The following updates have been added: the coverage studies study for the δ_{CP} vs mass hierarchy fit is shown in section 8. The main goal of this analysis is to search for CP violation in the context of the PMNS matrix and new confidence intervals, produced with the Feldman-Cousins method [64], are shown in section 8. These sections as well as appendix C have been added in this new version.

Confidence intervals of the other oscillation parameters are set using the constant $\Delta\chi^2$ method. Sensitivity studies are shown in section 7. The results of the measurement using the Run 1-7c dataset are given in section 9 of this note.

Finally, a summary is given in section 11.

2. Additional ν_e $CC(1\pi^+)$ sample at Super-K

Another sample in addition to the four single ring samples used in the previous VALOR joint $\nu/\bar{\nu}$ fit analysis [11] is included in the present analysis, this new sample is termed ν_e CC$1\pi^+$. This is the first Super-K sample included in the analysis which includes a non CCQE-like signal topology. For this new sample the signal selection cuts allow events with one decay electron in the ν_e appearance sample, whereas the single e-like ring samples in previous analyses only allowed events with no decay electrons. This new cut introduces a large ν_μ background, since the resulting decay electron ring may have been the result of a ν_μ CCQE interaction and subsequent μ decay. The neutrino energy reconstruction is performed by modification of the CCQE prescription and assumes that the single pion in the final state is produced by an outgoing Δ^{++} baryon, without considering nuclear effects. Further sample selection details, and Super-K error evaluations, can be found in [45].

3. Predictions of single ring event spectra in Super-K

3.1. Methodology

In this analysis a comparison of the reconstructed energy (E_r) and reconstructed lepton angle (θ) spectrum of single ring events, including the ν_e CC1π^+-like sample, observed at Super-K with the predicted spectrum is used to build the test statistic to quote confidence intervals for neutrino oscillation parameters. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in previous analyses, and a single θ bin is used. The predicted number $N_{SK;r,\theta}$ of single ring events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \quad (1)$$

In eq. (1), $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and reconstructed lepton angle bin θ. The construction of the MC templates is discussed in section 3.2. Both flux-tuning weights based on NA61/SHINE hadron-production data (see section 3.3) and flux and cross section tuning weights based on ND280 data (see section 3.4) are applied to the nominal MC templates. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on a vector of nuisance (systematic) parameters $\vec{f}$, which is a function of the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, and the reconstructed lepton angle bin θ. A complete description of the nuisance parameters considered in this analysis is given in section 4. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in section 3.5.

3.2. Construction of the nominal Super-K Monte Carlo templates

The nominal Super-K single ring and ν_e CC1π^+ MC analysis templates $N^{MC}_{SK;m;r\prime,\theta;t}$ are constructed by applying the single e-like ring selection cuts, the ν_e CC1π^+ selection cuts, or the single μ-like ring selection cuts to the official Super-K MC samples. The selections are identical for both neutrino and antineutrino beam mode. In total 5 samples are obtained:

- neutrino mode single μ-like ring sample
- antineutrino mode single μ-like ring sample
- neutrino mode single e-like ring sample
- antineutrino mode single e-like ring sample
- neutrino mode ν_e CC1π^+-like sample

Details of the MC samples are given in section 3.2.1. The calculation of the normalisation of each MC sample is described in section 3.2.2. The official single μ-like ring selection cuts are listed in section 3.2.3. while the official single e-like ring selection cuts are listed in section 3.2.4, and the official ν_e CC1π^+-like ring selection cuts are listed in section 3.2.5. The list of MC templates used in this analysis is shown in section 3.2.6, while the true and reconstructed energy as well as the reconstructed lepton angle binning of the templates are discussed in section 3.2.7.

3.2.1. Input Monte Carlo samples

This analysis uses the official 14a Super-K MC samples. These were generated using the 13a JNUBEAM flux [40, 41], NEUT 5.3.2 [44], skdetsim v13p90 and the 14b Super-K software [37] with nominal tunings. The CCQE model is reweighted from the Spectral Function to a Relative Fermi Gas model with a relativistic Random Phase Approximation on an event by event basis. The NC1γ events are scaled up by a factor of two in accordance with improved calculations of single photon emission in NC interactions [38]. Lastly, a

Running mode	Neutrino type	Number of predicted events (flux averaged) per 22.5 kt fiducial per 10^{21} POT	Number of generated events in true fiducial volume in all available MC
Neutrino	$\bar{\nu}_\mu$	50.964	634600
	ν_μ	1396.245	634881
	$\bar{\nu}_e$	2.884	126728
	ν_e	26.663	126899
	Oscillated $\bar{\nu}_e$	52.939	126984
	Oscillated ν_e	1483.994	126916
Antineutrino	$\bar{\nu}_\mu$	353.789	633988
	ν_μ	208.369	634659
	$\bar{\nu}_e$	6.811	127013
	ν_e	8.941	127262
	Oscillated $\bar{\nu}_e$	376.420	126812
	Oscillated ν_e	215.099	127278
CCπ^+	$\bar{\nu}_\mu$	0.166	2064
	ν_μ	5.489	2496
	$\bar{\nu}_e$	0.076	3316
	ν_e	2.073	9866
	Oscillated $\bar{\nu}_e$	0.931	2233
	Oscillated ν_e	101.177	8653

Table 1: Normalisation of input Super-K MC samples [37]. Oscillations are applied only to "Oscillated" samples which correspond to fully oscillated samples, e.g. assuming all ν_μ oscillated into ν_e.

simplified reweighting, as a function of the outgoing pion energy, is used to scale the CCCoherent events from the Rein Sehgal model to the Berger-Sehgal model prediction [42, 43]. Six MC samples were generated for both neutrino and antineutrino beam mode running (horn current ±250kA). Four of the samples ($\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$ and ν_e) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample. The sixth sample is ν_e interactions generated with the ν_μ flux, and will be denoted the "oscillated ν_e" sample.

3.2.2. Normalisation of Monte Carlo samples

The normalisation (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{2}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section on water for the given neutrino species, I is the beam exposure in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The Super-K MC samples are normalised to the Run 1-7c POT using the numbers in table 1 Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with the ratio of the number of events per 22.5 kt per 10^{21} POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to 7.482×10^{20} POT for neutrino and 7.471×10^{20} for antineutrino mode, by weighting the result respectively by 0.7482 and 0.7471 (the Run 1-7c POT in units of 10^{21} POT).

3.2.3. Single μ-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single μ-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than 16 (`nhitac < 16`),

- the reconstructed vertex is at least 2 metres away from the ID wall (`dwall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

The same selection is applied to both neutrino and antineutrino mode samples [35].

3.2.4. Single e-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single e-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than 16 (`nhitac < 16`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`dwall > 200`),
- the visible energy in the event is above 100 MeV (`evis > 100`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is e-like (`ip[0] = 2`),
- number of decay electrons is zero (`nmue = 0`),
- the reconstructed neutrino energy is below 1250 MeV
- fiTQun π^0 rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

The same selection is applied to both neutrino and antineutrino mode samples [35].

3.2.5. CC$1\pi^+$-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for ν_e CC$1\pi^+$-like ring events. These cuts are identical to those for the e-like ring events apart from allowing a decay electron and reconstruction of neutrino energy assuming a Δ^{++} These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than 16 (`nhitac < 16`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`dwall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is e-like (`ip[0] = 2`),
- number of decay electrons is one (`nmue = 1`),
- the reconstructed neutrino energy is below 1250 MeV, using the assumption of a Δ^{++} in the modified reconstruction equation
- fiTQun π^0 rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

This selection is only applied to neutrino mode samples [35].

3.2.6. List of Monte Carlo templates

For each Super-K MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template number depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6]. 50 templates were used for our 2016 Run 3-flavour joint analysis [11], and the same templates are used for the current analysis

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC 2p-2h,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC 1γ,
- ν_μ CCQE,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC 2p-2h,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC 1γ,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC 2p-2h,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,

- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC 1γ,
- ν_e CCQE,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC 2p-2h,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC$1\pi^0$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC$1\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC 1γ,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC 2p-2h,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC1π,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC 2p-2h,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC other.

In tables 2 to 6 the predicted number of events is shown for all the selected samples and each flavour and reaction mode.

3.2.7. Binning of Monte Carlo templates

This analyses uses an updated binning scheme for reconstructed variables in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ and $\nu_\mu \rightarrow \nu_e$, fundamental in measuring δ_{CP}. The ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increases the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to maintain sensitivity to the oscillations. For μ-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

μ-like templates have 84 true energy bins and 73 reconstructed energy bins. e-like templates have 84 true energy bins, 25 reconstructed energy bins and 15 reconstructed lepton angle bins.

For all templates the 84 true energy bins of constant width are the following:

- 6 50-MeV bins from 0-0.3 GeV,

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	71.26079	0.03725	4.63401	0.00187	0.24312	0.00114	76.17818
CC1pi	18.30962	0.01864	1.69079	0.00103	0.07190	0.00042	20.09241
CCcoherent	0.35206	0.00027	0.08627	0.00003	0.00119	0.00003	0.43985
2p-2h	20.53202	0.01096	0.91267	0.00032	0.05392	0.00024	21.51012
CCother	8.34849	0.00913	0.53311	0.00057	0.00195	0.00005	8.89329
NC1pi0	0.57972	0.01777	0.01866	0.00125	N/A	N/A	0.61740
NC1piPM	4.25309	0.09626	0.15291	0.00889	N/A	N/A	4.51115
NCcoherent	0.01705	0.00019	0.00155	0.00010	N/A	N/A	0.01889
NCother	3.22888	0.13029	0.17780	0.01319	N/A	N/A	3.55017
NC1gamma	0.00390	0.00000	0.00012	0.00000	N/A	N/A	0.00402
Total	126.88563	0.32075	8.20790	0.02725	0.37208	0.00187	135.81549

Table 2: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 7.482×10^{20} POT. The Asimov data set A in table 13 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	13.19436	0.00664	23.83240	0.00856	0.01155	0.02589	37.07941
CC1pi	5.10184	0.00490	5.65512	0.00367	0.00470	0.00485	10.77509
CCcoherent	0.10170	0.00006	0.31428	0.00020	0.00001	0.00060	0.41685
2p-2h	4.71879	0.00225	3.83018	0.00155	0.00321	0.00366	8.55964
CCother	2.41002	0.00254	1.64472	0.00088	0.00038	0.00015	4.05870
NC1pi0	0.11340	0.00467	0.09275	0.00249	N/A	N/A	0.21331
NC1piPM	0.71338	0.02835	0.70828	0.01935	N/A	N/A	1.46937
NCcoherent	0.00251	0.00012	0.01193	0.00012	N/A	N/A	0.01468
NCother	0.99673	0.04648	0.55030	0.02256	N/A	N/A	1.61607
NC1gamma	0.00064	0.00000	0.00000	0.00000	N/A	N/A	0.00064
Total	27.35338	0.09601	36.63997	0.05937	0.01986	0.03516	64.20374

Table 3: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 7.471×10^{20} POT. The Asimov data set A in table 13 is used.

- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV

For μ-like samples the 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.06080	2.29713	0.00085	0.09533	17.12951	0.10726	19.69088
CC1pi	0.01720	0.37587	0.00027	0.02697	1.82143	0.01847	2.26021
CCcoherent	0.00012	0.00263	0.00000	0.00193	0.01628	0.00182	0.02278
2p-2h	0.00860	0.76550	0.00008	0.02370	4.20736	0.02055	5.02579
CCother	0.00142	0.04860	0.00005	0.00351	0.03130	0.00123	0.08611
NC1pi0	0.56512	0.01248	0.01835	0.00096	N/A	N/A	0.59691
NC1piPM	0.10698	0.00251	0.00352	0.00023	N/A	N/A	0.11323
NCcoherent	0.18816	0.00386	0.01990	0.00096	N/A	N/A	0.21288
NCother	0.13569	0.00547	0.00832	0.00061	N/A	N/A	0.15009
NC1gamma	0.50065	0.00800	0.02283	0.00108	N/A	N/A	0.53256
Total	1.58473	3.52204	0.07418	0.15529	23.20588	0.14933	28.69144

Table 4: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 7.482×10^{20} POT. The Asimov data set A in table 13 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.00392	0.35323	0.00567	0.62877	0.71080	2.14728	3.84968
CC1pi	0.00310	0.07557	0.00191	0.10984	0.10810	0.24865	0.54716
CCcoherent	0.00005	0.00057	0.00004	0.01123	0.00091	0.03515	0.04795
2p-2h	0.00146	0.12956	0.00062	0.11748	0.21603	0.34387	0.80902
CCother	0.00205	0.01557	0.00004	0.00786	0.00674	0.00426	0.03651
NC1pi0	0.08545	0.00364	0.09084	0.00237	N/A	N/A	0.18230
NC1piPM	0.02157	0.00080	0.01494	0.00037	N/A	N/A	0.03770
NCcoherent	0.03855	0.00136	0.15037	0.00234	N/A	N/A	0.19262
NCother	0.04144	0.00168	0.02214	0.00069	N/A	N/A	0.06595
NC1gamma	0.06918	0.00360	0.15039	0.00316	N/A	N/A	0.22633
Total	0.26678	0.58559	0.43695	0.88411	1.04258	2.77920	5.99522

Table 5: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 7.471×10^{20} POT. The Asimov data set A in table 13 is used.

	ν_μ	ν_e	$\bar{\nu}_\mu$	$\bar{\nu}_e$	Osc. ν_e	Osc. $\bar{\nu}_e$	Total
CCQE	0.15777	0.00192	0.00412	0.00002	0.01434	0.00002	0.17819
CC1pi	0.05822	0.30131	0.00263	0.00096	1.92990	0.00054	2.29356
CCcoherent	0.00041	0.04269	0.00039	0.00003	0.29643	0.00007	0.34002
2p-2h	0.05610	0.00225	0.00047	0.00005	0.00963	0.00002	0.06852
CCother	0.01207	0.03733	0.00080	0.00094	0.03633	0.00054	0.08800
NC1pi0	0.00610	0.00012	0.00019	0.00000	N/A	N/A	0.00641
NC1piPM	0.03533	0.00134	0.00222	0.00021	N/A	N/A	0.03910
NCother	0.08631	0.00471	0.00588	0.00034	N/A	N/A	0.09725
NC1gamma	0.01690	0.00000	0.00023	0.00000	N/A	N/A	0.01713
Total	0.42923	0.39166	0.01693	0.00254	2.28662	0.00120	3.12818

Table 6: Predicted number of events in the neutrino mode ν_e CC1π^+-like sample obtained after applying the BANFF weight for a total exposure of 7.482×10^{20} POT. The Asimov data set A in table 13 is used.

- 1 bin from 10-30 GeV.

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 25 50-MeV bins from 0-1.25 GeV.

which is identical to the past analyses.

The θ binning used for e-like samples is

- 14 10° bins from 0° to 140°,
- 1 bin for the range 140° − 180°.

that is the same as used by the $p-\theta$ analysis group [17].

For μ-like samples a single θ bin, that covers the whole range 0° - 180°, is used. The predicted e-like 2-dimensional spectra are were studied in detail in [11] for each flavour component and were compared with the 1-dimensional reconstructed energy spectra. It is clear that the components $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ are better separated by the reconstructed energy vs θ templates.

A study was performed to decide the optimum binning scheme for reconstructing neutrino energy, balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4]. The reconstructed neutrino energy binning described here is unchanged with respect to the Run 1+2+3+4 joint fit analysis. The reconstructed lepton angle binning is the same as used by the $p-\theta$ group for all their analyses [17].

3.3. Applying flux tuning based on NA61/SHINE data

In this analysis, the nominal MC templates, which are constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which is tuned on NA61/SHINE hadron production measurements [40][54]. The reweighted MC templates are denoted the *NA61/SHINE-tuned* MC templates in this document. The applied weights are obtained from [40] and are shown as a function of the true neutrino energy in fig. 1 and 4 for neutrino mode. and in fig. 5 and 8 for antineutrino mode. The effect of this flux tuning on the predicted spectra events is discussed in section 3.6

3.4. Applying flux and cross section (BANFF) tuning based on ND280 data

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples to find the best-fit values of the flux normalisation and cross section parameters [30]. This analysis uses these best-fit values to reweight the NA61/SHINE-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their 1σ prefit and postfit errors is given in table 7, 8 and 9.

The effects of the BANFF tune on the predicted spectra is discussed in section 3.6.

3.5. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\overset{(-)}{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}. The $\overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_e \to \overset{(-)}{\nu}_e)$, while the oscillated $\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e)$ (the sample was generated assuming 100% of $\overset{(-)}{\nu}_\mu$ transform to $\overset{(-)}{\nu}_e$). All these oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code are first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [80], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of $\overset{(-)}{\nu}_e$ and $\overset{(-)}{\nu}_\mu$ can yield $\overset{(-)}{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ Super-K MC samples are available).

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like Super-K sample. A Super-K MC study is required to determine whether the Super-K efficiency assumptions made there are valid.

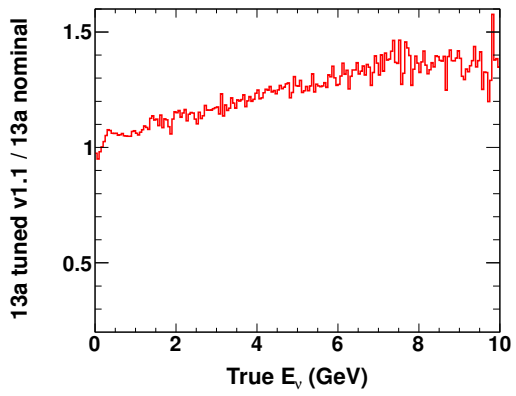


Figure 1: ν_μ 13a_v1.1 Run 1-6a flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [40] (neutrino mode).

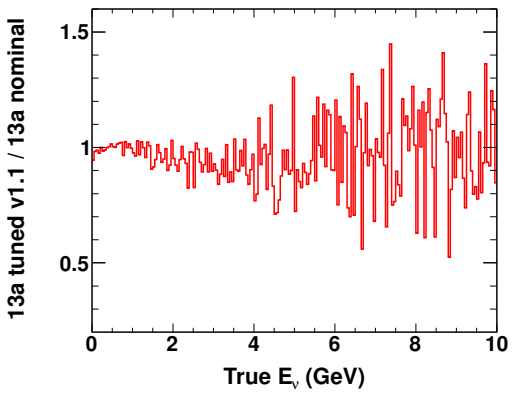


Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 1-6a flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [40] (neutrino mode).

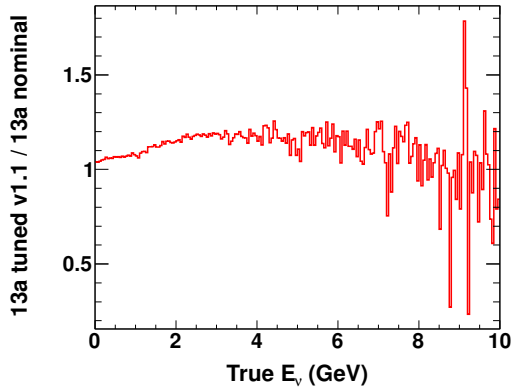


Figure 3: ν_e 13a_v1.1 Run 1-6a flux tuning weights applied to nominal ν_e MC templates [40] (neutrino mode).

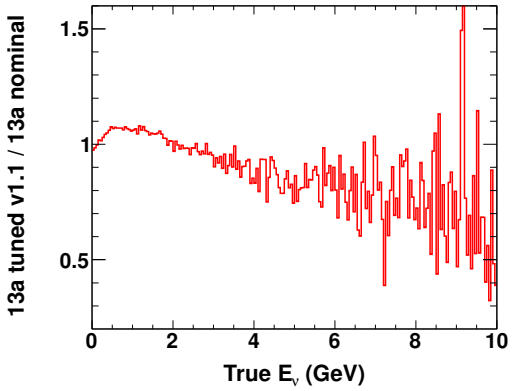


Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 1-6a flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [40] (neutrino mode).

It should be emphasised here that the ν_μ NC MC templates for a mode m contain the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

3.6. Predictions of nominal and tuned-MC spectra

In this section, we present the expected spectra in Super-K for various oscillation scenarios. Unless explicitly stated otherwise, the oscillation parameters are set to the Asimov data set A shown in table 13 and the normal mass hierarchy is assumed. All plots are generated for an integrated exposure corresponding to Run 1-7c.

Predicted single ring Super-K spectra are shown in fig. 9. By comparing neutrino and antineutrino mode e-like plots it is clear that E_{reco} vs θ distributions are differently populated by $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events.

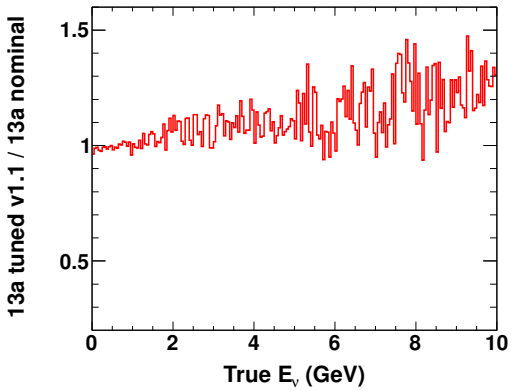


Figure 5: ν_μ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [40] (antineutrino mode).

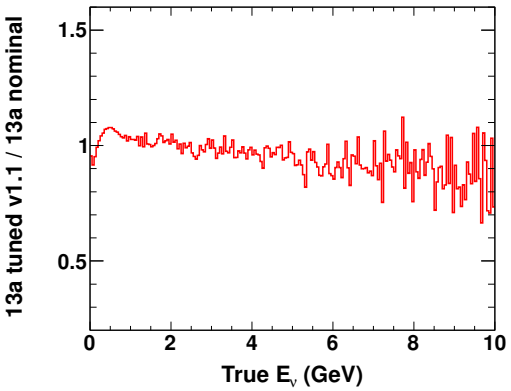


Figure 6: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [40] (antineutrino mode).

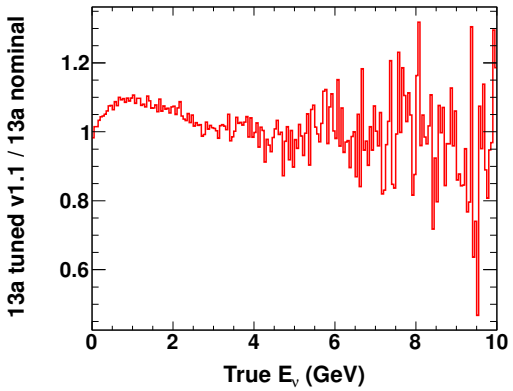


Figure 7: ν_e 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_e MC templates [40] (antineutrino mode).

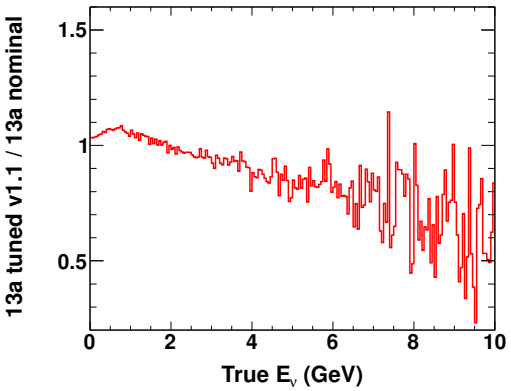
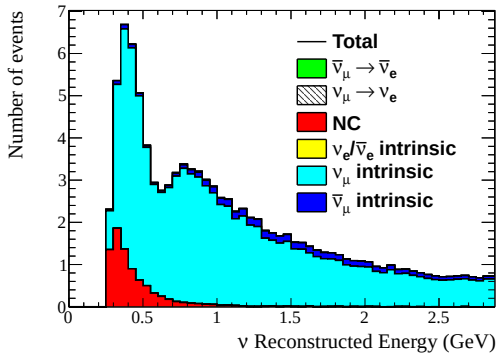
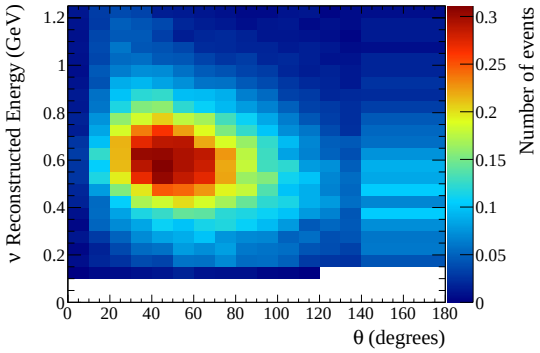


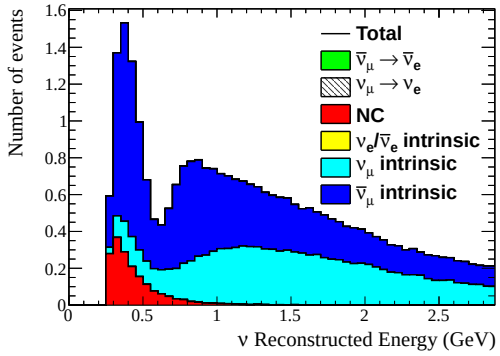
Figure 8: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [40] (antineutrino mode).



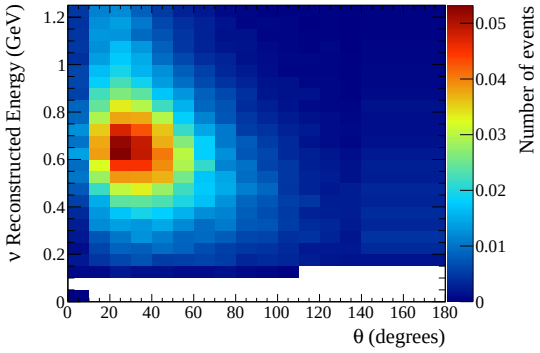
(a) Muon-like, neutrino mode



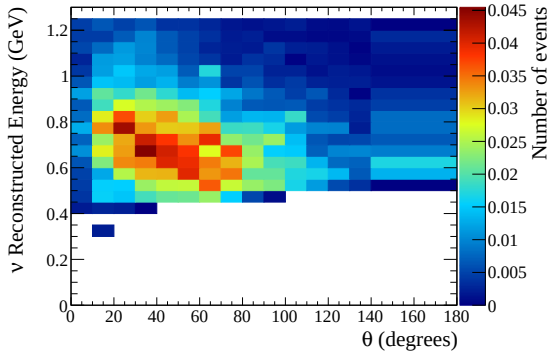
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode



(e) ν_e CC1π^+-like, neutrino mode

Figure 9: Predicted spectra, muon-like distributions are function of the reconstructed neutrino energy, while the electron-like, including ν_e CC1π^+, distributions are function of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-7c data set. The muon-like distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra are generated with the systematic parameters measured with the ND280 data shown in table 7, 8, 9 and the nominal Super-K systematic parameters shown in table 10 and the oscillation parameters corresponding to the Asimov data set A shown in table 13.

4. Systematic effects in the prediction of the single μ-like ring event energy spectrum

4.1. Input systematic parameters

This analysis considers 110 flux, cross section and Super-K detector systematic parameters plus 7 oscillation parameters, which can be treated as either systematic parameters or as parameters of interest, depending on the analysis that is performed.

1. **Flux and cross-section parameters** (65 parameters).

   In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit. However, since ND280 is not sensitive to the electron neutrino cross section and the NC1γ parameters they remain unconstrained. The parameters fit are 11 ν_μ, 5 $\bar{\nu}_\mu$, 7 ν_e and 2 $\bar{\nu}_e$ neutrino mode (FHC) flux parameters, 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ antineutrino mode (RHC) flux parameters and 15 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples as described in Ref [30]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors, taken from TN-230 [31], is given in table 7, table 8, table 9.

   The best-fit values are used to reweight the templates as described in section 3.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis. The Super-K systematic parameters are shown in table 10. All the cross section parameters set to 1 correspond to the NEUT nominal value, except "CC other shape" for which the NEUT nominal value is 0. For all the normalisation parameters, 1 is the nominal pre-banff fit value.

   The effects of these correlated uncertainties are evaluated using the Cholesky method, that allows throws of the systematic parameters according to their multivariate distribution taking into account existing correlations.

   Some of the cross-section parameters have complicated, often non-linear, effects on true and reconstructed neutrino energy bin normalisations; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3σ to +3σ in steps of 0.5σ, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7σ to +3σ). Cubic splines are made for the following cross-section uncertainties:

   - CCQE axial-mass scaling factor (MaQE)
   - Fermi momentum for ^{16}O
   - Binding energy for ^{16}O
   - Nucleon to Δ transition axial form factor (C_5^A)
   - Resonance-production axial-mass scaling factor (MaRes)
   - Scale of isospin 1/2 nonresonant background
   - CC other shape

   The splines are made in bins of theta, true and reconstructed energy, and the binning is given in Ref. [39]. The θ binning is the same as for the analysis templates, shown in section 3.2.7.

   The effects of the flux and cross-section normalisation uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalisation for a limited range of true energy). The cross section parameters which simply normalise the overall spectrum component are:

   - 2p-2h, applied to all the 2p-2h events (both neutrino and antineutrino)
   - 2p-2h-$\bar{\nu}$, applied to all the antineutrino 2p-2h events
   - CC coherent for ^{16}O normalisation
   - NC coherent normalisation

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
0	$f^{banff}_{0;t,r}$	FHC ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.128	0.099 / 0.064
1	$f^{banff}_{1;t,r}$	FHC ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.156	0.103 / 0.061
2	$f^{banff}_{2;t,r}$	FHC ν_μ flux normalisation, E = 0.5 - 0.6 GeV	1.148	0.096 / 0.051
3	$f^{banff}_{3;t,r}$	FHC ν_μ flux normalisation, E = 0.6 - 0.7 GeV	1.128	0.087 / 0.043
4	$f^{banff}_{4;t,r}$	FHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.104	0.113 / 0.046
5	$f^{banff}_{5;t,r}$	FHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.100	0.092 / 0.045
6	$f^{banff}_{6;t,r}$	FHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.127	0.070 / 0.044
7	$f^{banff}_{7;t,r}$	FHC ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.124	0.074 / 0.048
8	$f^{banff}_{8;t,r}$	FHC ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.121	0.087 / 0.049
9	$f^{banff}_{9;t,r}$	FHC ν_μ flux normalisation, E = 5.0 - 7.0 GeV	1.075	0.098 / 0.053
10	$f^{banff}_{10;t,r}$	FHC ν_μ flux normalisation, E = 7.0 - 30.0 GeV	1.064	0.114 / 0.065
11	$f^{banff}_{11;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.7 GeV	1.100	0.103 / 0.081
12	$f^{banff}_{12;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.112	0.079 / 0.048
13	$f^{banff}_{13;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.111	0.084 / 0.060
14	$f^{banff}_{14;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.116	0.086 / 0.070
15	$f^{banff}_{15;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 30.0 GeV	1.162	0.086 / 0.069
16	$f^{banff}_{16;t,r}$	FHC ν_e flux normalisation, E = 0.0 - 0.5 GeV	1.134	0.090 / 0.052
17	$f^{banff}_{17;t,r}$	FHC ν_e flux normalisation, E = 0.5 - 0.7 GeV	1.135	0.090 / 0.049
18	$f^{banff}_{18;t,r}$	FHC ν_e flux normalisation, E = 0.7 - 0.8 GeV	1.135	0.086 / 0.047
19	$f^{banff}_{19;t,r}$	FHC ν_e flux normalisation, E = 0.8 - 1.5 GeV	1.119	0.081 / 0.043
20	$f^{banff}_{20;t,r}$	FHC ν_e flux normalisation, E = 1.5 - 2.5 GeV	1.115	0.079 / 0.046
21	$f^{banff}_{21;t,r}$	FHC ν_e flux normalisation, E = 2.5 - 4.0 GeV	1.111	0.084 / 0.050
22	$f^{banff}_{22;t,r}$	FHC ν_e flux normalisation, E = 4.0 - 30.0 GeV	1.118	0.094 / 0.067
23	$f^{banff}_{23;t,r}$	FHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 2.5 GeV	1.121	0.074 / 0.057
24	$f^{banff}_{24;t,r}$	FHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 30.0 GeV	1.153	0.128 / 0.117

Table 7: Summary of neutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [30].

- NC other normalisation
- NC 1γ normalisation
- CC ν_e normalisation
- CC $\bar{\nu}_e$ normalisation

Uncertainties due to "second class currents" are implemented as an additional uncertainty on CC $\bar{\nu}_e$ and CC ν_e parameters with a 2% anti-correlated constraint between them. The BANFF parameters are applied to the MC templates as specified in appendix B in tables 39 to 42 The normalisation parameter of the NC1γ component was not used in the previous analyses, since it was included in NC other.

In the BANFF fit a bias on the best-fit flux parameters is present due to the degeneracy between the neutrino flux and ND280 detector parameters. For this reason a covariance matrix that accounts for the bias of the flux as well as all the other parameters is summed in quadrature with the total post-fit covariance. Furthermore the best-fit values have been corrected in order to remove the bias. A full description of the procedure is given in [30]. As shown in [11] the impact of the bias on the systematic parameters fitted by the BANFF analysis is small compared to the other systematic uncertainties.

Index	Parameter	Description	Best fit	1σ fractional error
25	$f^{banff}_{0;t,r}RHC$	RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	1.098	0.094 / 0.072
26	$f^{banff}_{1;t,r}RHC$	RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.121	0.079 / 0.052
27	$f^{banff}_{2;t,r}RHC$	RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.130	0.077 / 0.048
28	$f^{banff}_{3;t,r}RHC$	RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.155	0.081 / 0.054
29	$f^{banff}_{4;t,r}RHC$	RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.111	0.080 / 0.055
30	$f^{banff}_{5;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	1.118	0.104 / 0.071
31	$f^{banff}_{6;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	1.127	0.102 / 0.060
32	$f^{banff}_{7;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	1.117	0.096 / 0.052
33	$f^{banff}_{8;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	1.121	0.085 / 0.044
34	$f^{banff}_{9;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.155	0.125 / 0.066
35	$f^{banff}_{10;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.132	0.105 / 0.057
36	$f^{banff}_{11;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.139	0.080 / 0.053
37	$f^{banff}_{12;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	1.141	0.074 / 0.054
38	$f^{banff}_{13;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	1.151	0.094 / 0.071
39	$f^{banff}_{14;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	1.133	0.093 / 0.070
40	$f^{banff}_{15;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	1.082	0.130 / 0.110
41	$f^{banff}_{16;t,r}RHC$	RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.117	0.069 / 0.051
42	$f^{banff}_{17;t,r}RHC$	RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.112	0.085 / 0.071
43	$f^{banff}_{18;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	1.126	0.095 / 0.058
44	$f^{banff}_{19;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	1.127	0.091 / 0.051
45	$f^{banff}_{20;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	1.133	0.091 / 0.052
46	$f^{banff}_{21;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	1.132	0.084 / 0.046
47	$f^{banff}_{22;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	1.125	0.080 / 0.056
48	$f^{banff}_{23;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	1.119	0.089 / 0.071
49	$f^{banff}_{24;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	1.166	0.156 / 0.141

Table 8: Summary of antineutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [30].

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
50	$f^{banff}_{Norm_{2p-2h}}$	Meson exchange current normalisation for ^{16}O	1.546	- / 0.343
51	$f^{banff}_{C_5^A}$	C_5^A nucleon to Δ transition axial form factor	0.789	0.119 / 0.062
52	$f^{banff}_{Bg_{RES}}$	Scale of isospin 1/2 nonresonant background	1.042	0.154 / 0.134
53	$f^{banff}_{M_A^{QE}}$	CCQE axial-mass scaling factor	0.931	- / 0.028
54	$f^{banff}_{M_A^{RES}}$	Resonance-production axial-mass scaling factor	0.889	0.158 / 0.041
55	$f^{banff}_{p_f}$	Fermi momentum for ^{16}O	1.041	- / 0.105
56	$f^{banff}_{Shape_{CCoth}}$	CC other shape	-0.022	0.400 / 0.208
57	$f^{banff}_{E_B}$	Binding energy for ^{16}O	0.882	0.333 / 0.282
58	$f^{banff}_{Norm_{CCcoh}}$	CC coherent for ^{16}O normalisation	0.858	0.300 / 0.228
59	$f^{banff}_{Norm_{NCcoh}}$	NC coherent normalisation	0.931	0.300 / 0.298
60	$f^{banff}_{Norm_{NCother}}$	NC other normalisation	1.000	0.300 / 0.300
61	$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	CC ν_e normalisation	1.000	0.028 / 0.028
62	$f^{banff}_{Norm_{NC1\gamma}}$	NC 1γ normalisation	1.000	1.000 / 1.000
63	$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	CC $\bar{\nu}_e$ normalisation	1.000	0.028 / 0.028
64	$f^{banff}_{Norm_{2p-2h-Bar}}$	Antineutrino Meson exchange current normalisation for ^{16}O	0.578	- / 0.177

Table 9: Summary of cross section systematics included in the *VALOR* joint fit analysis. Parameters with no prefit error were unconstrained. A description of how the BANFF fit constrained these systematics can be found in Ref. [30].

Index	Parameter	Description	1σ frac. error
0	$f^{SK+FSI}_{0;t,r}$	SKDet + FSI/SI 0; E_{reco} 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
1	$f^{SK+FSI}_{1;t,r}$	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.017
2	$f^{SK+FSI}_{2;t,r}$	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
3	$f^{SK+FSI}_{3;t,r}$	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu)	0.077
4	$f^{SK+FSI}_{4;t,r}$	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu)	1.004
5	$f^{SK+FSI}_{5;t,r}$	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu)	0.598
6	$f^{SK+FSI}_{6;t,r}$	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re)	0.100
7	$f^{SK+FSI}_{7;t,r}$	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re)	0.030
8	$f^{SK+FSI}_{8;t,r}$	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re)	0.033
9	$f^{SK+FSI}_{9;t,r}$	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.225
10	$f^{SK+FSI}_{10;t,r}$	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.166
11	$f^{SK+FSI}_{11;t,r}$	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.028
12	$f^{SK+FSI}_{12;t,r}$	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.063
13	$f^{SK+FSI}_{13;t,r}$	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.040
14	$f^{SK+FSI}_{14;t,r}$	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.061
15	$f^{SK+FSI}_{15;t,r}$	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re)	0.488
16	$f^{SK+FSI}_{16;t,r}$	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re)	0.194
17	$f^{SK+FSI}_{17;t,r}$	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re)	0.590
18	$f^{SK+FSI}_{0;t,r}RHC$	SKDet + FSI/SI 0; Ereco 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.018
19	$f^{SK+FSI}_{1;t,r}RHC$	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.016
20	$f^{SK+FSI}_{2;t,r}RHC$	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$CCQE (1Rmu); RHC	0.016
21	$f^{SK+FSI}_{3;t,r}RHC$	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu); RHC	0.093
22	$f^{SK+FSI}_{4;t,r}RHC$	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu); RHC	1.004
23	$f^{SK+FSI}_{5;t,r}RHC$	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu); RHC	0.597
24	$f^{SK+FSI}_{6;t,r}RHC$	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re); RHC	0.060
25	$f^{SK+FSI}_{7;t,r}RHC$	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re); RHC	0.027
26	$f^{SK+FSI}_{8;t,r}RHC$	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re); RHC	0.056
27	$f^{SK+FSI}_{9;t,r}RHC$	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.244
28	$f^{SK+FSI}_{10;t,r}RHC$	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.123
29	$f^{SK+FSI}_{11;t,r}RHC$	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	0.904
30	$f^{SK+FSI}_{12;t,r}RHC$	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.044
31	$f^{SK+FSI}_{13;t,r}RHC$	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.032
32	$f^{SK+FSI}_{14;t,r}RHC$	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.068
33	$f^{SK+FSI}_{15;t,r}RHC$	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re); RHC	0.482
34	$f^{SK+FSI}_{16;t,r}RHC$	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re); RHC	0.171
35	$f^{SK+FSI}_{17;t,r}RHC$	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re); RHC	0.538
0	$f^{SK}_{E;r}$	SKDet Energy Scale; Ereco 0.00 - 30.00 GeV; All	0.024

Table 10: Summary of SK detector + FSI +SI + PN systematics included in the *VALOR* joint fit analysis.

2. **Super-K efficiencies & effects of intranuclear and secondary re-interactions** (44 parameters).

The Super-K detector uncertainties include the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1632.2 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples from atmospheric neutrinos. The NC PID error is evaluated using a NC-enriched control sample, obtained by not using the PID information during the selection. All the Monte Carlo control samples are fitted simultaneously to the data with a χ^2 function using the selection efficiencies as fit parameters [32] [33].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [22]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [22]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the Super-K detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [25]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single e-like or μ-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [21]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [22]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the Super-K detector efficiency and FSI+SI systematics together, and add their covariance matrices linearly. Each of these covariance matrices has 44 correlated parameters, corresponding to different modes and reconstructed energy bins, as shown in table 11. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **Super-K energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the Super-K reconstructed energy scale. This uncertainty is estimated to be 2.4% [34]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in tables 39 to 43 and is common to all the Super-K samples.

4. **Oscillation parameters and mass hierarchy** (7 parameters).

The 3-flavour oscillation probability is used. All the nuisance oscillation parameters, i.e. those parameters which are not shown in a given contour, are marginalised (see section 6) except the mass hierarchy which is fixed to either normal or inverted hierarchy. The treatment of the oscillation parameters in the analysis is shown in table 12 while the nominal values are shown in table 13. Two different sets of oscillation parameters were chosen. The Asimov data set A is the best-fit of the Run 1-4 data Mach3 analysis [49]. The Asimov data set B is a variation of Asimov data set A, which is useful for testing the sensitivity to the opposite $\sin^2\theta_{23}$ octant, which reduces the number of expected events in the e-like samples, as well as the sensitivity to δ_{CP}.

Parameter	Sample	Category	Energy range (GeV)
0 | 18	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
1 | 19	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
2 | 20	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
3 | 21	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$E_{reco} < 0.35$
4 | 22	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.35 < E_{reco} < 0.80$
5 | 23	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.80 < E_{reco} < 1.25$
6 | 24	electron-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
7 | 25	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
8 | 26	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
9 | 27	electron-like	all NC	$E_{reco} < 0.35$
10 | 28	electron-like	all NC	$0.35 < E_{reco} < 0.80$
11 | 29	electron-like	all NC	$0.80 < E_{reco} < 1.25$
12 | 30	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$E_{reco} < 0.40$
13 | 31	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$0.40 < E_{reco} < 1.10$
14 | 32	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$1.10 < E_{reco} < 30.0$
15 | 33	muon-like	all $\nu_\mu/\bar{\nu}_\mu$ CCnQE	$E_{reco} < 30.0$
16 | 34	muon-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 30.0$
17 | 35	muon-like	all NC	$E_{reco} < 30.0$
36 | –	ν_e CC1π^+-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.3 < E_{reco} < 0.80$
37 | –	ν_e CC1π^+-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
38 | –	ν_e CC1π^+-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.3 < E_{reco} < 0.80$
39 | –	ν_e CC1π^+-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.80 < E_{reco} < 1.25$
40 | –	ν_e CC1π^+-like	$\nu_e/\bar{\nu}_e$ CC	$0.3 < E_{reco} < 0.80$
41 | –	ν_e CC1π^+-like	$\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
42 | –	ν_e CC1π^+-like	all NC	$0.3 < E_{reco} < 0.80$
43 | –	ν_e CC1π^+-like	all NC	$0.80 < E_{reco} < 1.25$

Table 11: Binning of Super-K selection efficiency systematic parameters in covariance matrices for electron-, muon-, and ν_e CC1π^+-like samples for each category and reconstructed neutrino energy range. The same parametrization is used for both neutrino and antineutrino mode samples. The parameter number is shown for both neutrino (0-17) and antineutrino (18-35) mode for the 0π samples, and neutrino (36-43) mode for the ν_e CC1π^+ sample.

For $\sin^2\theta_{13}$ two different priors can be used, depending on whether the reactor constraint is applied, denoted as "reactors", when applied and "T2K only" when not applied. In the first case a gaussian prior given by the combined measurement of reactor experiments is used, while in the latter case a flat prior which covers a large range of values is adopted. With respect to the older global fit of 2013 [75], the value of $\sin^2\theta_{13}$ has been reduced by about 10% and the associated error is also reduced [76]. Since an excess of events was seen in the T2K Run 1-4 neutrino mode e-like sample and not much data are collected in neutrino mode during the Run 5+6 (only 0.34×10^{20} POT), this change is expected to have an impact on the search for CP violation, making the confidence intervals tighter.

Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	gauss	0.085 ± 0.005
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) T2K only	uniform	$[0; 0.4]$
$\sin^2 2\theta_{12}$	gauss	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	uniform	$[2; 3] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	gauss	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	uniform	$[-\pi; +\pi]$
Mass Hierarchy	fixed	NH or IH

Table 12: Treatment of the oscillation parameters in the $\nu/\bar{\nu}$ joint analysis. All the gaussian priors are from [76]. The parameter $\sin^2 2\theta_{13}$ can be constrained with the measurement of reactor experiments using the gaussian prior, otherwise a uniform prior is used and the measurement relies only on the T2K data. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [76], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [15]. The mass hierarchy is not marginalised but fixed to either NH or IH.

Parameter(s)	Asimov A	Asimov B	Asimov C	Asimov D	Asimov E
$\sin^2\theta_{23}$	0.528	0.45	0.528	0.528	0.528
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	0.085 (0.0217)				
$\sin^2 2\theta_{12}$	0.846				
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4				
Δm^2_{21}	7.53×10^{-5} eV^2/c^4				
δ_{CP}	-1.601	0	0	π	$\frac{\pi}{2}$
Mass Hierarchy	Normal				

Table 13: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [76], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [15]. For the other Asimov data sets, a selection of other reasonable and interesting values for $\sin^2\theta_{23}$ and δ_{CP} were chosen. The mass hierarchy is not marginalised but fixed to either NH or IH.

5. Validation of the $\bar{\nu}/\nu$ joint analysis with Run 1-7c statistics

The major update to the analysis is the number of POT, i.e. 7.482×10^{20} POT for neutrino mode and 7.471×10^{20} POT for antineutrino mode and the inclusion of the new ν_e CC1π^+ sample.

Other minor updates are the new BANFF covariance matrix. The BANFF fit was not performed again on the full ND280 Run 1-7c data set. Instead the Run 1-6 BANFF fit was used with the additional flux uncertainty due to the extra POT added in quadrature. Furthermore the flux tuning was updated in order to account for the Run 7c. However both updates give a very small effect and will not change significantly the systematic uncertainties and event rate in the analysis.

5.1. Prediction of event rate and effect of systematic uncertainties

In tables 2 to 6 the expected number of events for the Run 1-7c collected statistic is shown for the Asimov data set A.

5.2. Evaluation of the effects of systematic parameters on the predicted reconstructed energy spectrum

In this section, we present the effects of the systematics described in section 4.1 on the predicted number of single ring events and the predicted spectra in Super-K. All results are generated for an integrated exposure corresponding to Run 1-7c. and the BANFF-tuned MC templates are used.

In table 14, table 15, table 16, table 17 and table 18 the effect of each single systematic parameter obtained by varying its value by $\pm 1\sigma$ is shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	120.349	125.914	126.429	0.40884 %	125.399	-0.40884 %
$f^{banff}_{1;t,r}$	120.349	125.914	126.409	0.39338 %	125.418	-0.39338 %
$f^{banff}_{2;t,r}$	120.349	125.914	126.033	0.094558 %	125.795	-0.094558 %
$f^{banff}_{3;t,r}$	120.349	125.914	126.012	0.078072 %	125.815	-0.078072 %
$f^{banff}_{4;t,r}$	120.349	125.914	126.834	0.73082 %	124.994	-0.73082 %
$f^{banff}_{5;t,r}$	120.349	125.914	126.771	0.68095 %	125.056	-0.68095 %
$f^{banff}_{6;t,r}$	120.349	125.914	126.733	0.65073 %	125.094	-0.65073 %
$f^{banff}_{7;t,r}$	120.349	125.914	126.452	0.42768 %	125.375	-0.42768 %
$f^{banff}_{8;t,r}$	120.349	125.914	126.471	0.44277 %	125.356	-0.44277 %
$f^{banff}_{9;t,r}$	120.349	125.914	126.126	0.16892 %	125.701	-0.16892 %
$f^{banff}_{10;t,r}$	120.349	125.914	125.96	0.036964 %	125.867	-0.036964 %
$f^{banff}_{11;t,r}$	120.349	125.914	125.941	0.021242 %	125.887	-0.021242 %
$f^{banff}_{12;t,r}$	120.349	125.914	125.934	0.015671 %	125.894	-0.015671 %
$f^{banff}_{13;t,r}$	120.349	125.914	126.002	0.069974 %	125.826	-0.069974 %
$f^{banff}_{14;t,r}$	120.349	125.914	126.08	0.13209 %	125.747	-0.13209 %
$f^{banff}_{15;t,r}$	120.349	125.914	126.063	0.1185 %	125.765	-0.1185 %
$f^{banff}_{16;t,r}$	120.349	125.914	125.914	0.00034323 %	125.913	-0.00034323 %
$f^{banff}_{17;t,r}$	120.349	125.914	125.915	0.00058236 %	125.913	-0.00058236 %
$f^{banff}_{18;t,r}$	120.349	125.914	125.914	0.00041762 %	125.913	-0.00041762 %
$f^{banff}_{19;t,r}$	120.349	125.914	125.916	0.0021262 %	125.911	-0.0021262 %
$f^{banff}_{20;t,r}$	120.349	125.914	125.917	0.0028071 %	125.91	-0.0028071 %
$f^{banff}_{21;t,r}$	120.349	125.914	125.917	0.0024707 %	125.911	-0.0024707 %
$f^{banff}_{22;t,r}$	120.349	125.914	125.916	0.0015978 %	125.912	-0.0015978 %
$f^{banff}_{23;t,r}$	120.349	125.914	125.914	0.00041569 %	125.913	-0.00041569 %
$f^{banff}_{24;t,r}$	120.349	125.914	125.915	0.0011942 %	125.912	-0.0011942 %
$f^{banff}_{Norm_{2p-2h}}$	120.349	125.914	130.333	3.5095 %	121.495	-3.5095 %
$f^{banff}_{C_5^A}$	120.349	125.914	127.621	1.3562 %	124.311	-1.2726 %
$f^{banff}_{Bg_{RES}}$	120.349	125.914	126.985	0.85109 %	124.971	-0.74846 %
$f^{banff}_{M_A^{QE}}$	120.349	125.914	127.832	1.5237 %	123.975	-1.5395 %
$f^{banff}_{M_A^{RES}}$	120.349	125.914	127.008	0.86933 %	124.837	-0.85554 %
$f^{banff}_{p_f}$	120.349	125.914	123.763	-1.7078 %	127.899	1.5763 %
$f^{banff}_{Shape_{CCoth}}$	120.349	125.914	126.435	0.41379 %	125.393	-0.41379 %
$f^{banff}_{E_B}$	120.349	125.914	125.899	-0.01212 %	125.93	0.012715 %
$f^{banff}_{Norm_{CCcoh}}$	120.349	125.914	126.022	0.086181 %	125.805	-0.086181 %
$f^{banff}_{Norm_{NCcoh}}$	120.349	125.914	125.919	0.0044519 %	125.908	-0.0044519 %
$f^{banff}_{Norm_{NCother}}$	120.349	125.914	126.901	0.78393 %	124.927	-0.78393 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	120.349	125.914	125.926	0.0093354 %	125.902	-0.0093354 %
$f^{banff}_{Norm_{NC1\gamma}}$	120.349	125.914	125.918	0.0029615 %	125.91	-0.0029615 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	120.349	125.914	125.914	0.00011837 %	125.914	-0.00011837 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	120.349	125.914	126.172	0.20522 %	125.655	-0.20522 %
$f^{SK}_{E;r}$	120.349	125.914	125.914	-4.5145e-14 %	125.914	-7.9003e-14 %
$f^{SK+FSI}_{0;t,r}$	120.349	125.914	126.074	0.1271 %	125.754	-0.1271 %
$f^{SK+FSI}_{1;t,r}$	120.349	125.914	126.556	0.5104 %	125.271	-0.5104 %
$f^{SK+FSI}_{2;t,r}$	120.349	125.914	126.675	0.60427 %	125.153	-0.60427 %
$f^{SK+FSI}_{3;t,r}$	120.349	125.914	128.014	1.6677 %	123.814	-1.6677 %
$f^{SK+FSI}_{4;t,r}$	120.349	125.914	126.336	0.33551 %	125.491	-0.33551 %
$f^{SK+FSI}_{5;t,r}$	120.349	125.914	130.738	3.8311 %	121.09	-3.8311 %

Table 14: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 7.482×10^{20} POT for neutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	Nominal	BANFF tuned	BANFF tuned +1σ		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	24.2563	26.5935	26.6217	0.10583 %	26.5654	-0.10583 %
$f^{banff}_{1;t,r}$	24.2563	26.5935	26.7186	0.47018 %	26.4685	-0.47018 %
$f^{banff}_{2;t,r}$	24.2563	26.5935	26.8418	0.93345 %	26.3453	-0.93345 %
$f^{banff}_{3;t,r}$	24.2563	26.5935	26.8365	0.91353 %	26.3506	-0.91353 %
$f^{banff}_{4;t,r}$	24.2563	26.5935	26.8768	1.0652 %	26.3102	-1.0652 %
$f^{banff}_{5;t,r}$	24.2563	26.5935	26.6316	0.14302 %	26.5555	-0.14302 %
$f^{banff}_{6;t,r}$	24.2563	26.5935	26.6004	0.025826 %	26.5867	-0.025826 %
$f^{banff}_{7;t,r}$	24.2563	26.5935	26.5978	0.015909 %	26.5893	-0.015909 %
$f^{banff}_{8;t,r}$	24.2563	26.5935	26.5992	0.021436 %	26.5878	-0.021436 %
$f^{banff}_{9;t,r}$	24.2563	26.5935	26.5962	0.010031 %	26.5909	-0.010031 %
$f^{banff}_{10;t,r}$	24.2563	26.5935	26.595	0.0054078 %	26.5921	-0.0054078 %
$f^{banff}_{11;t,r}$	24.2563	26.5935	26.5975	0.014836 %	26.5896	-0.014836 %
$f^{banff}_{12;t,r}$	24.2563	26.5935	26.5963	0.010428 %	26.5907	-0.010428 %
$f^{banff}_{13;t,r}$	24.2563	26.5935	26.596	0.0093637 %	26.591	-0.0093637 %
$f^{banff}_{14;t,r}$	24.2563	26.5935	26.595	0.00542 %	26.5921	-0.00542 %
$f^{banff}_{15;t,r}$	24.2563	26.5935	26.5948	0.0046444 %	26.5923	-0.0046444 %
$f^{banff}_{16;t,r}$	24.2563	26.5935	26.6185	0.093913 %	26.5685	-0.093913 %
$f^{banff}_{17;t,r}$	24.2563	26.5935	26.628	0.12954 %	26.5591	-0.12954 %
$f^{banff}_{18;t,r}$	24.2563	26.5935	26.6109	0.065228 %	26.5762	-0.065228 %
$f^{banff}_{19;t,r}$	24.2563	26.5935	26.6464	0.19869 %	26.5407	-0.19869 %
$f^{banff}_{20;t,r}$	24.2563	26.5935	26.598	0.016936 %	26.589	-0.016936 %
$f^{banff}_{21;t,r}$	24.2563	26.5935	26.594	0.0017641 %	26.5931	-0.0017641 %
$f^{banff}_{22;t,r}$	24.2563	26.5935	26.5937	0.00071333 %	26.5933	-0.00071333 %
$f^{banff}_{23;t,r}$	24.2563	26.5935	26.6008	0.027288 %	26.5863	-0.027288 %
$f^{banff}_{24;t,r}$	24.2563	26.5935	26.5937	0.00079831 %	26.5933	-0.00079831 %
$f^{banff}_{Norm_{2p-2h}}$	24.2563	26.5935	27.6259	3.882 %	25.5612	-3.882 %
$f^{banff}_{C^A_5}$	24.2563	26.5935	26.8014	0.78157 %	26.3972	-0.73812 %
$f^{banff}_{Bg_{RES}}$	24.2563	26.5935	26.6753	0.30745 %	26.5216	-0.27037 %
$f^{banff}_{M^{QE}_A}$	24.2563	26.5935	27.0363	1.6648 %	26.1369	-1.7172 %
$f^{banff}_{M^{RES}_A}$	24.2563	26.5935	26.6925	0.37237 %	26.4929	-0.37839 %
$f^{banff}_{p_f}$	24.2563	26.5935	25.9799	-2.3076 %	27.1501	2.093 %
$f^{banff}_{Shape_{CCoth}}$	24.2563	26.5935	26.6063	0.048094 %	26.5807	-0.048083 %
$f^{banff}_{E_B}$	24.2563	26.5935	26.5533	-0.15134 %	26.626	0.12201 %
$f^{banff}_{Norm_{CCcoh}}$	24.2563	26.5935	26.5991	0.021135 %	26.5879	-0.021135 %
$f^{banff}_{Norm_{NCcoh}}$	24.2563	26.5935	26.6567	0.23764 %	26.5303	-0.23764 %
$f^{banff}_{Norm_{NCother}}$	24.2563	26.5935	26.6352	0.15691 %	26.5518	-0.15691 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	24.2563	26.5935	27.2933	2.6315 %	25.8937	-2.6315 %
$f^{banff}_{Norm_{NC1\gamma}}$	24.2563	26.5935	27.0873	1.8568 %	26.0997	-1.8568 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	24.2563	26.5935	26.6014	0.029817 %	26.5856	-0.029817 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	24.2563	26.5935	26.6061	0.047159 %	26.581	-0.047159 %
$f^{SK}_{E;r}$	24.2563	26.5935	26.5935	1.3359e-14 %	26.5935	-1.3359e-14 %
$f^{SK+FSI}_{6;t,r}$	24.2563	26.5935	26.8445	0.94359 %	26.3426	-0.94359 %
$f^{SK+FSI}_{7;t,r}$	24.2563	26.5935	27.0622	1.7624 %	26.1248	-1.7624 %
$f^{SK+FSI}_{8;t,r}$	24.2563	26.5935	26.7121	0.44605 %	26.4749	-0.44605 %
$f^{SK+FSI}_{9;t,r}$	24.2563	26.5935	26.6525	0.22178 %	26.5345	-0.22178 %
$f^{SK+FSI}_{10;t,r}$	24.2563	26.5935	26.6244	0.116 %	26.5627	-0.116 %
$f^{SK+FSI}_{11;t,r}$	24.2563	26.5935	26.6021	0.0321 %	26.585	-0.0321 %
$f^{SK+FSI}_{12;t,r}$	24.2563	26.5935	26.6184	0.093385 %	26.5687	-0.093385 %
$f^{SK+FSI}_{13;t,r}$	24.2563	26.5935	26.6657	0.27133 %	26.5214	-0.27133 %
$f^{SK+FSI}_{14;t,r}$	24.2563	26.5935	26.6662	0.27323 %	26.5209	-0.27323 %
$f^{SK+FSI}_{15;t,r}$	24.2563	26.5935	26.8604	1.0036 %	26.3266	-1.0036 %
$f^{SK+FSI}_{16;t,r}$	24.2563	26.5935	26.7358	0.53512 %	26.4512	-0.53512 %
$f^{SK+FSI}_{17;t,r}$	24.2563	26.5935	26.7166	0.46264 %	26.4705	-0.46264 %

Table 15: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 7.482×10^{20} POT for neutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}RHC$	33.354	34.2841	34.3212	0.1084 %	34.2469	-0.1084 %
$f^{banff}_{1;t,r}RHC$	33.354	34.2841	34.3316	0.13851 %	34.2366	-0.13851 %
$f^{banff}_{2;t,r}RHC$	33.354	34.2841	34.4224	0.4034 %	34.1458	-0.4034 %
$f^{banff}_{3;t,r}RHC$	33.354	34.2841	34.5137	0.66983 %	34.0544	-0.66983 %
$f^{banff}_{4;t,r}RHC$	33.354	34.2841	34.5214	0.69216 %	34.0468	-0.69216 %
$f^{banff}_{5;t,r}RHC$	33.354	34.2841	34.4278	0.41934 %	34.1403	-0.41934 %
$f^{banff}_{6;t,r}RHC$	33.354	34.2841	34.3655	0.23759 %	34.2026	-0.23759 %
$f^{banff}_{7;t,r}RHC$	33.354	34.2841	34.301	0.049526 %	34.2671	-0.049526 %
$f^{banff}_{8;t,r}RHC$	33.354	34.2841	34.2964	0.036085 %	34.2717	-0.036085 %
$f^{banff}_{9;t,r}RHC$	33.354	34.2841	34.4627	0.52119 %	34.1054	-0.52119 %
$f^{banff}_{10;t,r}RHC$	33.354	34.2841	34.4614	0.51727 %	34.1067	-0.51727 %
$f^{banff}_{11;t,r}RHC$	33.354	34.2841	34.4688	0.53874 %	34.0994	-0.53874 %
$f^{banff}_{12;t,r}RHC$	33.354	34.2841	34.3882	0.30381 %	34.1799	-0.30381 %
$f^{banff}_{13;t,r}RHC$	33.354	34.2841	34.3968	0.32868 %	34.1714	-0.32868 %
$f^{banff}_{14;t,r}RHC$	33.354	34.2841	34.3136	0.086075 %	34.2546	-0.086075 %
$f^{banff}_{15;t,r}RHC$	33.354	34.2841	34.2898	0.016631 %	34.2784	-0.016631 %
$f^{banff}_{16;t,r}RHC$	33.354	34.2841	34.2853	0.0035678 %	34.2828	-0.0035678 %
$f^{banff}_{17;t,r}RHC$	33.354	34.2841	34.2856	0.0045846 %	34.2825	-0.0045846 %
$f^{banff}_{18;t,r}RHC$	33.354	34.2841	34.2841	0.00019776 %	34.284	-0.00019776 %
$f^{banff}_{19;t,r}RHC$	33.354	34.2841	34.2841	0.00023211 %	34.284	-0.00023211 %
$f^{banff}_{20;t,r}RHC$	33.354	34.2841	34.2841	0.00018231 %	34.284	-0.00018231 %
$f^{banff}_{21;t,r}RHC$	33.354	34.2841	34.2843	0.0007317 %	34.2838	-0.0007317 %
$f^{banff}_{22;t,r}RHC$	33.354	34.2841	34.2845	0.0011599 %	34.2837	-0.0011599 %
$f^{banff}_{23;t,r}RHC$	33.354	34.2841	34.2846	0.0015682 %	34.2835	-0.0015682 %
$f^{banff}_{24;t,r}RHC$	33.354	34.2841	34.2846	0.0016844 %	34.2835	-0.0016844 %
$f^{banff}_{Norm_{2p-2h}}$	33.354	34.2841	35.2949	2.9484 %	33.2732	-2.9484 %
$f^{banff}_{C_5^A}$	33.354	34.2841	34.7665	1.4073 %	33.8496	-1.2673 %
$f^{banff}_{Bg_{RES}}$	33.354	34.2841	34.6114	0.95474 %	33.9962	-0.8396 %
$f^{banff}_{M_A^{QE}}$	33.354	34.2841	34.7782	1.4412 %	33.7964	-1.4226 %
$f^{banff}_{M_A^{RES}}$	33.354	34.2841	34.557	0.79619 %	34.0229	-0.76184 %
$f^{banff}_{p_f}$	33.354	34.2841	33.7369	-1.596 %	34.8188	1.5598 %
$f^{banff}_{Shape_{CCoth}}$	33.354	34.2841	34.4278	0.41939 %	34.1403	-0.41939 %
$f^{banff}_{E_B}$	33.354	34.2841	34.2825	-0.0045403 %	34.2856	0.0043363 %
$f^{banff}_{Norm_{CCcoh}}$	33.354	34.2841	34.3434	0.17311 %	34.2247	-0.17311 %
$f^{banff}_{Norm_{NCcoh}}$	33.354	34.2841	34.2866	0.0073745 %	34.2815	-0.0073745 %
$f^{banff}_{Norm_{NCother}}$	33.354	34.2841	34.5427	0.75445 %	34.0254	-0.75445 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	33.354	34.2841	34.2846	0.0015892 %	34.2835	-0.0015892 %
$f^{banff}_{Norm_{NC1\gamma}}$	33.354	34.2841	34.2844	0.00099164 %	34.2837	-0.00099164 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	33.354	34.2841	34.2848	0.0022231 %	34.2833	-0.0022231 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	33.354	34.2841	34.9123	1.8323 %	33.6559	-1.8323 %
$f^{SK}_{E;r}$	33.354	34.2841	34.2841	-2.0725e-14 %	34.2841	-2.0725e-14 %
$f^{SK+FSI}_{0;t,r}RHC$	33.354	34.2841	34.3274	0.12641 %	34.2407	-0.12641 %
$f^{SK+FSI}_{1;t,r}RHC$	33.354	34.2841	34.4079	0.36115 %	34.1602	-0.36115 %
$f^{SK+FSI}_{2;t,r}RHC$	33.354	34.2841	34.5168	0.67883 %	34.0513	-0.67883 %
$f^{SK+FSI}_{3;t,r}RHC$	33.354	34.2841	35.0403	2.2058 %	33.5278	-2.2058 %
$f^{SK+FSI}_{4;t,r}RHC$	33.354	34.2841	34.3304	0.13529 %	34.2377	-0.13529 %
$f^{SK+FSI}_{5;t,r}RHC$	33.354	34.2841	35.3403	3.0809 %	33.2278	-3.0809 %

Table 16: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 7.471×10^{20} POT for antineutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	Nominal	BANFF tuned	BANFF tuned +1σ		BANFF tuned -1σ	
$f^{banff}_{0;t,r}RHC$	3.11707	3.21831	3.23425	0.49537 %	3.20237	-0.49537 %
$f^{banff}_{1;t,r}RHC$	3.11707	3.21831	3.22891	0.32934 %	3.20771	-0.32934 %
$f^{banff}_{2;t,r}RHC$	3.11707	3.21831	3.22364	0.16581 %	3.21297	-0.16581 %
$f^{banff}_{3;t,r}RHC$	3.11707	3.21831	3.22045	0.066577 %	3.21617	-0.066577 %
$f^{banff}_{4;t,r}RHC$	3.11707	3.21831	3.22085	0.078879 %	3.21577	-0.078879 %
$f^{banff}_{5;t,r}RHC$	3.11707	3.21831	3.22043	0.065975 %	3.21619	-0.065975 %
$f^{banff}_{6;t,r}RHC$	3.11707	3.21831	3.22509	0.21083 %	3.21152	-0.21083 %
$f^{banff}_{7;t,r}RHC$	3.11707	3.21831	3.23439	0.49965 %	3.20223	-0.49965 %
$f^{banff}_{8;t,r}RHC$	3.11707	3.21831	3.2365	0.56517 %	3.20012	-0.56517 %
$f^{banff}_{9;t,r}RHC$	3.11707	3.21831	3.25195	1.0453 %	3.18467	-1.0453 %
$f^{banff}_{10;t,r}RHC$	3.11707	3.21831	3.2239	0.17371 %	3.21272	-0.17371 %
$f^{banff}_{11;t,r}RHC$	3.11707	3.21831	3.21966	0.041857 %	3.21696	-0.041857 %
$f^{banff}_{12;t,r}RHC$	3.11707	3.21831	3.21896	0.020249 %	3.21766	-0.020249 %
$f^{banff}_{13;t,r}RHC$	3.11707	3.21831	3.21907	0.023667 %	3.21755	-0.023667 %
$f^{banff}_{14;t,r}RHC$	3.11707	3.21831	3.21857	0.0081707 %	3.21805	-0.0081707 %
$f^{banff}_{15;t,r}RHC$	3.11707	3.21831	3.21855	0.0074537 %	3.21807	-0.0074537 %
$f^{banff}_{16;t,r}RHC$	3.11707	3.21831	3.23255	0.44247 %	3.20407	-0.44247 %
$f^{banff}_{17;t,r}RHC$	3.11707	3.21831	3.21849	0.0055449 %	3.21813	-0.0055449 %
$f^{banff}_{18;t,r}RHC$	3.11707	3.21831	3.22275	0.13805 %	3.21387	-0.13805 %
$f^{banff}_{19;t,r}RHC$	3.11707	3.21831	3.22312	0.14949 %	3.2135	-0.14949 %
$f^{banff}_{20;t,r}RHC$	3.11707	3.21831	3.22095	0.081949 %	3.21567	-0.081949 %
$f^{banff}_{21;t,r}RHC$	3.11707	3.21831	3.22701	0.2705 %	3.2096	-0.2705 %
$f^{banff}_{22;t,r}RHC$	3.11707	3.21831	3.21896	0.020273 %	3.21766	-0.020273 %
$f^{banff}_{23;t,r}RHC$	3.11707	3.21831	3.21839	0.002446 %	3.21823	-0.002446 %
$f^{banff}_{24;t,r}RHC$	3.11707	3.21831	3.21838	0.002219 %	3.21824	-0.002219 %
$f^{banff}_{Norm_{2p-2h}}$	3.11707	3.21831	3.31425	2.9812 %	3.12237	-2.9812 %
$f^{banff}_{C^A_5}$	3.11707	3.21831	3.25166	1.0364 %	3.19031	-0.86988 %
$f^{banff}_{Bg_{RES}}$	3.11707	3.21831	3.23459	0.5058 %	3.20399	-0.4448 %
$f^{banff}_{M^{QE}_A}$	3.11707	3.21831	3.26276	1.3812 %	3.17411	-1.3732 %
$f^{banff}_{M^{RES}_A}$	3.11707	3.21831	3.22955	0.34945 %	3.20849	-0.30505 %
$f^{banff}_{p_f}$	3.11707	3.21831	3.14236	-2.3599 %	3.29191	2.2869 %
$f^{banff}_{Shape_{CCoth}}$	3.11707	3.21831	3.22119	0.089515 %	3.21543	-0.089518 %
$f^{banff}_{E_B}$	3.11707	3.21831	3.21749	-0.025439 %	3.21891	0.018569 %
$f^{banff}_{Norm_{CCcoh}}$	3.11707	3.21831	3.22518	0.21355 %	3.21144	-0.21355 %
$f^{banff}_{Norm_{NCcoh}}$	3.11707	3.21831	3.2514	1.0282 %	3.18522	-1.0282 %
$f^{banff}_{Norm_{NCother}}$	3.11707	3.21831	3.22886	0.32779 %	3.20776	-0.32779 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	3.11707	3.21831	3.24258	0.75412 %	3.19404	-0.75412 %
$f^{banff}_{Norm_{NC1\gamma}}$	3.11707	3.21831	3.33932	3.76 %	3.0973	-3.76 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	3.11707	3.21831	3.27411	1.734 %	3.1625	-1.734 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	3.11707	3.21831	3.2943	2.3612 %	3.14232	-2.3612 %
$f^{SK}_{E;r}$	3.11707	3.21831	3.21831	-1.3799e-14 %	3.21831	-4.1397e-14 %
$f^{SK+FSI}_{6;t,r}RHC$	3.11707	3.21831	3.22805	0.30255 %	3.20857	-0.30255 %
$f^{SK+FSI}_{7;t,r}RHC$	3.11707	3.21831	3.25713	1.2064 %	3.17948	-1.2064 %
$f^{SK+FSI}_{8;t,r}RHC$	3.11707	3.21831	3.24365	0.78746 %	3.19297	-0.78746 %
$f^{SK+FSI}_{9;t,r}RHC$	3.11707	3.21831	3.22378	0.16988 %	3.21284	-0.16988 %
$f^{SK+FSI}_{10;t,r}RHC$	3.11707	3.21831	3.2219	0.11164 %	3.21472	-0.11164 %
$f^{SK+FSI}_{11;t,r}RHC$	3.11707	3.21831	3.22052	0.068625 %	3.2161	-0.068625 %
$f^{SK+FSI}_{12;t,r}RHC$	3.11707	3.21831	3.22172	0.10612 %	3.21489	-0.10612 %
$f^{SK+FSI}_{13;t,r}RHC$	3.11707	3.21831	3.23071	0.38524 %	3.20591	-0.38524 %
$f^{SK+FSI}_{14;t,r}RHC$	3.11707	3.21831	3.23982	0.66827 %	3.1968	-0.66827 %
$f^{SK+FSI}_{15;t,r}RHC$	3.11707	3.21831	3.28217	1.9844 %	3.15444	-1.9844 %
$f^{SK+FSI}_{16;t,r}RHC$	3.11707	3.21831	3.25076	1.0083 %	3.18586	-1.0083 %
$f^{SK+FSI}_{17;t,r}RHC$	3.11707	3.21831	3.2479	0.91957 %	3.18871	-0.91957 %

Table 17: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 7.471×10^{20} POT for antineutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	1 σ value	BANFF -3σ		BANFF tuned -1σ		BANFF tuned +1σ		BANFF tuned +3σ	
All syst		0.290	-91.8868 (%)	1.859	-48.079 (%)	6.241	74.28 (%)	15.768	340.265 (%)
BANFF 0, SK numu flux, 0.0 - 0.4 GeV	0.098 73	3.576	−0.152 20	3.579	−0.050 73	3.583	0.050 73	3.586	0.152 20
BANFF 1, SK numu flux, 0.4 - 0.5 GeV	0.103 49	3.519	−1.732 79	3.560	−0.577 60	3.602	0.577 60	3.643	1.732 79
BANFF 2, SK numu flux, 0.5 - 0.6 GeV	0.096 44	3.414	−4.664 33	3.525	−1.554 80	3.637	1.554 80	3.748	4.664 33
BANFF 3, SK numu flux, 0.6 - 0.7 GeV	0.086 70	3.352	−6.395 54	3.505	−2.131 80	3.657	2.131 80	3.810	6.395 54
BANFF 4, SK numu flux, 0.7 - 1.0 GeV	0.113 05	3.197	−10.735 00	3.453	−3.578 30	3.709	3.578 30	3.965	10.735 00
BANFF 5, SK numu flux, 1.0 - 1.5 GeV	0.091 75	3.535	−1.287 86	3.566	−0.429 29	3.596	0.429 29	3.627	1.287 86
BANFF 6, SK numu flux, 1.5 - 2.5 GeV	0.070 17	3.571	−0.270 70	3.578	−0.090 23	3.584	0.090 23	3.591	0.270 70
BANFF 7, SK numu flux, 2.5 - 3.5 GeV	0.073 68	3.577	−0.118 10	3.580	−0.039 37	3.582	0.039 37	3.585	0.118 10
BANFF 8, SK numu flux, 3.5 - 5.0 GeV	0.087 37	3.569	−0.323 13	3.577	−0.107 71	3.585	0.107 71	3.593	0.323 13
BANFF 9, SK numu flux, 5.0 - 7.0 GeV	0.097 94	3.577	−0.120 03	3.580	−0.040 01	3.582	0.040 01	3.585	0.120 03
BANFF 10, SK numu flux, 7.0 - 30.0 GeV	0.114 36	3.579	−0.041 46	3.580	−0.013 82	3.581	0.013 82	3.582	0.041 46
BANFF 11, SK numubar flux, 0.0 - 0.7 GeV	0.102 58	3.580	−0.023 56	3.581	−0.007 85	3.581	0.007 85	3.582	0.023 56
BANFF 12, SK numubar flux, 0.7 - 1.0 GeV	0.078 53	3.580	−0.018 35	3.581	−0.006 12	3.581	0.006 12	3.582	0.018 35
BANFF 13, SK numubar flux, 1.0 - 1.5 GeV	0.084 45	3.580	−0.030 79	3.581	−0.010 27	3.581	0.010 27	3.582	0.030 79
BANFF 14, SK numubar flux, 1.5 - 2.5 GeV	0.085 57	3.580	−0.027 20	3.581	−0.009 07	3.581	0.009 07	3.582	0.027 20
BANFF 15, SK numubar flux, 2.5 - 30.0 GeV	0.086 43	3.580	−0.034 63	3.581	−0.011 55	3.581	0.011 55	3.582	0.034 63
BANFF 16, SK nue flux, 0.0 - 0.5 GeV	0.089 70	3.576	−0.148 19	3.579	−0.049 40	3.583	0.049 40	3.586	0.148 19
BANFF 17, SK nue flux, 0.5 - 0.7 GeV	0.089 95	3.554	−0.746 64	3.572	−0.248 88	3.590	0.248 88	3.608	0.746 64
BANFF 18, SK nue flux, 0.7 - 0.8 GeV	0.085 96	3.564	−0.464 61	3.575	−0.154 87	3.587	0.154 87	3.598	0.464 61
BANFF 19, SK nue flux, 0.8 - 1.5 GeV	0.080 92	3.521	−1.684 75	3.561	−0.561 58	3.601	0.561 58	3.641	1.684 75
BANFF 20, SK nue flux, 1.5 - 2.5 GeV	0.078 97	3.577	−0.117 65	3.580	−0.039 22	3.582	0.039 22	3.585	0.117 65
BANFF 21, SK nue flux, 2.5 - 4.0 GeV	0.083 85	3.580	−0.023 40	3.581	−0.007 80	3.581	0.007 80	3.582	0.023 40
BANFF 22, SK nue flux, 4.0 - 30.0 GeV	0.093 89	3.581	−0.011 81	3.581	−0.003 94	3.581	0.003 94	3.581	0.011 81
BANFF 23, SK nuebar flux, 0.0 - 2.5 GeV	0.074 03	3.581	−0.013 18	3.581	−0.004 39	3.581	0.004 39	3.581	0.013 18
BANFF 24, SK nuebar flux, 2.5 - 30.0 GeV	0.128 42	3.581	−0.005 75	3.581	−0.001 92	3.581	0.001 92	3.581	0.005 75
BANFF; Norm; MEC	0.352 28	3.539	−1.173 79	3.567	−0.391 26	3.595	0.391 26	3.623	1.173 79
BANFF; CA5 RES	0.118 81	2.621	−26.796 00	3.231	−9.769 40	3.961	10.607 00	4.811	34.332 50
BANFF; Norm; BgRES Isospin 1/2	0.153 85	3.410	−4.780 65	3.512	−1.912 70	3.661	2.231 40	3.855	7.647 92
BANFF, Ma QE	0.058 01	3.556	−0.694 38	3.573	−0.220 64	3.588	0.209 43	3.602	0.594 85
BANFF, Ma RES	0.157 90	2.482	−30.682 60	3.267	−8.761 70	3.842	7.292 60	4.238	18.334 30
BANFF; Fermi Momentum	0.054 67	3.587	0.169 57	3.584	0.087 06	3.578	−0.092 65	3.571	−0.287 50
BANFF; Shape; CC Oth	0.400 00	3.509	−2.005 74	3.552	−0.819 04	3.610	0.819 04	3.669	2.457 12
BANFF; Binding energy	0.333 33	3.580	−0.016 69	3.581	−0.010 36	3.581	0.008 74	3.581	0.012 61
BANFF; Norm, CC Coh	0.300 00	3.265	−8.823 97	3.476	−2.941 30	3.686	2.941 30	3.897	8.823 97
BANFF; Norm, NC Coh	0.300 00	3.581	0	3.581	0	3.581	0	3.581	0
BANFF; Norm, NC Oth	0.300 00	3.502	−2.193 68	3.555	−0.731 23	3.607	0.731 23	3.660	2.193 68
BANFF; Norm, ν_e To ν_μ	0.028 28	3.314	−7.449 33	3.492	−2.483 10	3.670	2.483 10	3.848	7.449 33
BANFF; Norm; NC 1 γ	1	3.535	−1.276 93	3.566	−0.425 64	3.596	0.425 64	3.627	1.276 93
BANFF; Norm, $\bar{\nu}_e$ To $\bar{\nu}_\mu$	0.028 28	3.581	−0.007 81	3.581	−0.002 60	3.581	0.002 60	3.581	0.007 81
BANFF; Norm; MECBar	1	3.579	−0.045 96	3.580	−0.015 32	3.582	0.015 32	3.583	0.045 96
SKDet + FSI/SI 0; E_{reco}(0.30 - 0.80)GeV; Osc. ν_e CC (ν_e CC1π^+)	0.156 15	2.711	−24.285 10	3.291	−8.095 00	3.871	8.095 00	4.451	24.285 10
SKDet + FSI/SI 1; E_{reco}(0.80 - 1.25)GeV; Osc. ν_e CC (ν_e CC1π^+)	0.137 57	3.234	−9.684 98	3.465	−3.228 30	3.697	3.228 30	3.928	9.684 98
SKDet + FSI/SI 2; E_{reco}(0.30 - 0.80)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC (ν_e CC1π^+)	0.565 55	3.242	−9.460 58	3.468	−3.153 50	3.694	3.153 50	3.920	9.460 58
SKDet + FSI/SI 3; E_{reco}(0.80 - 1.25)GeV; $\nu_\mu/\bar{\nu}_\mu$ CC (ν_e CC1π^+)	0.481 01	3.482	−2.777 10	3.548	−0.925 70	3.614	0.925 70	3.680	2.777 10
SKDet + FSI/SI 4; E_{reco}(0.30 - 0.80)GeV; $\nu_e/\bar{\nu}_e$ CC (ν_e CC1π^+)	0.148 36	3.498	−2.317 96	3.553	−0.772 65	3.609	0.772 65	3.664	2.317 96
SKDet + FSI/SI 5; E_{reco}(0.80 - 1.25)GeV; $\nu_e/\bar{\nu}_e$ CC (ν_e CC1π^+)	0.145 69	3.466	−3.219 56	3.543	−1.073 20	3.619	1.073 20	3.696	3.219 56
SKDet + FSI/SI 6; E_{reco}(0.30 - 0.80)GeV; all NC (ν_e CC1π^+)	1.168 26	3.153	−11.939 30	3.438	−3.979 80	3.724	3.979 80	4.009	11.939 30
SKDet + FSI/SI 7; E_{reco}(0.80 - 1.25)GeV; all NC (ν_e CC1π^+)	0.586 98	3.505	−2.130 22	3.556	−0.710 07	3.606	0.710 07	3.657	2.130 22
SK energy scale	0.024 00	3.581	0	3.581	-1.24e-14	3.581	-1.24e-14	3.581	0

Table 18: $\pm1\sigma$ effect of the systematic parameters on the total number of ν_e CC1π^+ Super-K events with $7.482\,00 \times 10^{20}$ POT for neutrino mode. Only the systematic parameters which effect this sample are shown.

The effects of all the combined systematic uncertainties on the reconstructed energy spectrum of events is shown in figs. 12 and 13 for the typical oscillation scenario with parameters of Asimov datasets A. These plots show the total error envelope for the reconstructed energy spectra; this is calculated as the $\pm 1\sigma$ spread with respect to the median of bin contents from 10k toy MC experiments generated with randomised systematic parameters. All systematic parameters are considered and their correlations are taken into account. The spectra were tuned with the systematic parameters measured with the ND280 data shown in tables 7 to 9, the nominal Super-K systematic parameters shown in table 10 and the oscillation parameter priors shown in table 12. In figs. 14 to 23 the error envelopes and fractional errors of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.

The effects of systematic uncertainties on the predicted number of events are summarised in tables 19 to 23 with the typical parameter values given in table 13. For the measurement of δ_{CP} it is important to understand the effect of the systematic uncertainties on the ratio of the number of events between neutrino and antineutrino mode samples, which is the consequence of correlations between systematic parameters. If correlations do not reproduce the physics correctly, a fake asymmetry between neutrino and antineutrino oscillations can be seen. On the other hand if correlations exist but are not taken into account, the measurement of δ_{CP} could be biased. Depending on the value of δ_{CP}, correlations or anti-correlations between parameters are more conservative. If true $\delta_{CP} = \pm\pi/2$ the most conservative case is given by anti-correlation in the e-like samples, since the oscillation asymmetry could be hidden by the systematic uncertainties. If true $\delta_{CP} \sim 0$ uncorrelated parameters reduce the sensitivity to δ_{CP}. The effect of the systematic parameters on the ratio between neutrino and antineutrino mode events are shown in table 24. The most critical systematic parameters are σ_{ν_e} and $\sigma_{\bar{\nu}_e}$ which are applied respectively to ν_e and $\bar{\nu}_e$ events. Their fractional error is 2.83% and are partially anti-correlated with a non-diagonal elements in the covariance matrix of -0.0004 [43].

Source of uncertainty	$\delta N_{SK}/N_{SK}$ Asimov A
SKDet+FSI+SI	4.13%
FSI+SI	1.47%
SKDet	3.89%
Flux	3.54%
MEC (corr)	3.52%
MECbar (corr)	0.21%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.01%
XSec Tot	4.07%
Flux+XSec	2.90%
Flux+XSec (Pre ND280)	10.90%
Oscillations	0.02%
All	5.08%
All (Pre ND280)	12.25%

Table 19: Effect of 1σ variations on the total number of events 1-ring muon-like events for neutrino mode given by all the systematic uncertainties, obtained by performing 10k toys MC. The RMS / Mean of the distribution is assumed to be the 1σ uncertainty. The systematic parameter values measured with the ND280 data shown in table 7, 8 9 and 10 and the oscillation parameters of the Asimov data set A shown in table 13 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ Asimov A
SKDet+FSI+SI	3.91%
FSI+SI	2.05%
SKDet	3.34%
Flux	3.79%
MEC (corr)	2.96%
MECbar (corr)	1.82%
NC 1gamma (uncorr)	0.00%
XSec nue/numu	0.00%
XSec Tot	4.16%
Flux+XSec	3.39%
Flux+XSec	11.84%
Oscillations	0.03%
All	5.19%
All (Pre ND280)	12.88%

Table 20: Effect of 1σ variations on the total number of events 1-ring muon-like events for antineutrino mode given by all the systematic uncertainties, obtained by performing 10k toys MC. The RMS / Mean of the distribution is assumed to be the 1σ uncertainty. The systematic parameter values measured with the ND280 data shown in table 7, 8 9 and 10 and the oscillation parameters of the Asimov data set A shown in table 13 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ Asimov A
SKDet+FSI+SI	3.47%
FSI+SI	2.50%
SKDet	2.45%
Flux	3.66%
MEC (corr)	3.90%
MECbar (corr)	0.05%
NC 1gamma (uncorr)	1.46%
XSec nue/numu	2.63%
XSec Tot	5.14%
Flux+XSec	4.12%
Flux+XSec (Pre ND280)	11.47%
Oscillations	0.49%
All	5.53%
All (Pre ND280)	12.06%

Table 21: Effect of 1σ variations on the total number of events 1-ring electron-like events for neutrino mode given by all the systematic uncertainties, obtained by performing 10k toys MC. The RMS / Mean of the distribution is assumed to be the 1σ uncertainty. The systematic parameter values measured with the ND280 data shown in table 7, 8 9 and 10 and the oscillation parameters of the Asimov data set A shown in table 13 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ Asimov A
SKDet+FSI+SI	3.85%
FSI+SI	2.45%
SKDet	3.00%
Flux	3.83%
MEC (corr)	3.03%
MECbar (corr)	2.35%
NC 1gamma (uncorr)	2.98%
XSec nue/numu	1.49%
XSec Tot	5.50%
Flux+XSec (Pre ND280)	4.67%
Flux+XSec	13.38%
Oscillations	0.27%
All	6.31%
All (Pre ND280)	14.06%

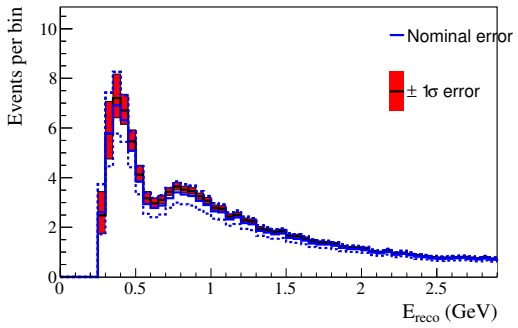
Table 22: Effect of 1σ variations on the total number of events 1-ring electron-like events for antineutrino mode given by all the systematic uncertainties, obtained by performing 10k toys MC. The RMS / Mean of the distribution is assumed to be the 1σ uncertainty. The systematic parameter values measured with the ND280 data shown in table 7, 8 9 and 10 and the oscillation parameters of the Asimov data set A shown in table 13 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ Asimov A
SKDet+FSI+SI	14.10%
FSI+SI	10.39%
SKDet	9.22%
Flux	3.58%
MEC (corr)	0.48%
MECbar (corr)	0.01%
NC 1gamma (uncorr)	0.44%
XSec nue/numu	2.43%
XSec Tot	4.80%
Flux+XSec	5.03%
Flux+XSec (Pre ND280)	16.26%
Oscillations	0.42%
All	14.84%
All (Pre ND280)	21.84%

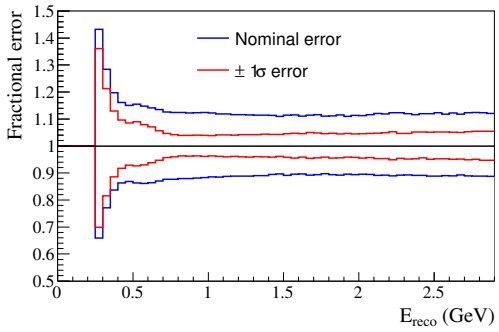
Table 23: Effect of 1σ variations on the total number of events CC1π^+-like events for neutrino mode given by all the systematic uncertainties, obtained by performing 10k toys MC. The RMS / Mean of the distribution is assumed to be the 1σ uncertainty. The systematic parameter values measured with the ND280 data shown in table 7, 8 9 and 10 and the oscillation parameters of the Asimov data set A shown in table 13 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ e-like	$\delta N_{SK}/N_{SK}$ μ-like
SKDet+FSI+SI	3.17%	2.67%
FSI+SI	2.96%	2.57%
SKDet	1.32%	0.69%
Flux	1.77%	1.76%
MEC (corr)	0.56%	0.56%
MECbar (corr)	2.30%	1.61%
NC 1gamma (uncorr)	1.57%	0.00%
XSec nue/numu	3.06%	0.01%
XSec Tot	4.27%	1.70%
Flux+XSec	4.12%	1.88%
Oscillations	0.00%	0.00%
All	5.32%	3.27%
All (Pre ND280)	8.48%	6.86%

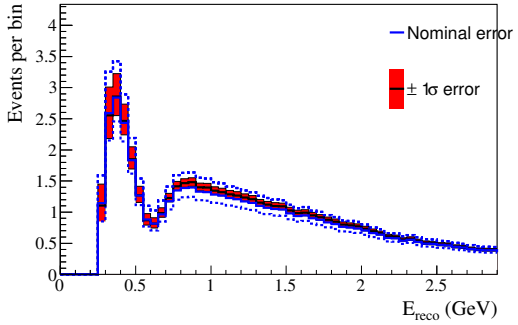
Table 24: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron-like and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 7, 8 9 and 10 and the oscillation parameters of the Asimov data set A shown in table 13 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



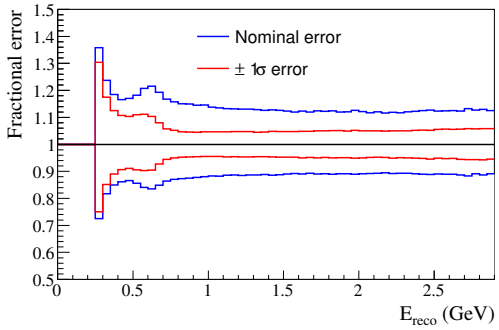
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

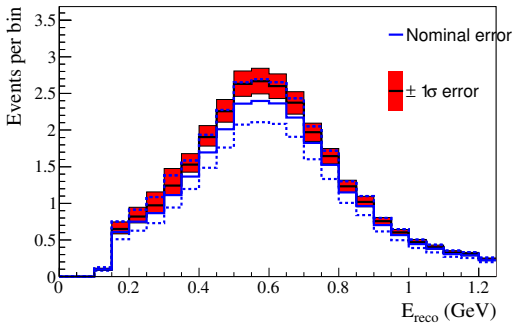


(c) Muon-like, antineutrino mode

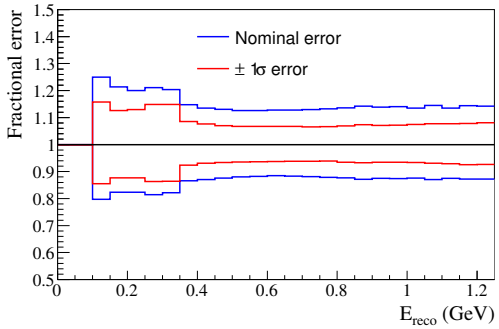


(d) Muon-like, antineutrino mode

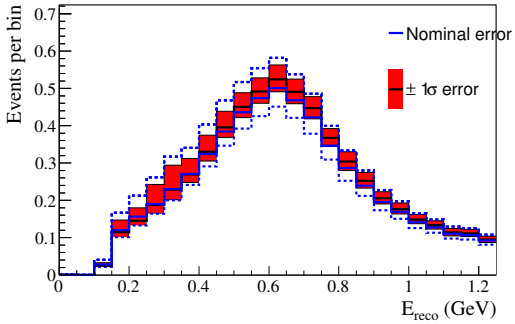
Figure 10: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all the systematic parameters. Both pre- (blue), assuming the CCQE parameters to be gaussian distributed instead of flat, and post- (red) BANFF fit errors are shown. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



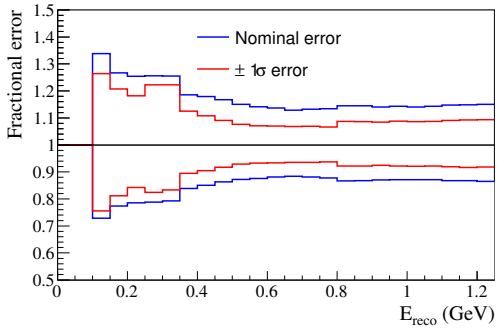
(a) Electron-like, neutrino mode



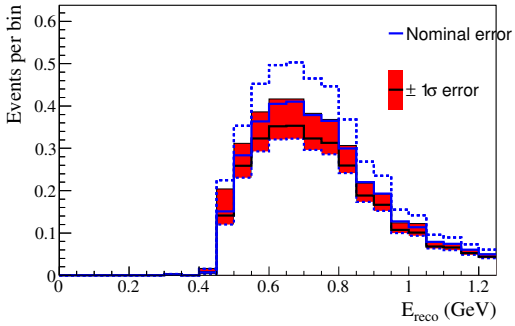
(b) Electron-like, neutrino mode



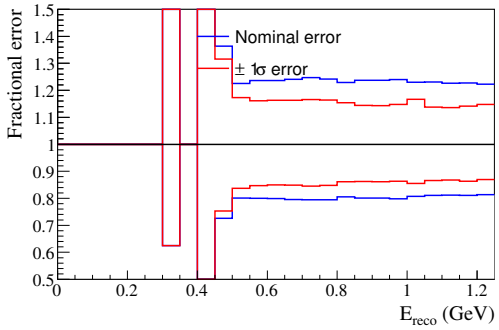
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode

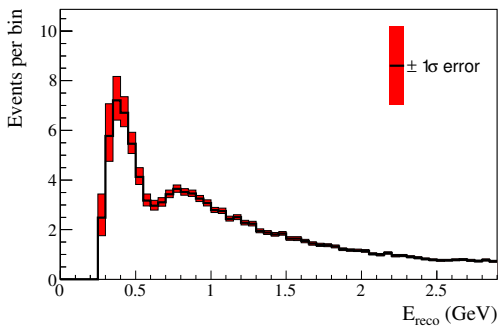


(e) ν_e CC1π^+, neutrino mode

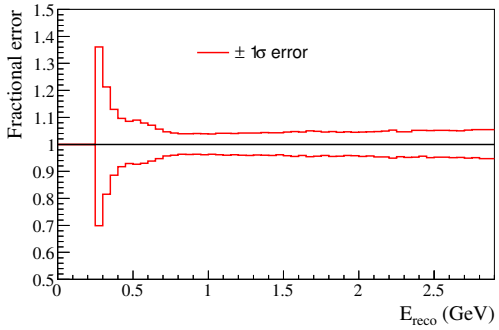


(f) ν_e CC1π^+, neutrino mode

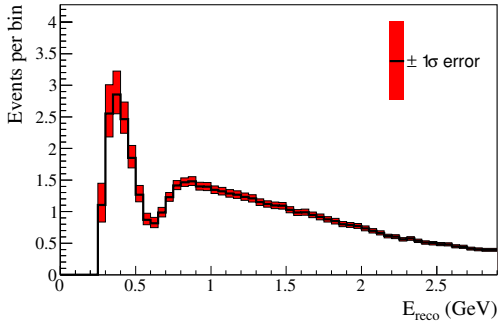
Figure 11: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all the systematic parameters. Both pre- (blue), assuming the CCQE parameters to be gaussian distributed instead of flat, and post- (red) BANFF fit errors are shown. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



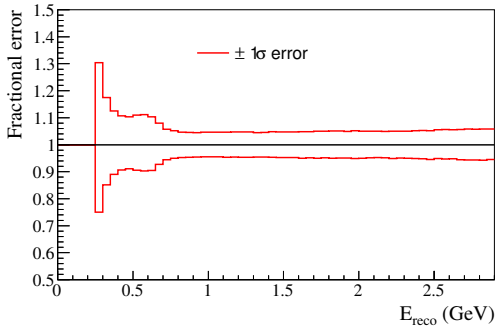
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

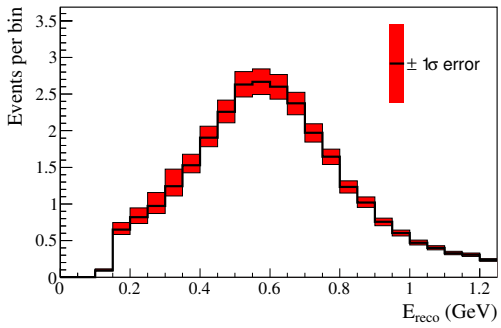


(c) Muon-like, antineutrino mode

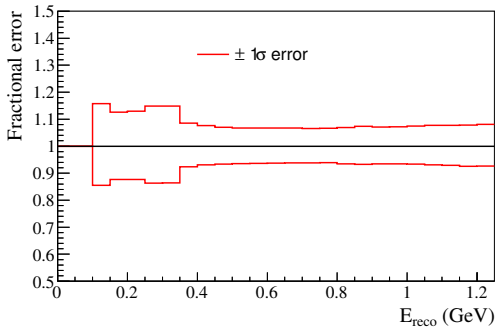


(d) Muon-like, antineutrino mode

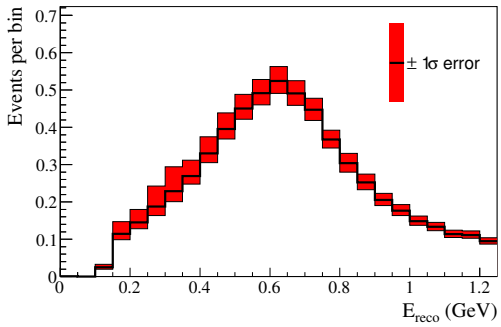
Figure 12: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all the systematic parameters. The errors shown are those once the ND280 data has been fitted. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



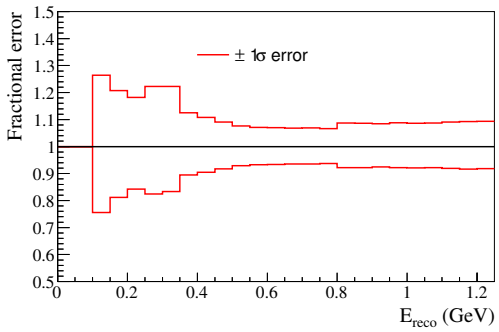
(a) Electron-like, neutrino mode



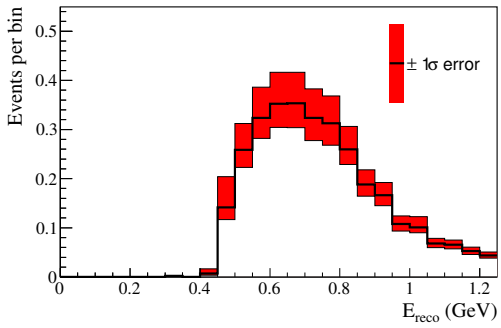
(b) Electron-like, neutrino mode



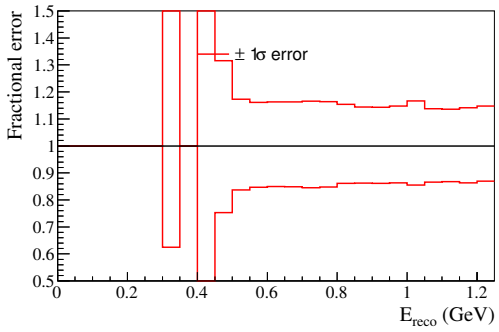
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode

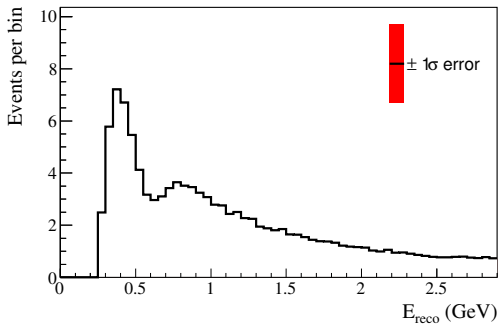


(e) ν_e CC1π^+, neutrino mode

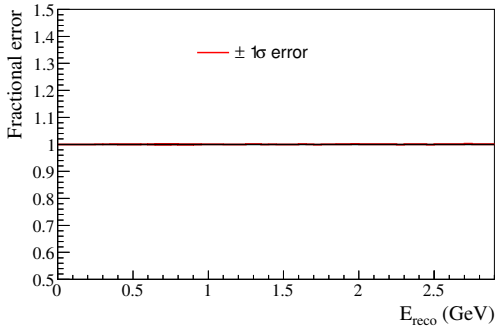


(f) ν_e CC1π^+, neutrino mode

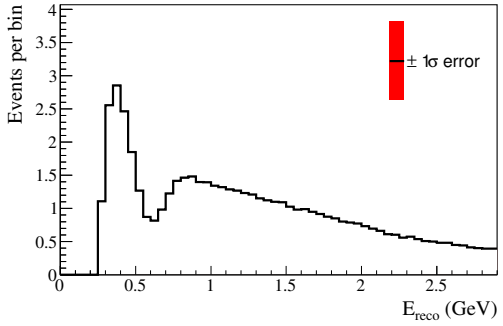
Figure 13: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all the systematic parameters. The errors shown are those once the ND280 data has been fitted. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



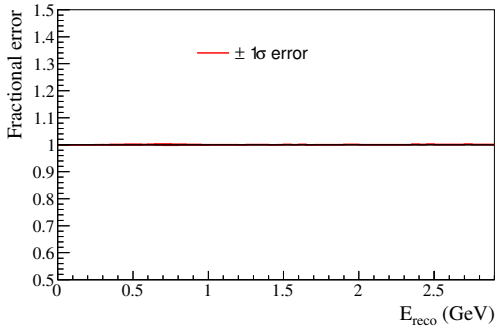
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

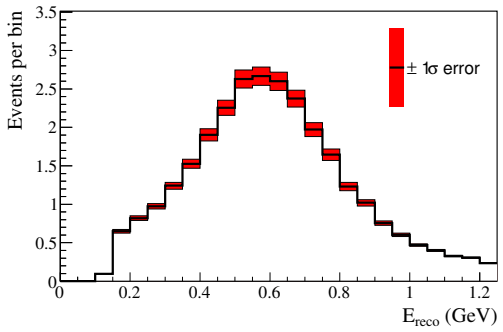


(c) Muon-like, antineutrino mode

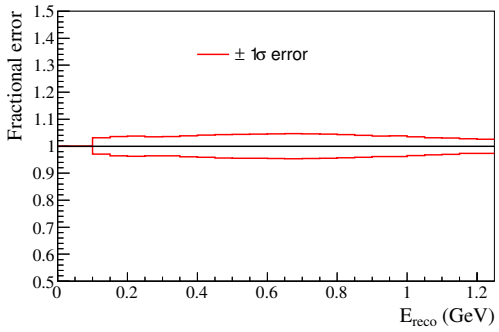


(d) Muon-like, antineutrino mode

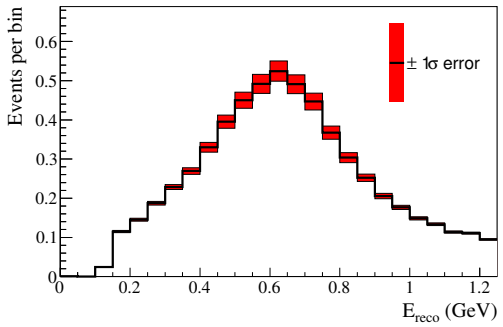
Figure 14: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



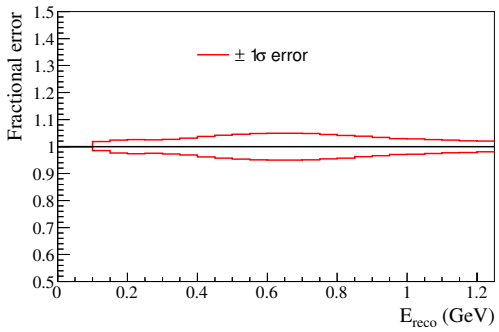
(a) Electron-like, neutrino mode



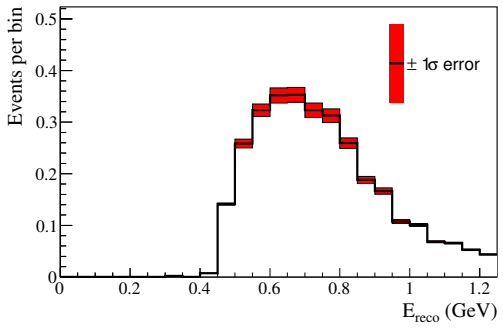
(b) Electron-like, neutrino mode



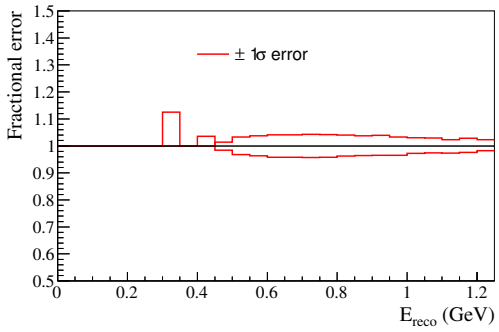
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode

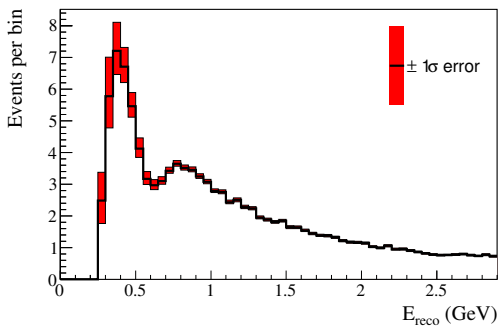


(e) ν_e CC1π^+, neutrino mode

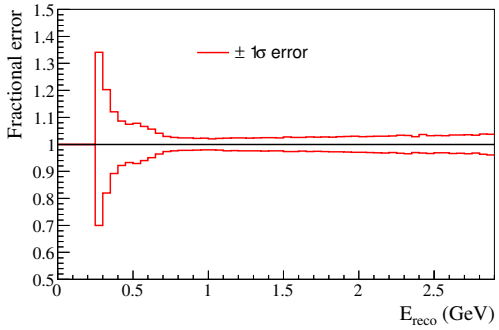


(f) ν_e CC1π^+, neutrino mode

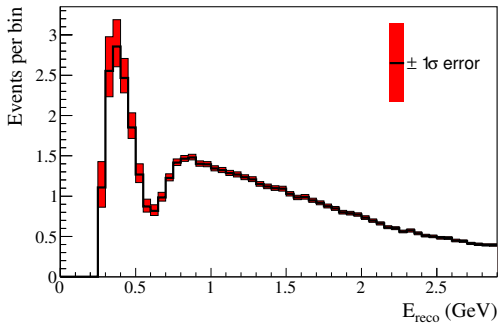
Figure 15: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



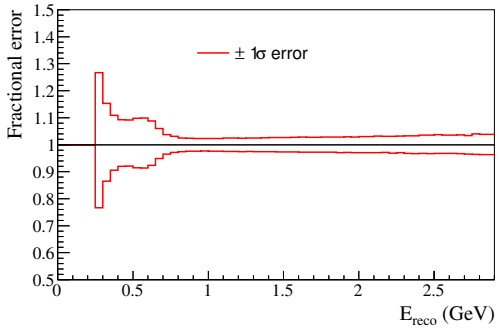
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

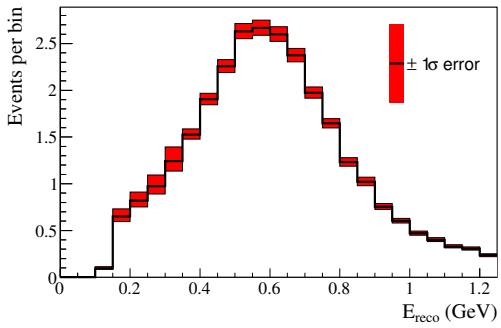


(c) Muon-like, antineutrino mode

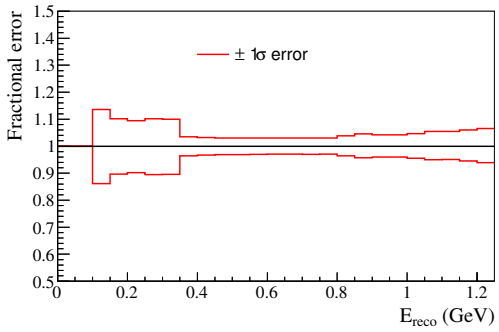


(d) Muon-like, antineutrino mode

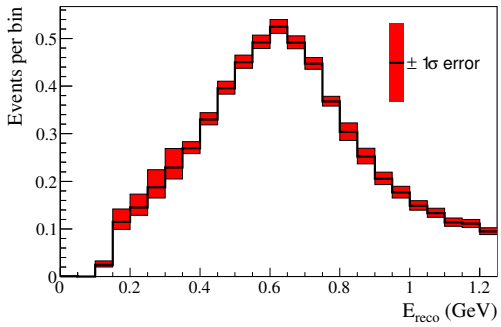
Figure 16: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the Super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



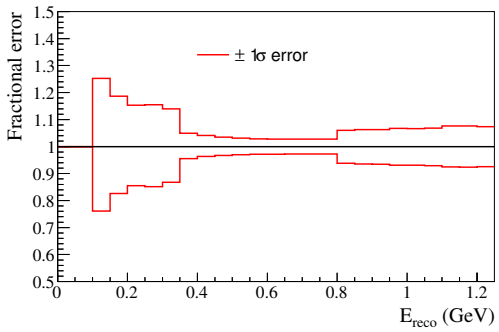
(a) Electron-like, neutrino mode



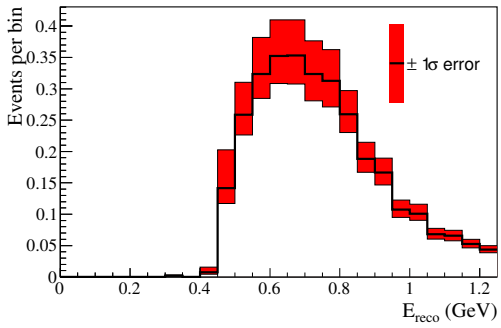
(b) Electron-like, neutrino mode



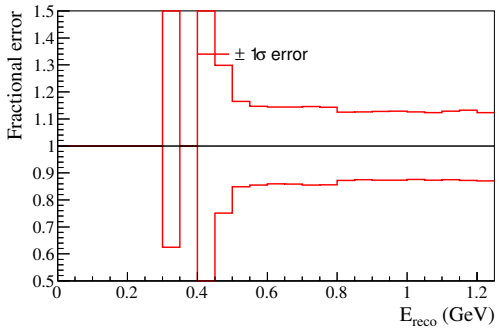
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode

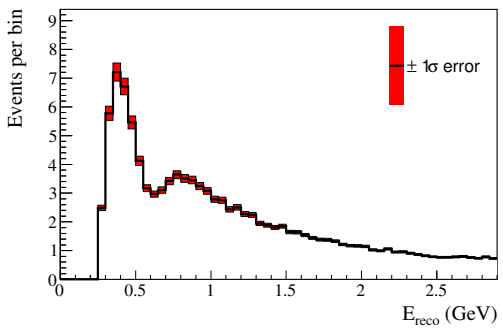


(e) ν_e CC1π^+, neutrino mode

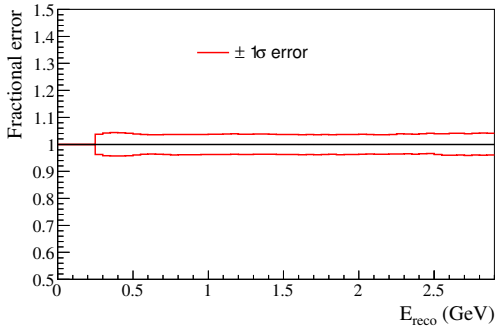


(f) ν_e CC1π^+, neutrino mode

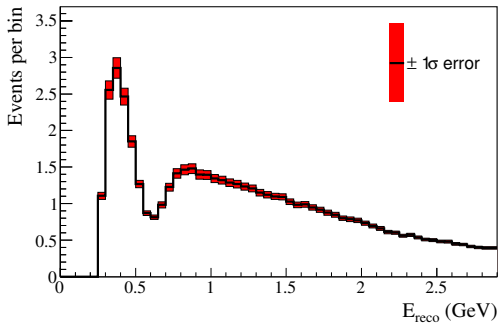
Figure 17: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the Super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



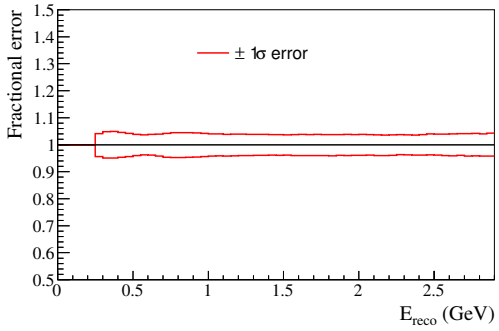
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

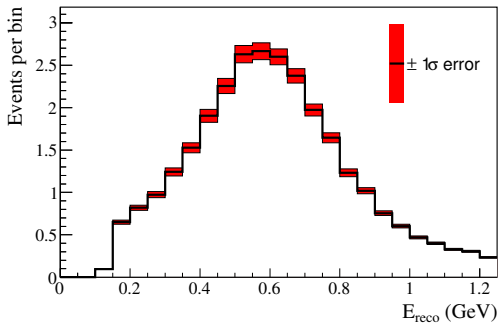


(c) Muon-like, antineutrino mode

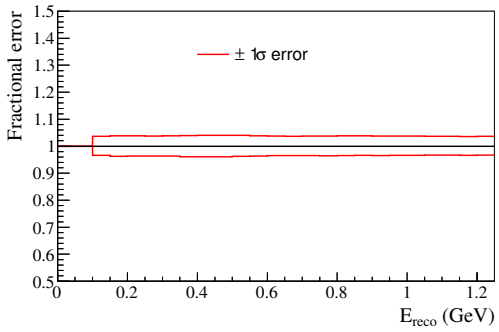


(d) Muon-like, antineutrino mode

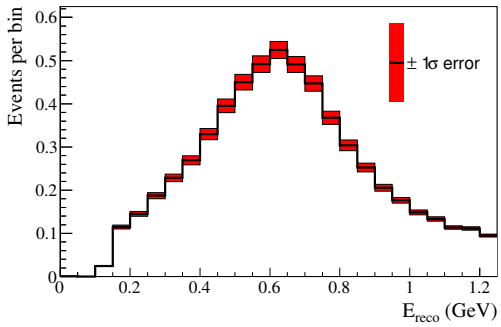
Figure 18: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



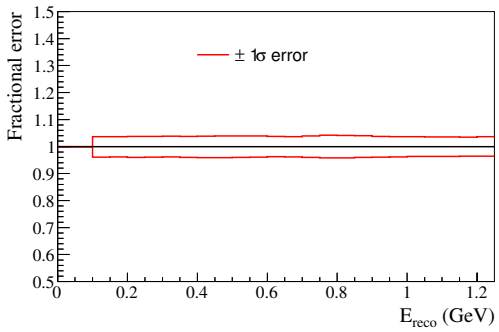
(a) Electron-like, neutrino mode



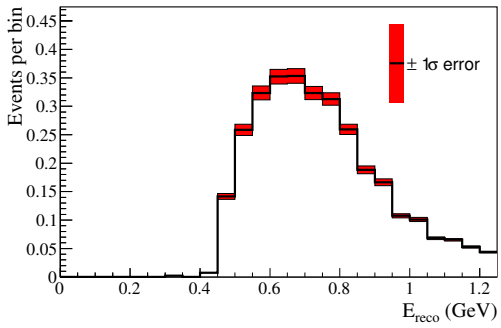
(b) Electron-like, neutrino mode



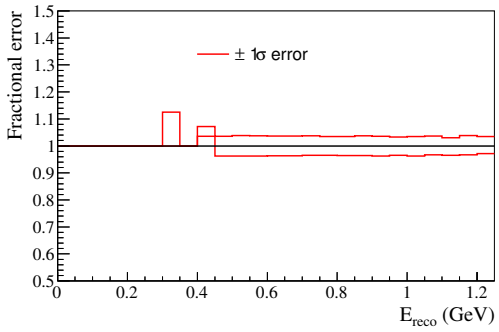
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode

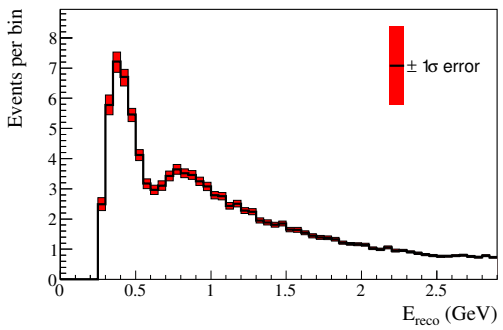


(e) ν_e CC1π^+, neutrino mode

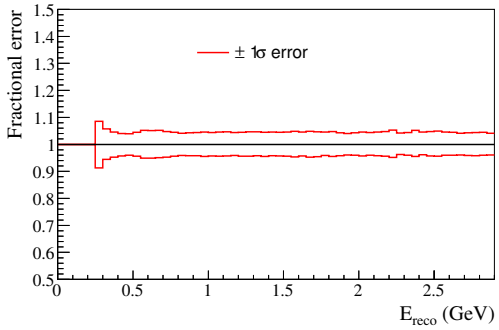


(f) ν_e CC1π^+, neutrino mode

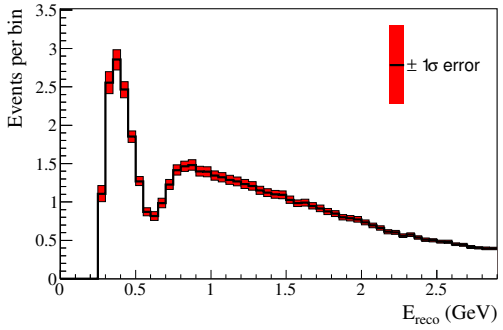
Figure 19: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



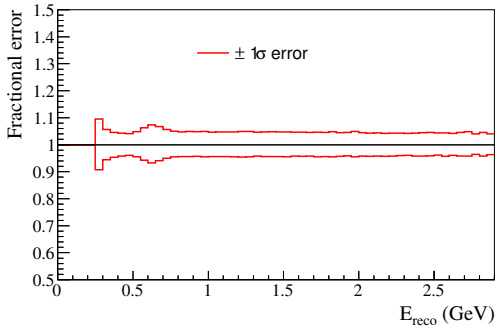
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

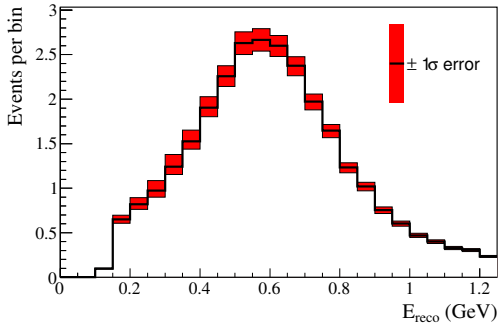


(c) Muon-like, antineutrino mode

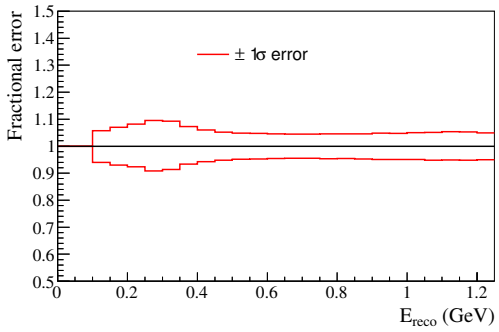


(d) Muon-like, antineutrino mode

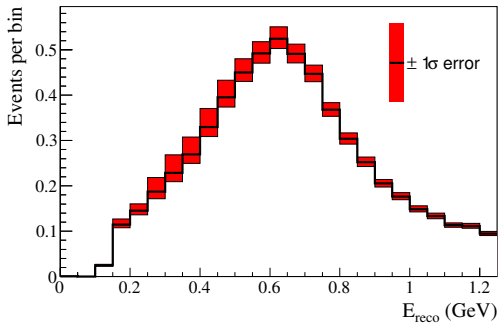
Figure 20: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



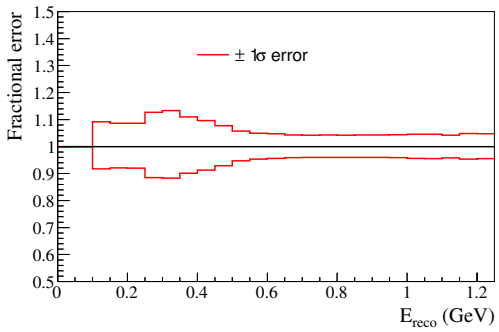
(a) Electron-like, neutrino mode



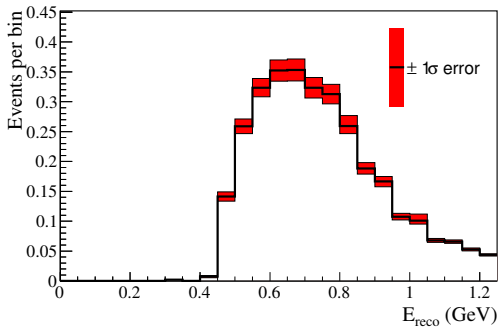
(b) Electron-like, neutrino mode



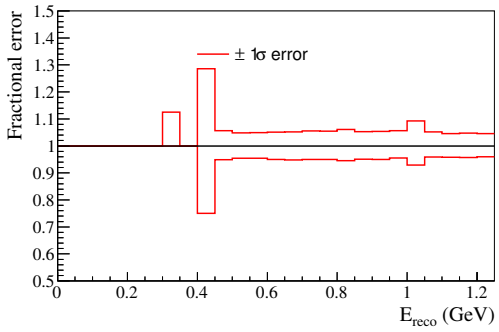
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode

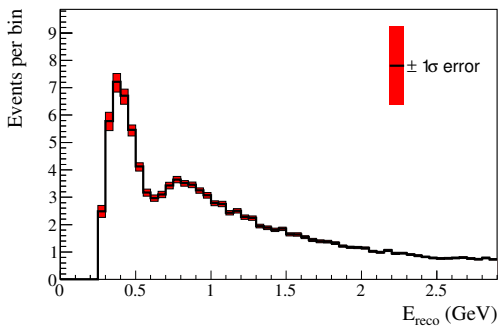


(e) ν_e CC1π^+, neutrino mode

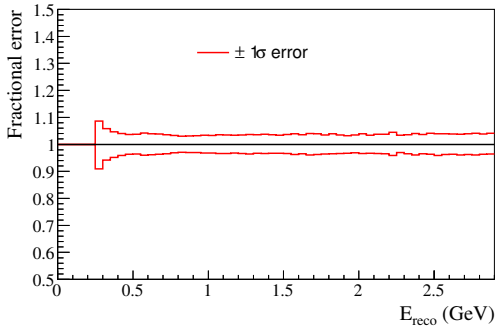


(f) ν_e CC1π^+, neutrino mode

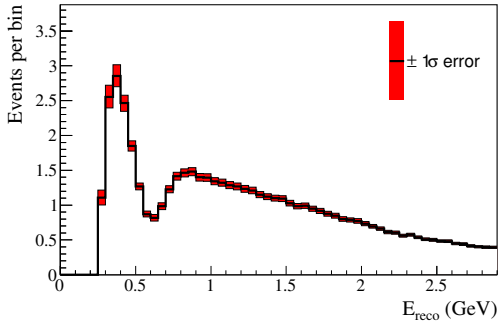
Figure 21: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



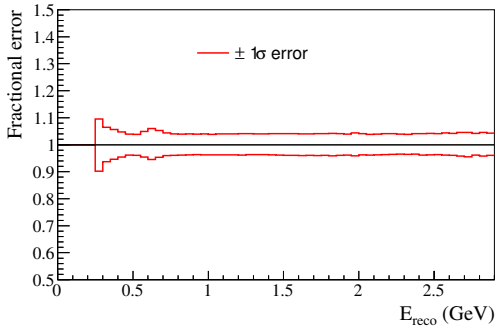
(a) Muon-like, neutrino mode



(b) Muon-like, neutrino mode

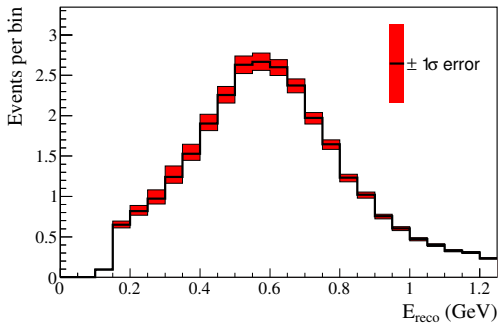


(c) Muon-like, antineutrino mode

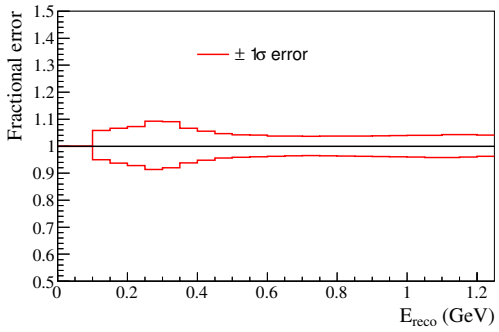


(d) Muon-like, antineutrino mode

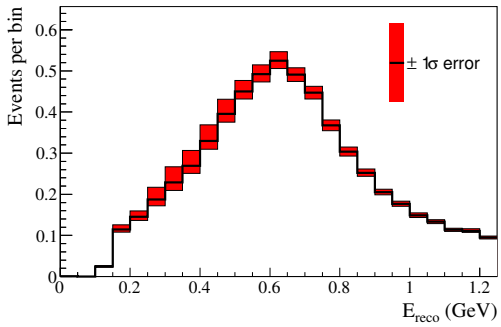
Figure 22: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the flux plus cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with random systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.



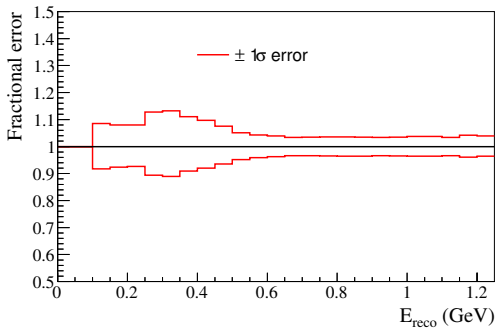
(a) Electron-like, neutrino mode



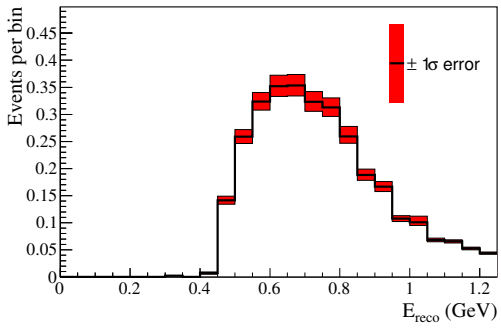
(b) Electron-like, neutrino mode



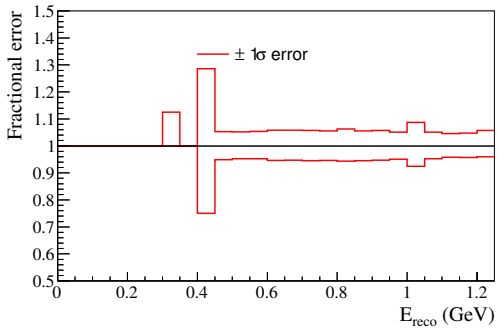
(c) Electron-like, antineutrino mode



(d) Electron-like, antineutrino mode



(e) ν_e CC1π^+, neutrino mode



(f) ν_e CC1π^+, neutrino mode

Figure 23: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the flux plus cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 13 with randomised systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-7c data set. The full analysis covariance matrix was used to produce these numbers, including the additional flux uncertainty from the addition of Run 7 data.

6. Analysis procedure

6.1. Fit method

The goal of this analysis is to measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} and search for CP violation by setting limits on δ_{CP} for a given hypothesis of mass hierarchy, which is fixed to either normal or inverted. All the systematics, as well as the oscillation parameters which are not measured in that particular analysis and treated as nuisance, are marginalised. The treatment of the oscillation parameters is shown in table 12. The parameters of interest will be denoted with the vector $\vec{\theta}$, since more than 1 parameter at a time can be measured. $\vec{\theta}$ can include the combination of all the oscillation parameters listed above.

The *VALOR* analysis uses a binned likelihood-ratio method where the estimation of $\vec{\theta}$ is obtained by comparing the observed and predicted Super-K E_{reco} vs θ_l spectra for 1-ring events.

Let N be the number of reconstructed energy and lepton angle bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the Super-K reconstructed energy spectrum prediction (e.g. parameters that control the beam, neutrino interaction simulations, detector response and oscillations).

The log-likelihood function used in the analysis is

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) \tag{3}$$

where n_i^{obs} is the observed number of events in the ith bin and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The log-likelihood function obtained by adding a gaussian penatly term can be written as

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \tag{4}$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematic parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The total likelihood is given by the product of the likelihood terms of each selected sample,

$$\lambda(\vec{\theta};\mathbf{a}) = \prod_{s=1}^{N_{samples}}\lambda_s(\vec{\theta};\mathbf{a}) \tag{5}$$

where $N_{samples}$ is the number of selected samples used in the analysis, or by the sum of $-2\ln\lambda(\vec{\theta},\mathbf{a})$ from all the Super-K samples

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s(\vec{\theta},\mathbf{a}) \tag{6}$$

The advantage of the used binned likelihood ratio method is that, in the large-sample limit, the quantity $-2\ln\lambda(\vec{\theta};\mathbf{a})$ in eq. (3) has a χ^2 distribution and it can therefore be used as a goodness-of-fit test.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

6.2. Marginalization

In the previous analyses, except [10], a fully frequentist approach, in which all nuisance parameters were profiled, was used. Now a hybrid Bayesian-frequentist method is used. The nuisance parameters are marginalised over while the parameters of interest are minimized with respect to the likelihood distribution. The marginalization of the nuisance parameters give a so-called marginal likelihood that depends on only on a vector of parameters of interest, $\vec{\theta}$.

The marginal likelihood is derived using the following equation:

$$\lambda_{marg}(\vec{\theta}) = \int_A \lambda(\vec{\theta};\mathbf{a})\pi(\mathbf{a})d\mathbf{a} = \frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i) \tag{7}$$

where $\mathbf{a}$ is the set of nuisance parameters in the parameter space A, $\pi(\mathbf{a})$ is the prior of the nuisance parameters. n is the number of toys MC produced for the marginalization and $\mathbf{a_i}$ is the single set of nuisance parameters extracted in the "i-th" toy MC. Many sets of $\mathbf{a}$ should be extracted in order to effectively sample the space of the nuisance parameters. The number of extractions depends on the number of nuisance parameters as well as the precision that is required by the measurement. In all the analyses shown in this note $n = 10k$ toys MC were used for the marginalisation of the systematic parameters.

We use the marginal log-likelihood, which will be denoted as χ^2_{marg} from now on, since it follows a χ^2 distribution when the Poisson statistic converges to the Gaussian:

$$\chi^2_{marg}(\vec{\theta}) = -2\ln\lambda_{marg}(\vec{\theta}) = -2\ln\left(\frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i)\right) \tag{8}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the n throws, not a weighted average of the log-likelihood. During the marginalization process, random values of the nuisance parameters are drawn according to the prior constraints which come from fits to near detector data and control samples at Super-K. Therefore, each throw is already weighted based on the prior information, and as a result the nuisance parameter values are not included in the calculation of the likelihood. For this reason eq. (3), which doesn't include the penalty term, is used.

Sometimes the free parameters, which are not marginalised, are constrained with a penalty term. In this case a modified version of eq. (8) is used:

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + (\vec{\theta} - \vec{\theta_0})^T V^{-1}(\vec{\theta} - \vec{\theta_0}) \tag{9}$$

where $\vec{\theta_0}$ is the vector of nominal values of the parameters of interest, $\vec{\theta}$ is the vector of its fitted values and V is the covariance matrix which relates all the free parameters. If the constrained parameter θ_{constr} is not correlated with the other parameters of interest, V is reduced to the uncertainty $\Delta\theta^{constr}$, and eq. (9) becomes

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + \frac{(\theta^{constr} - \theta_0^{constr})^2}{(\Delta\theta^{constr})^2} \tag{10}$$

This is the case of $\sin^2\theta_{13}$ when it is measured, i.e. it is parameter of interest, but at the same time it is constrained with the measurement of the reactor experiments. When a penalty term is applied to a parameter of interest it will be specified.

The toys MC used for the marginalization of the nuisance parameters are performed exactly in the same way as described in section 4.1, taking into account the correlations, but without statistical fluctuations. First $n = 10k$ distributions of E_{reco} vs θ_l for e-like and E_{reco} only for μ-like samples are produced using the extracted systematic parameters for a given point of the parameter of interest grid $\vec{\theta}$ for 1×10^{20} POT so that they may be scaled appropriately when comparing to datasets of differing POT. Finally they are stored in output files which can be used for whatever needed fit. A fit on a single dataset is performed as following:

- all the n toy MC distributions are read from the files,
- each distribution is normalised to the number of POT of the data set,
- for each toy distribution eq. (3) is calculated and $\lambda(\vec{\theta},\mathbf{a}) = e^{-(-2\ln\lambda(\vec{\theta},\mathbf{a}))/2}$ is obtained,
- the χ^2 value is finally obtained, using eq. (8), for a given set of parameter values of interest, $\vec{\theta}$. If $\vec{\theta}$ is constrained, eq. (10) is used.

This procedure is followed for each point in the $\vec{\theta}$ grid, where the binning is so fine that it does not impact on the precision of the measurement. Finally the best fit is found by looking for the $\vec{\theta}$ grid point with the minimum χ^2 value, χ^2_{bf}. This point corresponds to the best-fit parameters $\vec{\theta}_{bf}$. The minimization can be

also performed by using a discrete descent gradient algorithm [63] implemented in the VALOR framework that allows the minimum χ^2 to be found without the need of calculating the full χ^2 distribution across all grid points. This method is employed when a large number of fits is required, such as the calculation of critical values for Feldman-Cousins contours, as it drastically reduces the amount of CPU time required. A validation of the minimization algorithm will be shown in section 5 for the analyses where it is used.

6.3. Best-fit spectra with marginalization

When profiling, the best fit spectra are defined by the MC prediction that gives the largest likelihood values.

However, since the nuisance parameters are marginalised, there does not exist a single set of parameters which corresponds to the best fit. Therefore a new method has been developed to represent the best fit spectra which is consistent with the oscillation results.

The following procedure is used. From each toy MC i, used to marginalise the nuisance parameters, a different spectrum, with $N^j(\vec{\theta}_{bf})$ number of events in each bin j, is produced as well as a likelihood $\lambda(\vec{\theta}_{bf}, \mathbf{a_i})$ value, calculated using eq. (7), where $\vec{\theta}_{bf}$ is the best-fit value of the free parameters vector and $\mathbf{a_i}$ is the vector of nuisance parameters corresponding to the toy i. The best-fit number of events in the bin j of the spectrum is calculated as

$$N^j_{marg} = \frac{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i}) \cdot N^j(\vec{\theta}_{bf}; \mathbf{a_i})}{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i})} \tag{11}$$

where n is the number of toys MC.

One can easily find a relation between the profiled and marginalised best-fit spectra. In the limit of infinite number of toys MC, i.e. the full nuisance parameters space is scanned, the likelihood can be written as

$$\lambda_{marg}(\vec{\theta}_{bf}) = \frac{1}{n} \sum_{i=1,\ i\neq bf}^{n} \Big[\lambda(\vec{\theta}_{bf}; \mathbf{a_i}) + \lambda_{\mathbf{prof}}\Big] \tag{12}$$

where $\lambda_{prof} = \lambda(\vec{\theta}_{bf}; \mathbf{a_{bf}})$ is the profiled best-fit likelihood. If we denote as $N^j_{prof} = N^j_i(\vec{\theta}_{bf}; \mathbf{a_{bf}})$ the number of events in the bin j obtained with profiling, eq. (11) can be written as

$$N^j_{marg} = \frac{\sum_{i=1,\ i\neq bf}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i}) \cdot \mathbf{N^j}(\tilde{\theta}_{\mathbf{bf}}; \mathbf{a_i}) + \lambda_{\mathbf{prof}} \cdot \mathbf{N^j_{prof}}}{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i})} \tag{13}$$

6.4. Confidence intervals

After the fit is completed, the full χ^2 distribution is then shifted with respect to $\chi^2(\vec{\theta}_{bf})$:

$$\Delta\chi^2(\vec{\theta}) = \chi^2(\vec{\theta}) - \chi^2(\vec{\theta}_{bf}) \tag{14}$$

Confidence intervals are set on $\Delta\chi^2(\vec{\theta})$. If

$$\Delta\chi^2(\vec{\theta}) > \Delta\chi^2_{crit} \tag{15}$$

where $\Delta\chi^2_{crit}$ is the critical value for a certain confidence level X%(CL), that value of $\vec{\theta}$ is excluded at the X% CL. In the gaussian approximation constant values of $\Delta\chi^2_{crit}$ can be used. It will be called "constant $\Delta\chi^2$ method".

However in some cases the gaussian approximation is not reliable and the "Feldman-Cousins method" [64] must be used. In this method a different critical value for each point of the $\vec{\theta}$ grid $\Delta\chi^2(\vec{\theta})_{crit}$ is calculated. The procedure consists of the following steps:

1. many toys MC (e.g. 10k) are produced assuming the oscillation hypothesis of the grid point $\vec{\theta}$ and taking into account both the statistical and systematic uncertainties;
2. for each toy first $\chi^2(\vec{\theta}_{fix})$ is obtained by minimizing $\chi^2(\vec{\theta})$ where the oscillation parameters $\vec{\theta}$ are kept fixed to the values corresponding to the grid point and the nuisance parameters are marginalised. Then $\chi^2(\vec{\theta})$ is minimized, leaving the oscillation parameters $\vec{\theta}$ as free and $\chi^2(\vec{\theta}_{bf})$ is obtained.

3. the test statistic is calculated as

$$\Delta\chi^2 = \chi^2(\vec{\theta}_{fix}) - \chi^2(\vec{\theta}_{bf}) \tag{16}$$

4. The steps 2 and 3 are repeated for each toy MC and $f(\Delta\chi^2)$, the distribution of $\Delta\chi^2(\vec{\theta})$, is obtained.

5. Several critical values for $X\%$ CL (e.g. 68%, 90%...), $\Delta\chi^2(\vec{\theta})_{crit}$, can be defined as:

$$\Delta\chi^2_{crit} : \int_{-\infty}^{\Delta\chi^2_{crit}} f(\Delta\chi^2)d(\Delta\chi^2) = X\% \tag{17}$$

where $f(\Delta\chi^2)$ is the p.d.f. of $\Delta\chi^2$ and $d(\Delta\chi^2)$ is the differential of the integral.

6. This procedure is repeated for all the points of the $\vec{\theta}$ grid and several critical values are computed

Once a critical value, $\Delta\chi^2(\vec{\theta})_{crit}$, is computed for each oscillation hypothesis in the $\vec{\theta}$ grid, the confidence intervals are set by the condition of eq. (15).

Several analyses are performed and confidence intervals are computed for:

- 1D δ_{CP} with reactor constraint: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 12, is used.
- 1D δ_{CP} T2K only: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the uniform prior, shown in table 12, is used.
- 1D $\sin^2\theta_{13}$ T2K only: the parameter of interest is $\sin^2\theta_{13}$ and all the other oscillation parameters are marginalised;
- 1D Δm^2_{32} (Δm^2_{13}) with reactor constraint: the parameter of interest is Δm^2_{32} (Δm^2_{13}) and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 12, is used.
- 1D $\sin^2\theta_{23}$ with reactor constraint: the parameter of interest is $\sin^2\theta_{23}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 12, is used.
- 2D $\sin^2\theta_{13}$ vs δ_{CP} with reactor constraint: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is a free parameter, the gaussian prior of reactor measurements shown in table 12, is used, through a penalty term as in eq. (10).
- 2D $\sin^2\theta_{13}$ vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{23}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D Δm^2_{32} (Δm^2_{13}) vs δ_{CP} T2K only: the parameters of interest are Δm^2_{32} and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with reactor constraint: the parameters of interest are $\sin^2\theta_{23}$ and Δm^2_{32} (Δm^2_{13}) while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 12, is used.

Each confidence interval is calculated for both normal and inverted hierarchy.

The 1D χ^2 confidence intervals with "constant $\Delta\chi^2$ method" are used to set the $\pm 1\sigma$ ranges.

Each point of the $\vec{\theta}$ grid corresponds to the center of the histogram bins, i.e. the χ^2 values are calculated at each histogram bin center. The ranges and binnings used for the analyses listed above are shown in table 25. All the 1-dimensional analyses, where the precision on the best-fit point is important and must be quoted, are performed with a fine binning. Since 2-dimensional analyses are performed only to set contours with "constant $\Delta\chi^2$ method" and a fit with a very high precision is not needed to find the best-fit point, coarser binning are used, without affecting the contour. Furthermore a very fine 2-dimensional binning would require much more CPU time and storage space. In the future $\bar{\nu}/\nu$ joint analysis releases when the "Feldman-Cousins method" will be needed also for 2-dimensional fits, a finer binning will be used.

Parameter	Min	Max	Step	Bins
	1-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/200$	201
$\sin^2\theta_{13}$ (reactors)	0.015	0.036	0.021 / 81	201
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 200	201
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 200	201
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	3×10^{-3}	$10^{-3}/200$	201
	2-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/50$	51
$\sin^2\theta_{13}$ (reactors)	0.015	0.036	0.021 / 81	81
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 81	81
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 81	81
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	3×10^{-3}	$10^{-3}/21$	21

Table 25: Grid limits and spacing used for all the 1-dimensional and 2-dimensional marginalization fits and confidence regions. The minimum and maximum values correspond to the center of the χ^2 histogram first and last bin.

6.5. Contours with profiled mass hierarchy

As already pointed out, the constant $\Delta\chi^2$ method gives the correct coverage when the gaussian approximation is valid and is very useful because it does not require a large amount of CPU time. However, we know that in some cases the constant $\Delta\chi^2$ method fails to give the correct coverage and can provide only approximate confidence intervals. An example of this is the case when the constraint on a parameter is not precise enough to keep it away from a physical boundary. For cases such as these a toy MC study is required in order to estimate the correct critical values (see section 6.4).

Also, when considering discrete parameters, such as the mass hierarchy, that is distributed as binomial, the $\Delta\chi^2$ is not expected to follow a χ^2 distribution but more likely to follow a gaussian distribution as found in [69]. This is an important distinction as the exclusion significance one can interpret from a constant $\Delta\chi^2$ contour is heavily dependant on the distribution the $\Delta\chi^2$ is expected to take.

This means that when the constant $\Delta\chi^2$ method is used in order to set limits as a function of some oscillation parameters as well as the mass hierarchy, the constant critical values we usually use do not give the right coverage for the disfavoured mass hierarchy hypothesis contour.

For this reason the data fit 2-dimensional confidence intervals, which are calculated for both the hierarchy hypotheses, will be computed in two different ways:

- normal and inverted hierarchy contours are obtained independently with respect to their own minimum ("independent")
- the global minimum between the normal and inverted hierarchy hypotheses is used to set both the contours ("global");

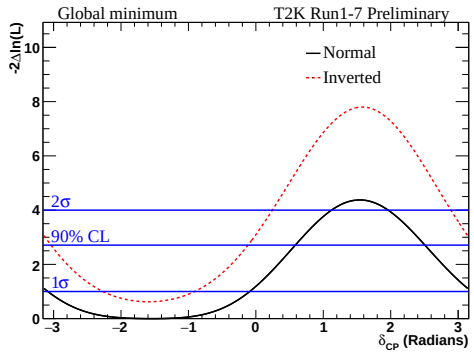
These different ways to show the confidence intervals correspond to the two extremes, where the "independent" and "global" contours provide respectively the most and the least conservative results.

Since for the 1-dimensional fits the $\Delta\chi^2$ distribution are shown, there is no need to provide results using both "independent" and "global" approaches. However the reader should remember that the constant critical

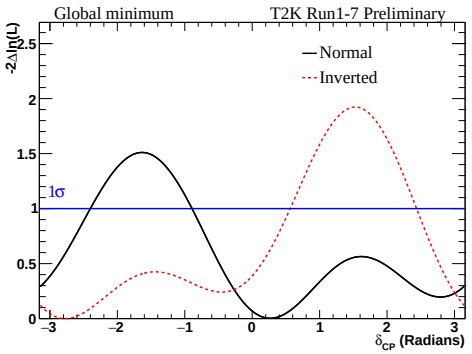
values shown in the figures hold only for the $\Delta\chi^2$ distribution corresponding to the best-fit mass hierarchy hypothesis. Indeed the constant $\Delta\chi^2$ method does not give the proper coverage for discrete parameters and consequently the contour of the disfavoured hierarchy has bad coverage.

7. Sensitivity studies

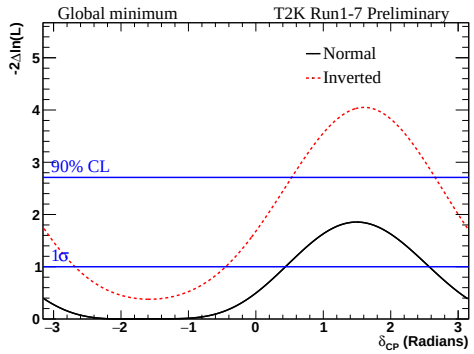
In fig. 24 the sensitivity to δ_{CP} with reactor constraint and T2K only is shown for the Asimov data sets A and B.



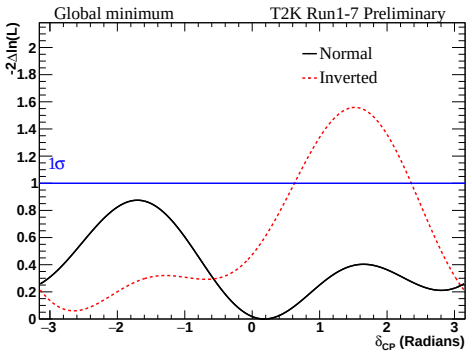
(a) Asimov A: with reactor constraint



(b) Asimov B: with reactor constraint



(c) Asimov A: with T2K only



(d) Asimov B: with T2K only

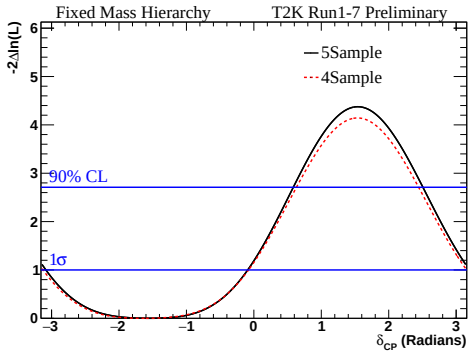
Figure 24: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} for the Asimov data sets A and B of table 13 corresponding to the Run 1-7c statistics is shown. The mass hierarchy is minimised: both the normal and inverted hierarchy χ^2 distributions are shifted to the same global best-fit χ^2 value, which is taken as the minimum between normal and inverted hierarchy. All 5 samples were used to produce these fits.

In fig. 25 the comparison of sensitivities to δ_{CP} and mass hierarchy for the 1Ring only and 1Ring + CC1π^+ fits Run 1-7c statistics is shown. We see there is a slight increase in expected sensitivity to δ_{cp} with the inclusion of the CC1π^+ sample. The effect of the new sample is also expected to be greater when the reactor constraint is applied.

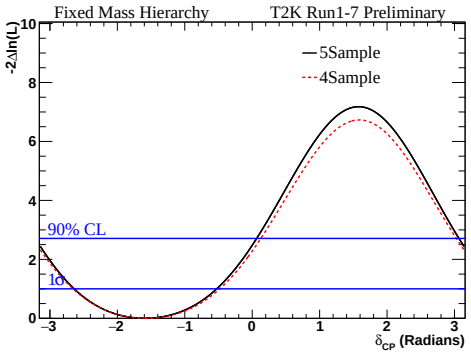
fig. 26 shows the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} and $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with the reactor constraint. fig. 27 shows the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} with T2K data only. In both fig. 26 and fig. 27, normal and inverted hierarchy contours are set independently, i.e. each calculated with respect to its own minimum χ^2 value and not the global minimum between the two mass hierarchies.

fig. 26 shows a comparison of the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} and $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with the reactor constraint when the ν_e CC1π^+ sample is included. fig. 27 shows a comparison of the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} with T2K data only. We see similar results as for the δ_{cp} only fits, a marginal improvement in constraining the parameters δ_{cp} and $\sin^2(\theta_{13})$. As expected there is little change in the $\sin^2\theta_{23}$ Vs Δm^2_{32} contour as the constraint is driven by the disappearance spectra which remains unchanged from the previous analysis.

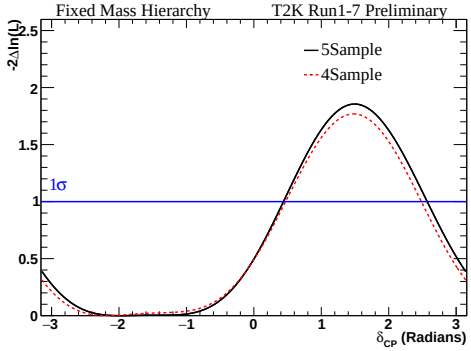
The comparison of the Asimov data set contours with the other oscillation analyses groups is shown in TN-269v3 [51]. The agreement with the VALOR and the other analysis groups sensitivities is very good.



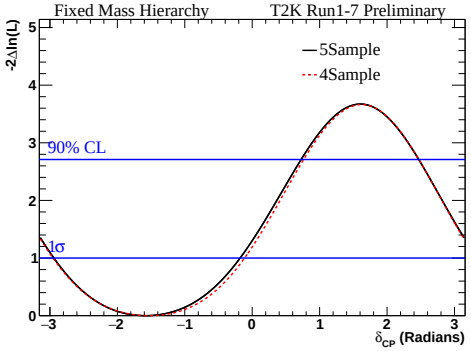
(a) Normal hierarchy with the reactor constraint



(b) Inverted hierarchy with the reactor constraint

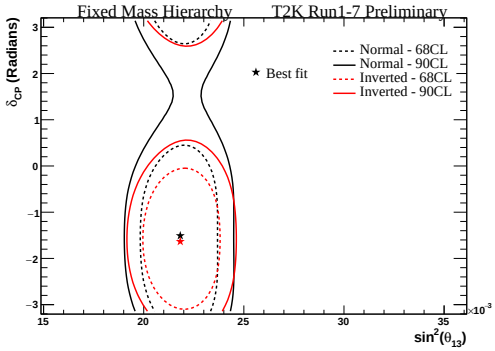


(c) Normal hierarchy without the reactor constraint

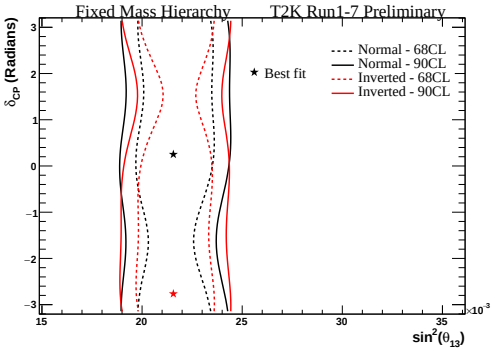


(d) Inverted hierarchy without the reactor constraint

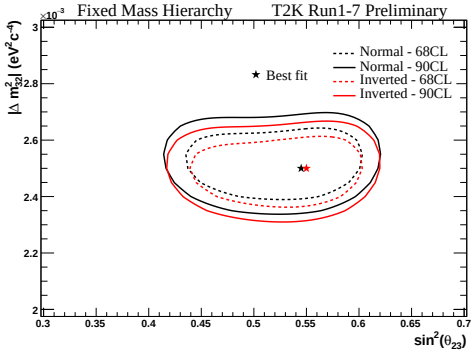
Figure 25: The comparison of sensitivities to δ_{CP} and mass hierarchy for the 1Ring and 1Ring+CC1π^+ analysis is shown. The expected $\Delta\chi^2$ distribution as a function of δ_{CP} and mass hierarchy is shown for Asimov data set A of table 13. The comparison is done both applying the reactor constraint (top) and with only T2K data (bottom)



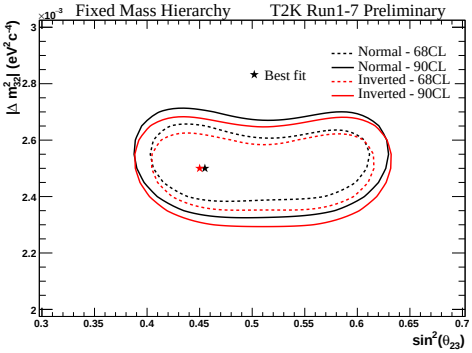
(a) Asimov A: $\sin^2\theta_{13}$ Vs δ_{CP}.



(b) Asimov B: $\sin^2\theta_{13}$ Vs δ_{CP}.

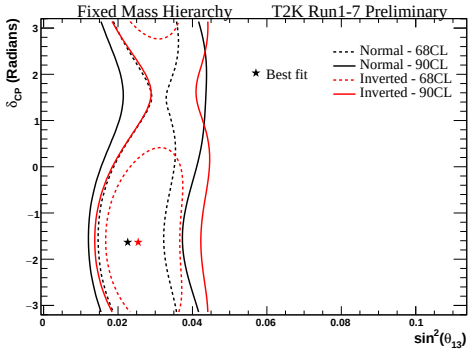


(c) Asimov A: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}).

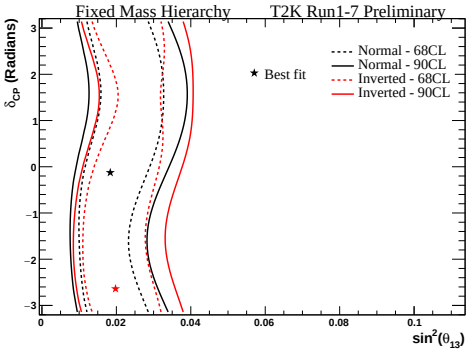


(d) Asimov B: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}).

Figure 26: The contours at 68% and 90% CL with reactor constraint for the Asimov data sets A and B of table 13 corresponding to the Run 1-7c statistics is shown. Normal and inverted hierarchy contours are independent. All 5 samples were used to produce these fits.

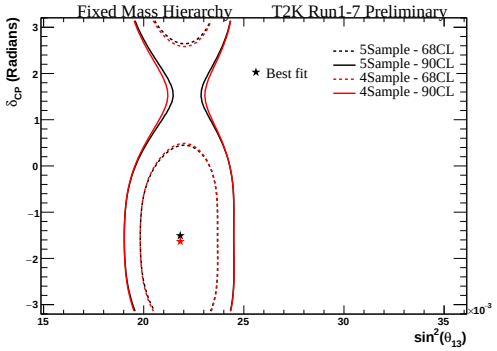


(a) Asimov A.

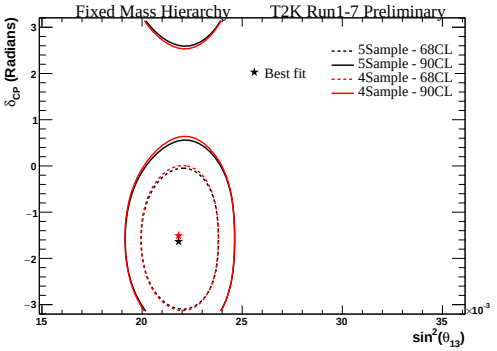


(b) Asimov B.

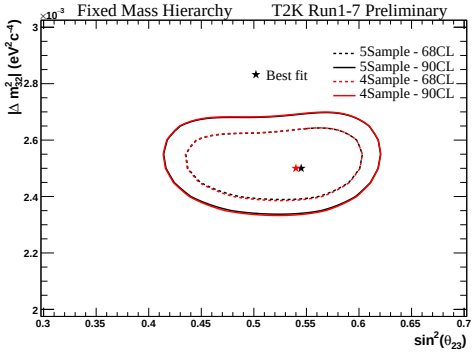
Figure 27: The contours at 68% and 90% CL for $\sin^2\theta_{13}$ Vs δ_{CP} for the Asimov data sets A and B of table 13 corresponding to the Run 1-7c statistics is shown. Normal and inverted hierarchy contours are independent. Only T2K information used. All 5 samples were used to produce these fits.



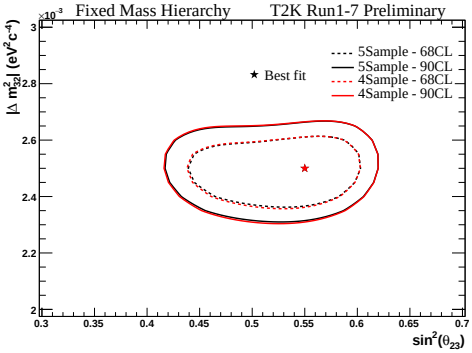
(a) Normal: $\sin^2\theta_{13}$ Vs δ_{CP}.



(b) Inverted: $\sin^2\theta_{13}$ Vs δ_{CP}.

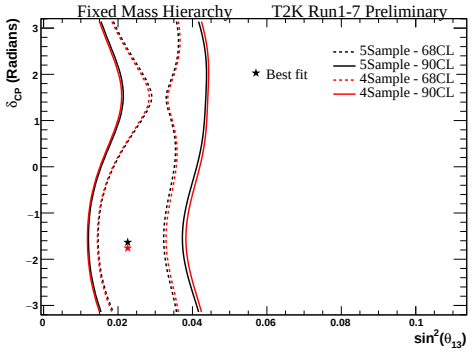


(c) Normal: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}).

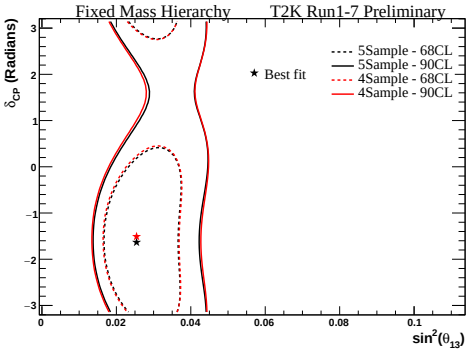


(d) Inverted: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}).

Figure 28: The comparison of sensitivities to δ_{CP} and mass hierarchy for the 1Ring and 1Ring+CC1π^+ analysis is shown. The contours at 68% and 90% CL with reactor constraint for Asimov data set A of table 13 corresponding to the Run 1-7c statistics is shown.



(a) Normal

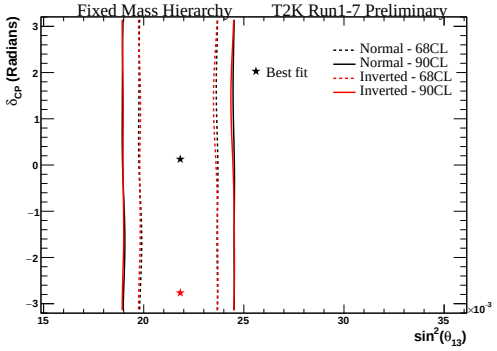


(b) Inverted

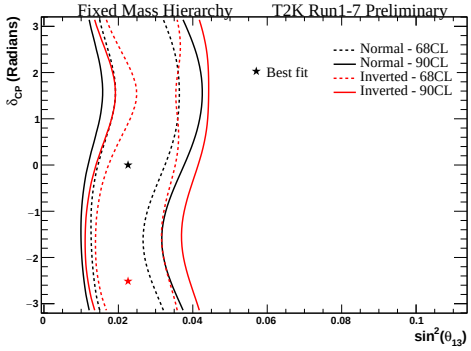
Figure 29: The comparison of sensitivities to δ_{CP} and mass hierarchy for the 1Ring and 1Ring+CC1π^+ analysis is shown. The contours at 68% and 90% CL for $\sin^2\theta_{13}$ Vs δ_{CP} for Asimov data set A of table 13 corresponding to the Run 1-7c statistics is shown. Only T2K information is used.

7.1. Additional sensitivity studies

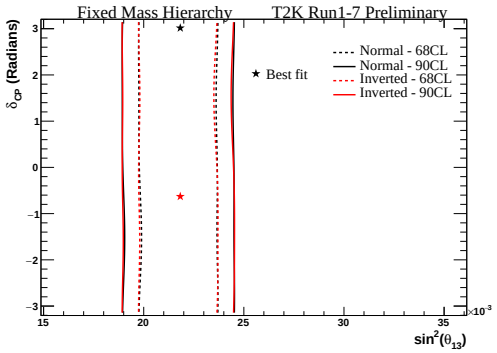
Sensitivity studies for various values of true δ_{CP} were performed to give some context to our data constraint on δ_{CP}. New Asimov data sets C, D and E are defined in table 13 . The effect of the true value of δ_{CP} was studied in the $\sin^2\theta_{13}$ Vs δ_{CP}, with and without the reactor constraint, and 1D δ_{CP} parameter spaces. The results are shown in figs. 30 and 31. Fits to Asimov data set C and D have similar results to Asimov data set B but provide slightly stronger constraint due to the additional events in the appearance samples due to more maximal disappearance. The fit to Asimov data set E is similar to that to Asimov data set A, both provide similar constraints on δ_{CP} however the more CP violating data set (E) naturally provides a stronger constraint.



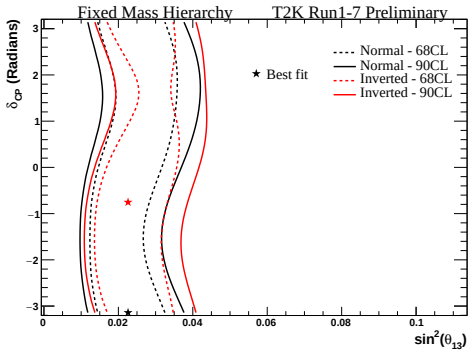
(a) Asimov C: $\sin^2\theta_{13}$ Vs δ_{CP}. Reactor constraint.



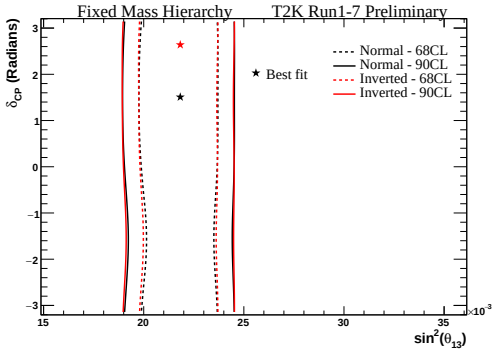
(b) Asimov C: $\sin^2\theta_{13}$ Vs δ_{CP}.



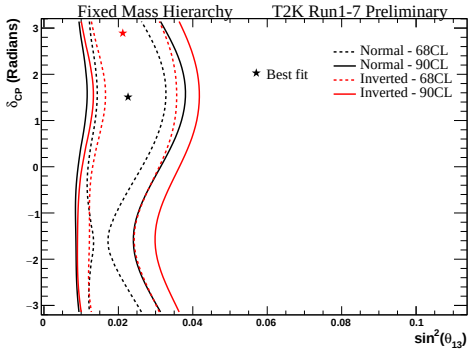
(c) Asimov D: $\sin^2\theta_{13}$ Vs δ_{CP}. Reactor constraint.



(d) Asimov D: $\sin^2\theta_{13}$ Vs δ_{CP}.

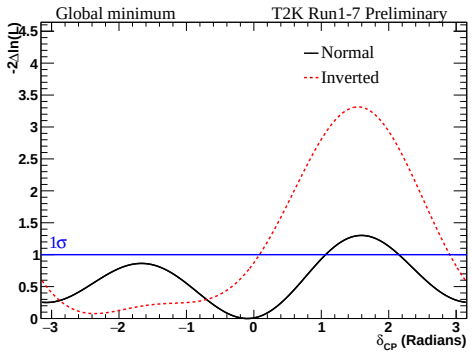


(e) Asimov E: $\sin^2\theta_{13}$ Vs δ_{CP}. Reactor constraint.

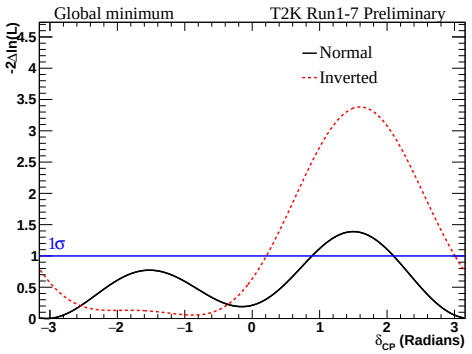


(f) Asimov E: $\sin^2\theta_{13}$ Vs δ_{CP}.

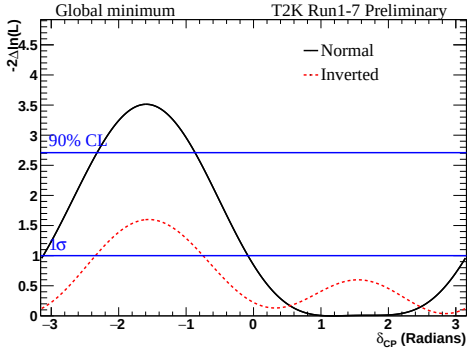
Figure 30: The contours at 68% and 90% CL with reactor constraint for the Asimov data sets C, D and E of table 13 corresponding to the Run 1-7c statistics is shown. Normal and inverted hierarchy contours are independent.



(a) Asimov C



(b) Asimov D



(c) Asimov E

Figure 31: The contours 1D δ_{CP} contours for the Asimov data sets C, D and E of table 13 with the reactor constraint corresponding to the Run 1-7c statistics is shown. Normal and inverted hierarchy contours are independent.

8. Coverage study of the simultaneous fit of δ_{CP} and mass hierarchy

The coverage study and critical value adjustment was performed using a similar technique to that in [11]. The only improvement to the method is that the fake data experiments are now generated taking into account the constraint on the parameters $\sin^2\theta_{23}$ and Δm^2_{32} from this analysis. To do this, the value of $\sin^2\theta_{23}$ and Δm^2_{32} were randomly thrown according to the posterior likelihood shown in fig. 32. A different posterior likelihood was used for each hierarchy hypothesis.

The coverage is defined as the number of times the true values of δ_{CP}^{true} and MH^{true}, used to generate the toys MC, is contained in the confidence interval defined by eq. (15):

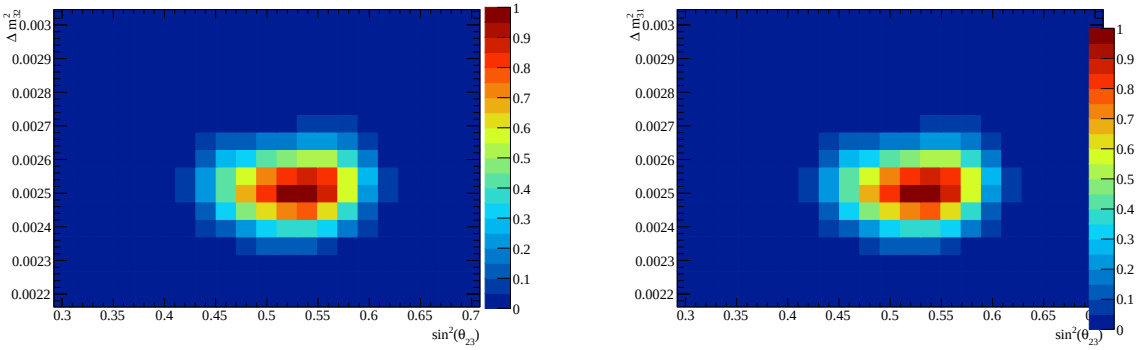
$$\Delta\chi^2(\delta_{CP},\mathrm{MH}) = \chi^2(\delta_{CP}^{true},\mathrm{MH}^{true}) - \chi^2(\delta_{CP}^{bf},\mathrm{MH}^{bf}) \tag{18}$$

where δ_{CP}^{bf} and MH^{bf} are the best-fit values. Adopting the "constant $\Delta\chi^2$" method, corresponding to the gaussian approximation, the critical value $\Delta\chi^2_{crit} = \Delta\chi^2_{const}$, pre-computed analytically, at a certain $X\%$ confidence level is used for any value of δ_{CP} and MH. Many toys MC are performed (10k) and for each one eq. (18) is tested. The condition used to estimate the coverage is

$$\Delta\chi^2(\delta_{CP},\mathrm{MH}) < \Delta\chi^2_{crit} \tag{19}$$

The fit has a good coverage if the number of toys MC for which eq. (19) is satisfied (N^X_{toys}) is equal or close to $X\%$ within the statistical uncertainty of the number of toys MC. If $N^X_{toys} > X\%$, then the fit has over-coverage, that means the confidence intervals should be tighter than what obtained with the "constant $\Delta\chi^2$ method", while if $N^X_{toys} < X\%$, then the fit has under-coverage and excludes a wider range of values than in the correct case. In fig. 34 the $\Delta\chi^2$ distribution of δ_{CP} 1-dimensional fit with reactor constraint, obtained by using eq. (18), is shown for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$. The critical values computed using the

Feldman-Cousins method are shown in fig. 35. It can be seen that the coverage given by the constant critical values is not correct, so, in order to provide correct confidence intervals, the Feldman-Cousins method is required.



(a) Normal (b) Inverted

Figure 32: The posterior likelihoods used to throw random values of the oscillation parameters $\sin^2\theta_{23}$ and Δm^2_{32} under the normal and inverted hierarchy hypothesis.

In table 44 the critical values and the coverage is shown for all studied values of δ_{CP} and mass hierarchy are shown and in fig. 33 the true coverage from the study using the critical values under the gaussian approximation is shown.

In fig. 34 the expected distributions of $\Delta\chi^2$ and best-fit δ_{CP} are shown for all points.

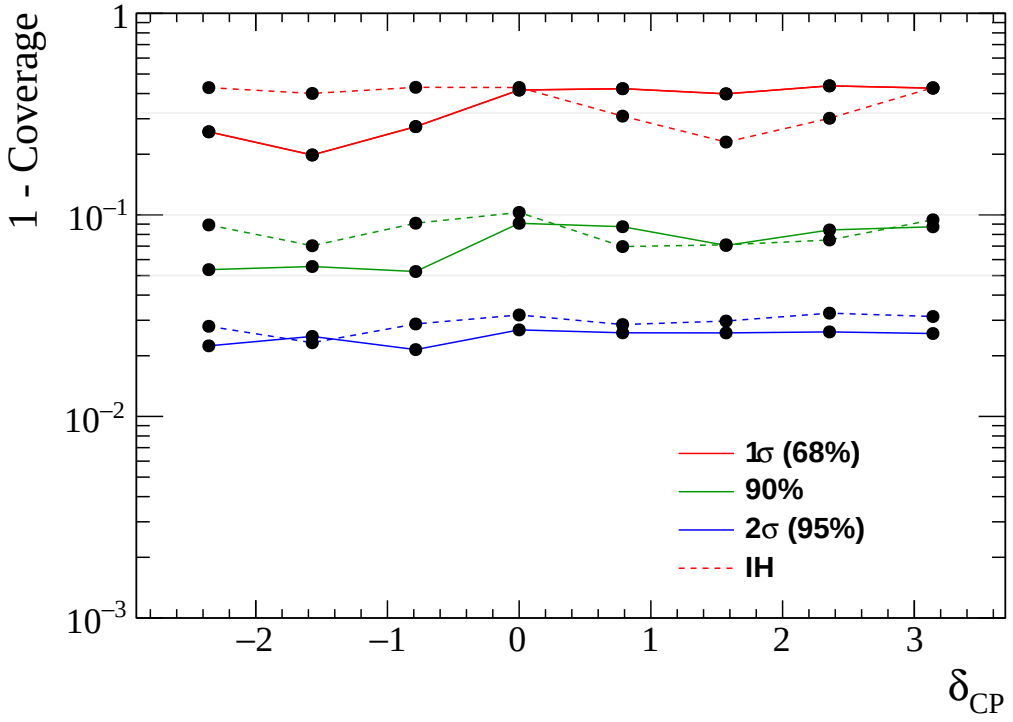
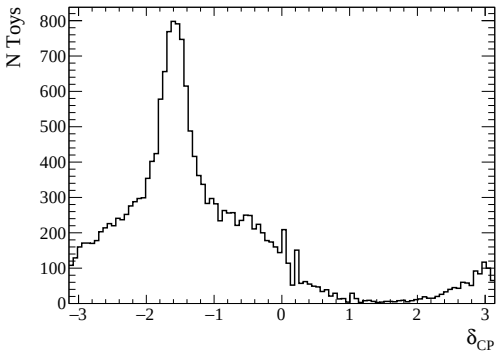
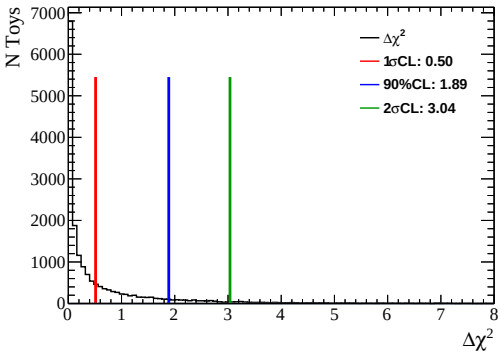


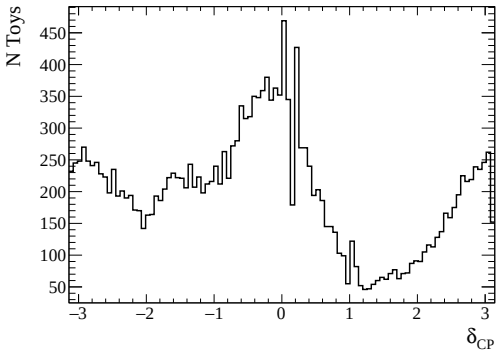
Figure 33: The coverage from an MC study when using the critical values for 1σ, 90% and 2σ (1,2.71,4) in the gaussian approximation. The expected coverage in the gaussian approximation is given in the legend and guide lines in the figure. For 90% and 2σ critical values, we see that the gaussian critical values give a smaller fraction of toy experiments than expected so we have under-coverage. For the 1σ



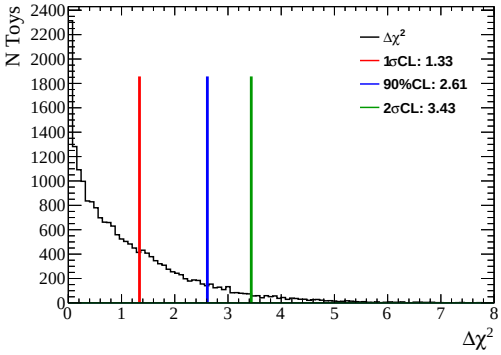
(a) Best-fit distribution of δ_{CP}. $\delta_{CP}^{true} = -\pi/2$.



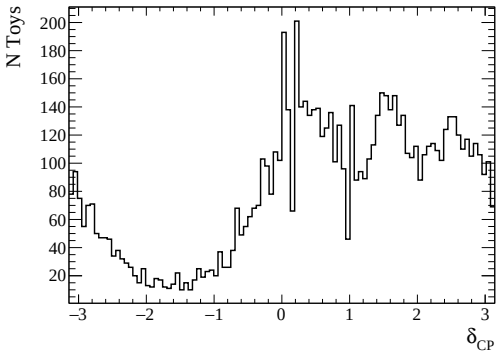
(b) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = -\pi/2$.



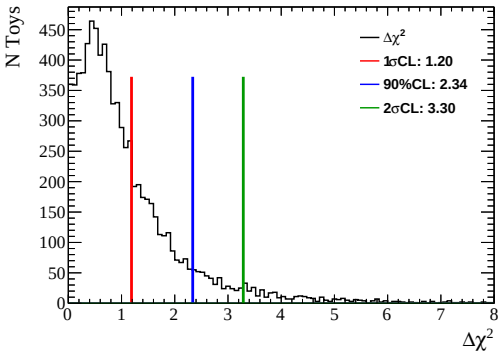
(c) Best-fit distribution of δ_{CP}. $\delta_{CP}^{true} = 0$.



(d) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = 0$.



(e) Best-fit distribution of δ_{CP}. $\delta_{CP}^{true} = +\pi/2$.



(f) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = +\pi/2$.

Figure 34: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). $10k$ toy experiments are performed with true values of $\delta_{CP} = -\pi/2, 0, +\pi/2$, and normal hierarchy, and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

In fig. 35 the critical values obtained with the Feldman-Cousins method for both 4 sample and 5 sample analyses are shown. The critical values at both 90% and 2σ CL are computed for both the mass hierarchies and the following true values of δ_{CP}: $-\pi$, -2.356, $-\pi/2$, -0.785, 0, +0.785, $+\pi/2$, -2.356, $+\pi$.

The inclusion of the new sample and use of the new toy generation method have a small impact on the critical values from this study. The critical values important for this analysis $\delta_{CP} \sim 0, \pi$ are very close to that from the previous analysis. There is a noticeable reduction in the critical value for this analysis compared to the previous Run1-7c joint analysis [11] which can be explained by the difference in the methods used.

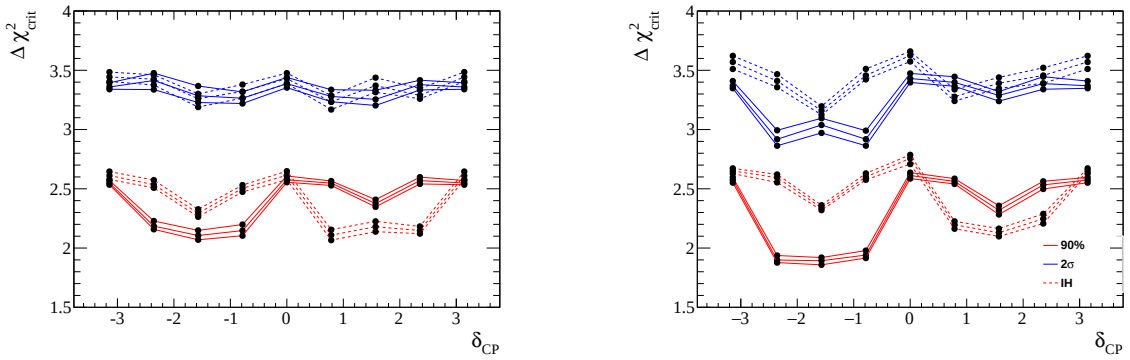
In the previous analysis, fake toy experiments were not chosen based on their likelihood with respect to the data samples. In the 5 sample analysis, the values of $\sin^2\theta_{23}$ and Δm^2_{32} for the fake data toy experiments were chosen based on the posterior likelihoods from a fit to T2K data. The new analysis therefore selects toy

experiments with data more similar to what is observed than the previous method.

This result at $\delta_{CP} \sim -\frac{\pi}{2}$ could be a result of the fact that the toy experiments used with the new method have on average more events in the FHC electron like sample than previously. The larger statistics mean that the effect of boundaries and the cyclic nature of δ_{CP} have a greater impact on the critical values. As the true value of δ_{CP} moves away from $-\frac{\pi}{2}$ the events int the FHC electron like sample decrease and statistical uncertainties become more significant.

This can be observed from the difference in both 90% and 2σ critical values.

The results of other points are shown in section E.



(a) Run 1-7c statistic, four 1 Ring samples

(b) Run 1-7c statistic, 5 samples

Figure 35: Critical values obtained with the Feldman-Cousins method for Run 1-7c for the 4 sample joint fit and the new 5 sample for several true values of δ_{CP} and both normal (solid line) and inverted (dashed line) hierarchy. Both 90% (red) and 2σ (blue) CL critical values are shown. Greater than $10k$ toy experiments are performed for each point. All the other oscillation parameters are marginalised. The $\pm 1\sigma$ error on the critical values is shown as well and is represented by the lowest and highest set of critical values, with the same colour of the central ones.

8.1. Validation of the minimisation algorithm

For each true value of δ_{CP}, where the critical value is performed using the Feldman-Cousins method, and normal hierarchy, 10000 fits were performed on fake data sets of the same type as used for Feldman-Cousins. About all the fits except 1 found the global minimum.

9. Results of the $\bar{\nu}/\nu$ joint analysis with the Run 1-7c dataset

The results of the Run 1-7c data set fit, for an exposure of 7.482×10^{20} POT in neutrino mode and of 7.471×10^{20} POT in antineutrino mode, are shown in this chapter. In fig. 36 the predicted spectra of μ-like and e-like samples are compared to the observed data. A comparison of the data and a breakdown of the predicted spectra by neutrino and interaction type is shown in section H. In table 26 the expected and observed number of events in Run 1-7c data set are shown. The predicted number of events are shown for the oscillation parameters in table 13 of Asimov data sets A and B.

In the new ν_e CC1π^+ sample there are five events observed with only $\sim$3 expected. However from fig. 36 we see that two of the five events observed fall in the outer edges of the distribution which isn't as well populated by the signal. A more detailed comparison of the signal and background distribution in the samples are given in section H.

The comparison of the confidence intervals obtained by the fit of the Run 1-7c data set among all the oscillation analysis groups is shown in section F.

Beam	Sample	Exp. Asimov A	Exp. Asimov A ($\delta_{CP} = 0$)	Exp. Asimov B	Exp. Not Osc	Obs.	Best fit (reactors)
ν	μ-like	135.815	135.459	140.724	521.777	135	137.212
ν	e-like	28.687	24.170	21.524	6.147	32	28.553
$\bar{\nu}$	μ-like	64.205	64.059	65.597	184.837	66	64.522
$\bar{\nu}$	e-like	6.004	6.902	6.252	2.335	4	6.120
ν	CC1π^+-like	3.126	2.744	2.511	3.258	5	3.330

Table 26: The observed, expected and best-fit number of events in Run 1-7c data set for an exposure of 7.482×10^{20}POT in neutrino mode and of 7.471×10^{20}POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning and the oscillation parameters shown in table 13. The total number of events of the best-fit spectra is obtained by using eq. (11).

9.1. 1-dimensional χ^2 distributions and best-fit

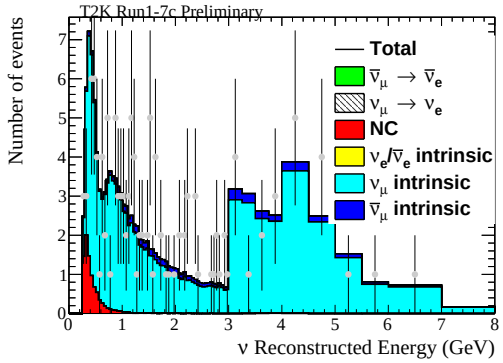
1-dimensional χ^2 distributions as in section 7 are produced and shown in fig. 37. The best-fit value of δ_{CP} when the reactor constraint is applied is -1.728 similar to that of the 1Ring sample analysis.

In table 27 the best-fit values as well as the 1σ ranges for each oscillation parameter are shown for the Run 1-7c data fit, for both normal and inverted hierarchy. The 1σ ranges are extracted for each oscillation parameter from the 1-dimensional $\Delta\chi^2$ distributions shown in fig. 37, where one oscillation parameter is free and all the other oscillation as well as the systematic parameters are marginalised. The mass hierarchy is minimised: for both normal and inverted hierarchy the 1σ ranges are calculated with respect to the same global minimum. Since this approach is not entirely correct and there does not exist any procedure to set more reliable confidence intervals which does not consist of toy MC studies (see section 6.5), the advice to the reader is to give more importance to the difference in χ^2_{marg} between normal and inverted hierarchies in the respective best-fit points, instead of the 1σ ranges of inverted hierarchy, which are smaller by construction. It is also worth to notice that the 1σ ranges for normal hierarchy (best-fit) are set correctly with the constant $\Delta\chi^2$ method in the gaussian approximation.

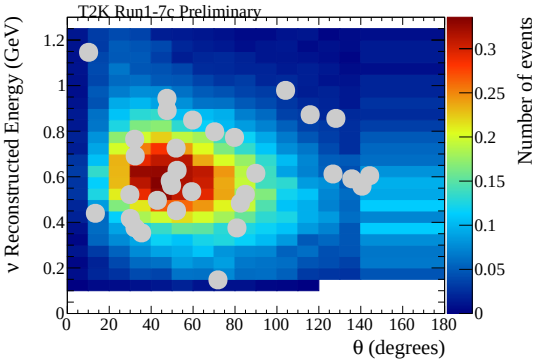
All the confidence intervals are very similar to those from the 1Ring sample analysis, full comparisons are detailed in section 9.4.

9.2. Feldman-Cousins and significance for the simultaneous fit of δ_{CP} and mass hierarchy for Run 1-7c

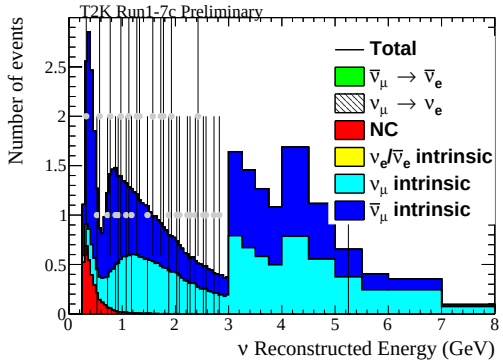
In fig. 38 the confidence intervals obtained with the "Feldman-Cousins" method are shown for δ_{CP} and mass hierarchy with reactor constraint for the Run 1-7c data set. The critical values at both 90% and 2σ CL are computed for both the mass hierarchies and the following true values of δ_{CP}: $-\pi$, -2.356, $-\pi/2$, -0.785, 0, +0.785, $+\pi/2$, -2.356, $+\pi$. The procedure described in section 8 is used. More than 10k toy experiments were used for all points and the uncertainty on each critical value was calculated as described in section D. This can be seen in fig. 38 where the $\pm 1\sigma$ error on the critical values are shown as well. Inverted hierarchy is almost excluded at 90% CL except for a small interval around $\delta_{CP} = -\pi/2$. Both $\delta_{CP} = 0$ and $\delta_{CP} = \pi$, are excluded at 90% CL but $\delta_{CP} = 0$ is excluded also at 2σ. In table 28 the 90% CL intervals for δ_{CP} vs mass hierarchy, obtained with the Feldman-Cousins method, are shown.



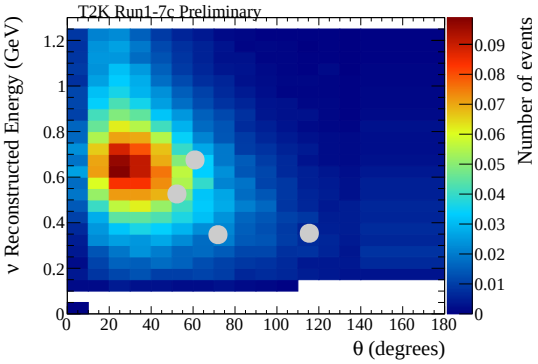
(a) Muon-like, neutrino mode



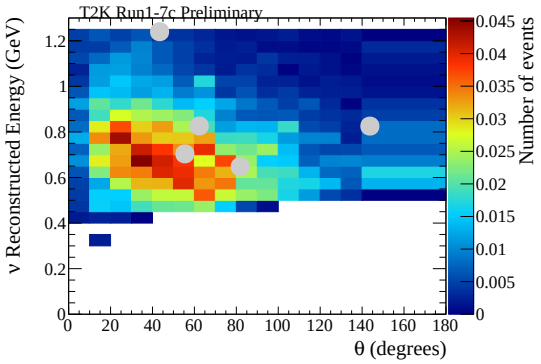
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode

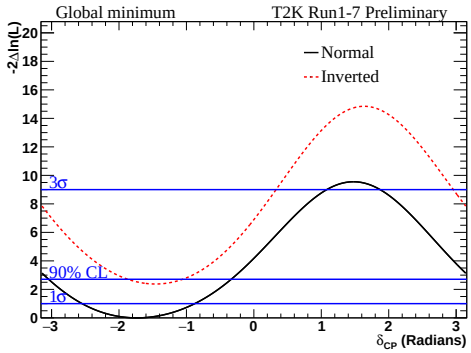


(d) Electron-like, antineutrino mode

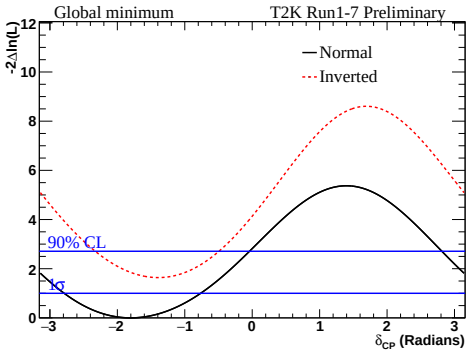


(e) ccπ^+-like, neutrino mode

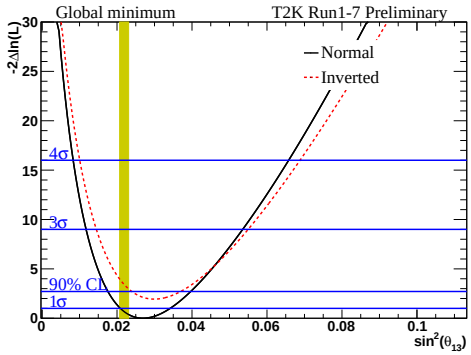
Figure 36: The predicted spectra compared to the Run 1-7c dataset (dots), which corresponds to an exposure of 7.482×10^{20}POT in neutrino mode and of 7.471×10^{20}POT in antineutrino mode, are shown. The expected distribution corresponds to the oscillation parameters of the Asimov data set A shown in table 13. The spectra were generated with the systematic parameters measured with the ND280 data shown in table 7, 8, 9 and the nominal Super-K systematic parameters shown in table 10.



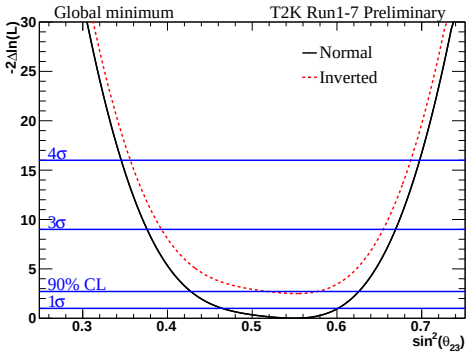
(a) δ_{CP} with reactor constraint



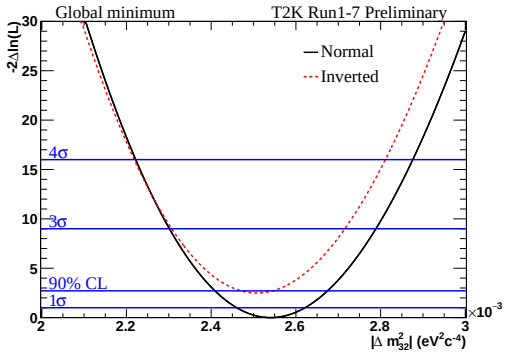
(b) δ_{CP} with T2K data only



(c) $\sin^2\theta_{13}$ with T2K data only



(d) $\sin^2\theta_{23}$ with reactor constraint

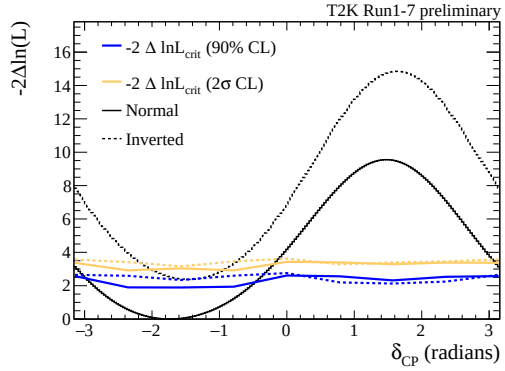


(e) Δm^2_{32} (Δm^2_{13}) with reactor constraint

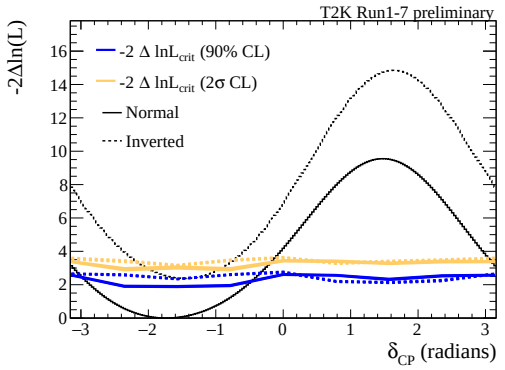
Figure 37: Measured $\Delta\chi^2$ distributions with the full Run 1-7c data set. Both normal (red) and inverted (blue) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 25. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [76], shown by the yellow bar.

	Reactors	Normal Hierarchy			Inverted Hierarchy		
Parameter		Best-fit	χ^2_{min}	$\pm1\sigma$	Best-fit	χ^2_{min}	$\pm1\sigma$ global (inverted)
δ_{CP}	YES	-1.728	370.61	[-2.538; -0.877]	-1.445	373.07	- ([-2.170; -0.768])
δ_{CP}	NO	-1.791	373.53	[-2.789; -0.764]	-1.382	375.17	- ([-2.296;-0.524])
$\sin^2\theta_{13}$	NO	0.0271	371.44	[0.0209; 0.0342]	0.0299	373.38	- ([0.0232; 0.0380])
$\sin^2\theta_{23}$	YES	0.550	369.80	[0.465; 0.601]	0.5525	372.30	- ([0.470; 0.601])
Δm^2_{32} or Δm^2_{13} (10^{-3} eV$^2/c^4$)	YES	2.54	369.05	[2.460; 2.621]	2.51	297.56	- ([2.429; 2.588])

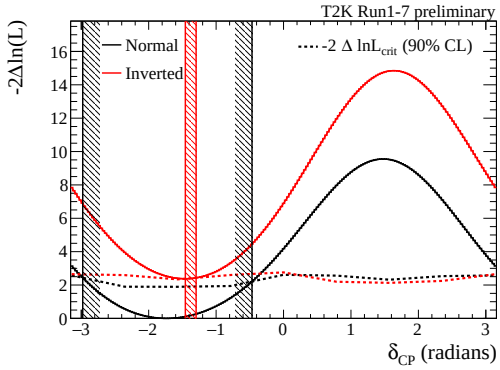
Table 27: The Run 1-7c measured oscillation parameter best-fit and minimum χ^2 values as well as the 1σ CL intervals are shown, for both normal and inverted hierarchy. The $\pm1\sigma$ interval is obtained by using the *constant* $\Delta\chi^2$ method. All the parameters are measured using the reactor constraint $\sin^2 2\theta_{13} = 0.085 \pm 0.005$ [76], except for δ_{CP} and $\sin^2\theta_{13}$. For the inverted hierarchy the intervals are calculated both with respect to the global minimum, i.e. normal hierarchy, as well as independently for inverted hierarchy (between parenthesis in second row). As explained in section 6.5, the gaussian approximation does not work well for discrete parameters like the mass hierarchy however, since there is not a clear way to compute 1σ ranges for both mass hierarchies without performing CPU intensive toy MC studies, the constant $\Delta\chi^2$ method is used.



(a) 90% and 2σ CL



(b) 90% and 2σ CL with uncertainty



(c) 90% CL with exclusion region

Figure 38: Measured $\Delta\chi^2$ distributions with the full Run 1-7c data set. The "Feldman-Cousins" method is used to set the $\Delta\chi^2$ critical values in order to provide confidence intervals with the proper coverage. The ranges of δ_{CP} which correspond to $\Delta\chi^2$ values larger than the critical values (blue for 90% CL and orange for 2σ CL) are excluded at a certain confidence level. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 25. In fig. 38b, the thickness of the lines which represent the critical values are set according to the one sigma variation in the critical values. These uncertainties in the critical values are previously shown in fig. 35.

Parameter	Reactors	Normal Hierarchy	Inverted Hierarchy
δ_{CP} (radians)	YES	[-2.978; -0.467]	[-1.466; -1.272]

Table 28: The confidence intervals at 90% CL obtained with the Feldman-Cousins method for the Run 1-7c data set is shown for the measurement of δ_{CP} vs mass hierarchy. The reactor constraint $\sin^2 2\theta_{13} = 0.085 \pm 0.005$ [76] is applied.

9.3. 2-dimensional confidence intervals

The same 2-dimensional confidence intervals as in section 7 are produced. In fig. 39 the corresponding 2-dimensional contours produced with the Run 1-7c data set are shown drawing the normal and inverted hierarchy contours independently, i.e. each contour is produced with respect to the global minimum under that hierarchy hypothesis.

In fig. 40 both normal and inverted hierarchy contours are produced with respect to the same global minimum, i.e. the normal hierarchy hypothesis. As already explained in section 6.5, "global" and "independent" contours correspond respectively to the least and most conservative approach.

In fig. 39a we see there is good agreement between the result of reactor measurements for $\sin^2(\theta_{13})$ and this analysis. The reactor best fit value is well within our 68% confidence contour. When the reactor constraint is applied in our analysis, shown in fig. 39b, we see almost all positive values of δ_{CP} are excluded at 90% confidence. The best fit of δ_{CP} is around $-\frac{\pi}{2}$ under both hierarchy hypotheses, as in the 1D fits shown in fig. 37. Also we see that $\delta_{CP} = 0$ is excluded slightly more than $\delta_{CP} = \pi$ which also appears in the 1D fits.

fig. 39d highlights the strong correlation that remains between δ_{CP} and $\sin^2(\theta_{23})$ in this analysis as it was seen in [11]. This is expected as we expect only additional statistics in the antineutrino mode samples to be able to break the degeneracy in the fit between these two parameters.

9.4. CC1π^+ vs 1 Ring comparison

This section includes a comparison of the data fits from the 1 Ring analysis from [11] and this analysis. Each contour has been drawn independently with respect to the minimum χ^2 in each respective analysis and only the $\Delta\chi^2$ is compared between the two analyses. fig. 41 shows the comparison of the 1D fits to δ_{CP} and 2D fits are shown in figs. 42 and 43 with and without the reactor constraint respectively.

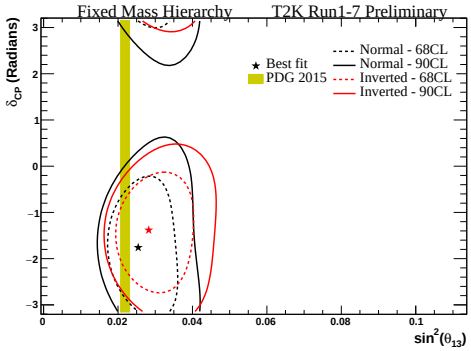
The improvement in the constraints are comparable to the results from the sensitivity study comparison shown in section 7. The constraint on δ_{CP} is slightly stronger than expected which is due to the upward fluctuation observed in the ν_e CC1π^+ sample where 5 events were observed with only 3 expected. With this, there is a noticeably greater exclusion of $\delta_{CP} = \frac{\pi}{2}$. The effect is also apparent in the $\sin^2\theta_{13}$ Vs δ_{CP} contours where the new analysis prefers a larger value of $\sin^2\theta_{13}$ which, when constrained by the reactor measurements, results in a stronger constraint on δ_{CP}. The comparison of the fits to $\sin^2\theta_{23}$ Vs Δm^2_{32} are slightly unexpected considering the sensitivity study, however the result where a greater portion of the lower octant of $\sin^2\theta_{23}$ is excluded at 90%CL and 68%CL can be explained again by the excess of events observed in the ν_e CC1π^+ which favour a much larger value of $\sin^2\theta_{23}$ and therefore push the parameter to the upper octant.

10. Comparison between ν and $\bar{\nu}$ only contours with the Run 1-7c dataset

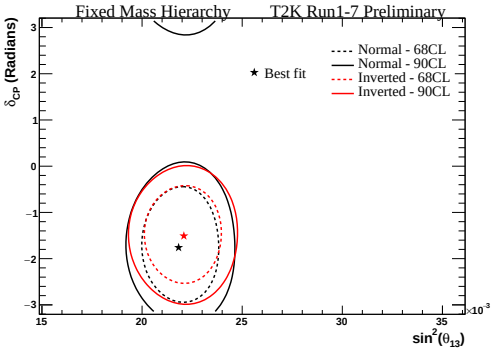
In fig. 44 the contours of $\sin^2\theta_{13}$ Vs δ_{CP} obtained by analysing either the neutrino mode or antineutrino mode electron like event samples in the Run 1-7c data set are compared. In each case, both the muon like samples are used in the fits. Both the ν-only and $\bar{\nu}$-only contours are in good agreement with the reactor measurement in the region of the parameters approximately close to $\delta_{CP} = -\pi/2$ and normal hierarchy. This is reasonable considering the constraint on δ_{CP} found when considering all constraints together. The region by $\delta_{CP} = -\pi/2$ is where all 3 constraints on $\sin^2\theta_{13}$ disagree the most.

It is also clear that the $\bar{\nu}$-only contour is not statistically powerful enough to observe non-zero θ_{13} with good significance. Without analysing the ν and $\bar{\nu}$ data simultaneously the region at $\delta_{CP} = \pi/2$ is not excluded at 90% CL using the constant $\Delta\chi^2$ method.

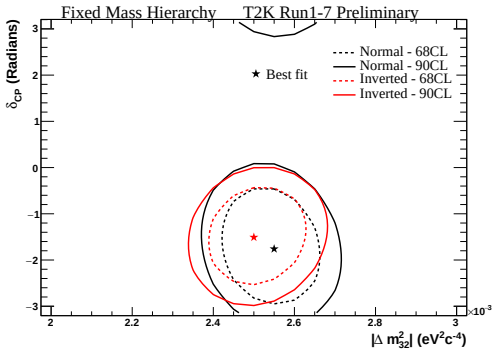
In fig. 45 the contours for $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with the full Run 1-7c data set are compared with the other experimental results.



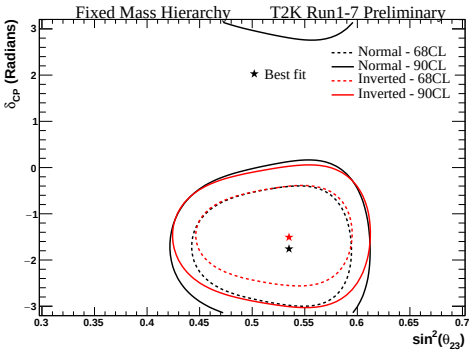
(a) $\sin^2\theta_{13}$ vs δ_{CP} with T2K data only



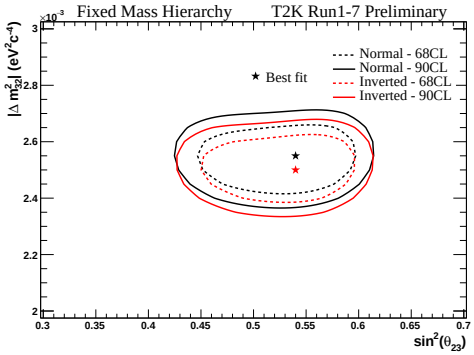
(b) $\sin^2\theta_{13}$ vs δ_{CP} with reactor constraint



(c) Δm^2_{32} (Δm^2_{13}) vs δ_{CP} with reactor constraint

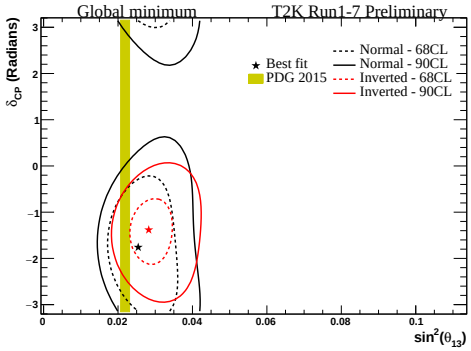


(d) $\sin^2\theta_{23}$ vs δ_{CP} with reactor constraint

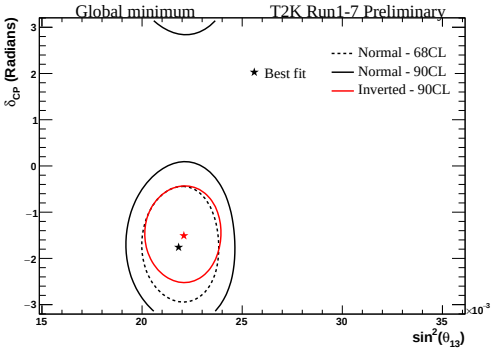


(e) $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with reactor constraint

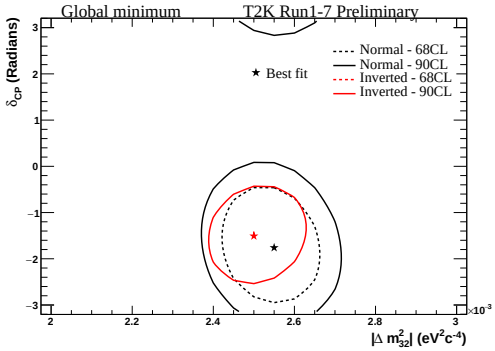
Figure 39: Measured confidence intervals with the full Run 1-7c data set. Both normal (black) and inverted (red) hierarchies are shown. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed independently. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 12. The measured contour of $\sin^2\theta_{13}$ vs δ_{CP} without reactor constraint, in fig. 39a is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [76], shown by the yellow bar.



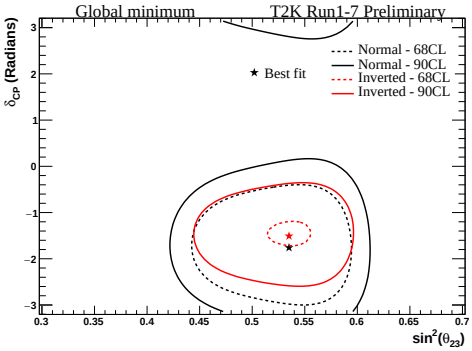
(a) $\sin^2\theta_{13}$ vs δ_{CP} with T2K data only



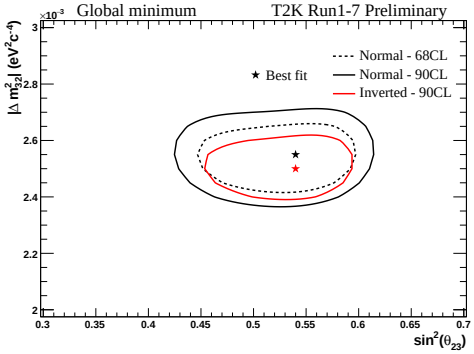
(b) $\sin^2\theta_{13}$ vs δ_{CP} with reactor constraint



(c) Δm^2_{32} (Δm^2_{13}) vs δ_{CP} with reactor constraint

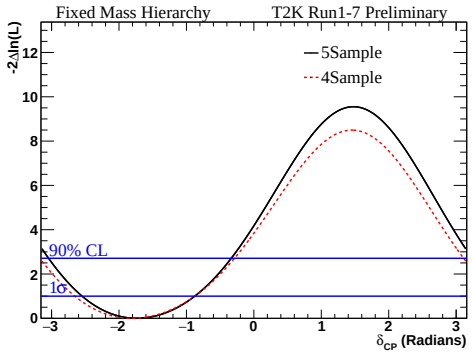


(d) $\sin^2\theta_{23}$ vs δ_{CP} with reactor constraint

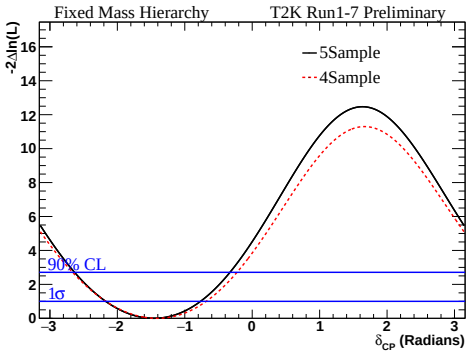


(e) $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with reactor constraint

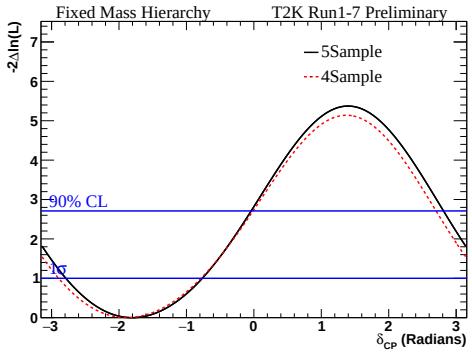
Figure 40: Measured confidence intervals with the full Run 1-7c data set. Both normal (black) and inverted (red) hierarchies are shown. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed with respect to the minimum χ^2 in both hierarchies. In all cases the best fit was found to be in the normal hierarchy hypothesis so this is defined as the best fit point, the best fit point in the inverted hierarchy hypothesis (shown in red) is for reference. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 12. The measured contour of $\sin^2\theta_{13}$ vs δ_{CP} without reactor constraint, in fig. 40a is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [76], shown by the yellow bar.



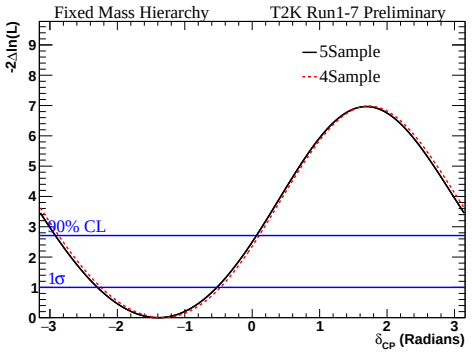
(a) Normal hierarchy with the reactor constraint



(b) Inverted hierarchy with the reactor constraint

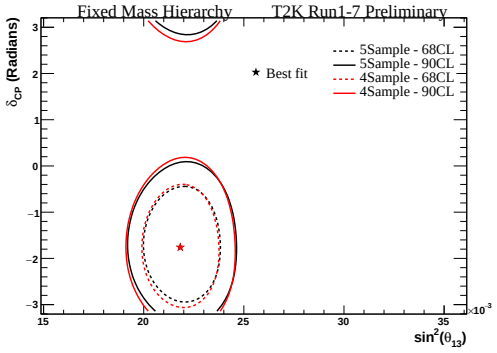


(c) Normal hierarchy without the reactor constraint

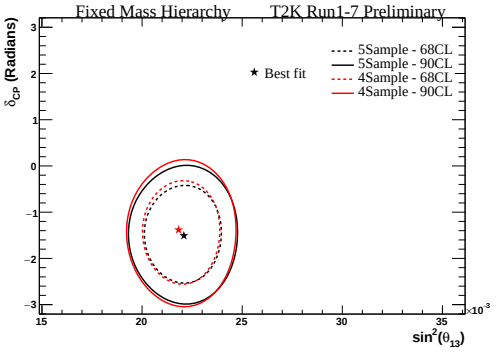


(d) Inverted hierarchy without the reactor constraint

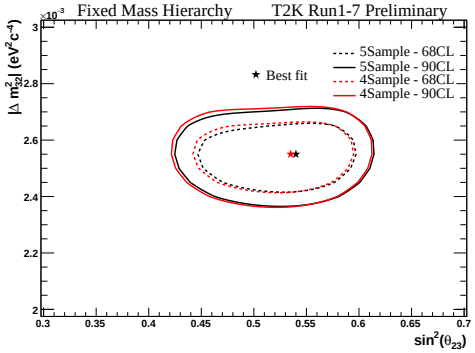
Figure 41: The comparison of the results of fits to δ_{cp} from the 1Ring and 1Ring+ccπ^+ analyses are shown. The comparison is done both applying the reactor constraint (top) and with only T2K data (bottom). The contours were produced with respect to the minimum in each respective analysis independently.



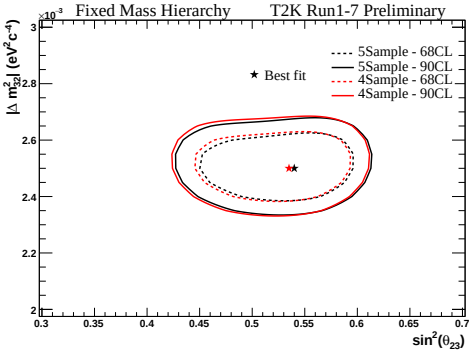
(a) Normal: $\sin^2\theta_{13}$ Vs δ_{CP}.



(b) Inverted: $\sin^2\theta_{13}$ Vs δ_{CP}.

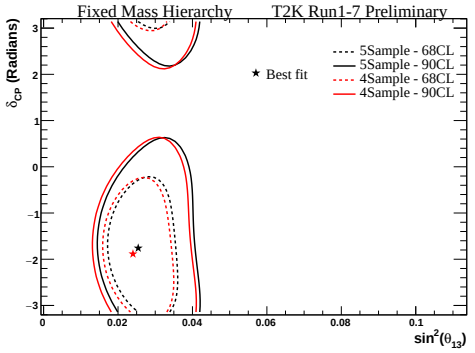


(c) Normal: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}).

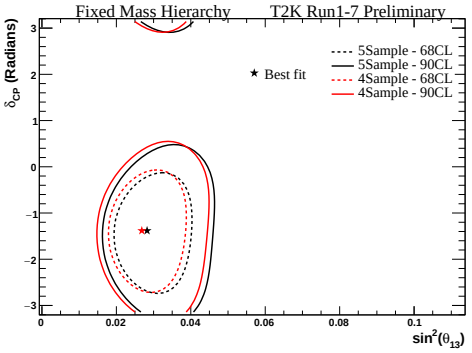


(d) Inverted: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}).

Figure 42: The comparison of the results of fits from the 1Ring and 1Ring+ccπ^+ analyses are shown. The contours at 68% and 90% CL with reactor constraint the Run 1-7c statistics are shown. The contours were produced with respect to the minimum in each respective analysis independently.

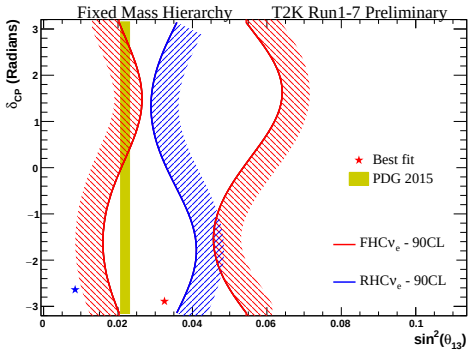


(a) Normal

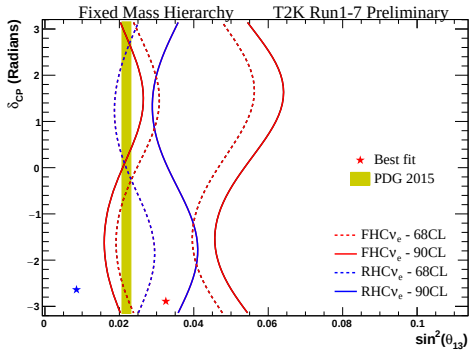


(b) Inverted

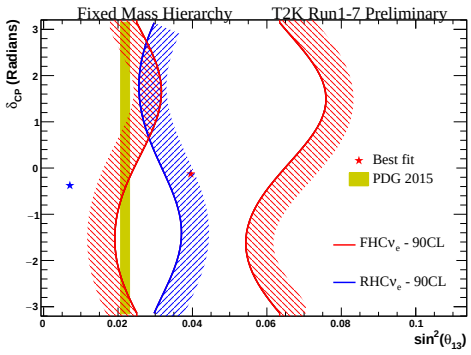
Figure 43: The comparison of the results of fits from the 1Ring and 1Ring+ccπ^+ analyses are shown. The contours at 68% and 90% CL for $\sin^2\theta_{13}$ Vs δ_{CP} corresponding to the Run 1-7c statistics are shown. The contours were produced with respect to the minimum in each respective analysis independently. Only T2K information is used.



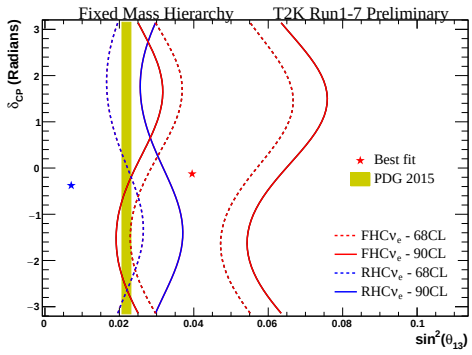
(a) Only 90% CL - Normal Hierarchy



(b) 68% and 90% CL - Normal Hierarchy

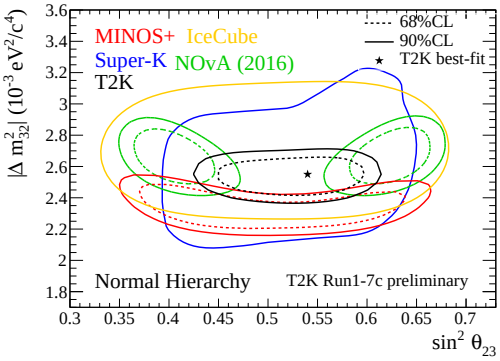


(c) Only 90% CL - Inverted Hierarchy

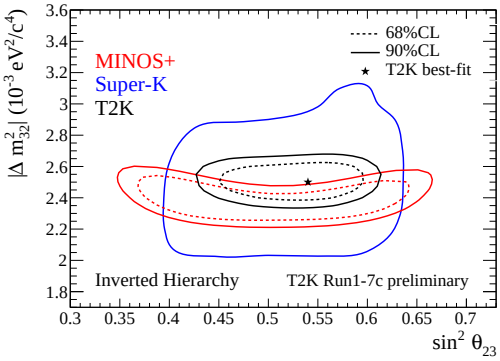


(d) 68% and 90% CL - Inverted Hierarchy

Figure 44: The contours of $\sin^2\theta_{13}$ Vs δ_{CP} T2K only, obtained by analysing only either the ν only or the $\bar{\nu}$ only Run 1-7c data set are compared for both hierarchies. The yellow band corresponds to the reactor value of the PDG 2015. Depending on the figure, either 68% and/or 90% CL contours are shown. In the left plots only 90% CL contours are shown. The hatches indicate the excluded region. In the right plots both the 68% and 90% CL contours are shown. For $\bar{\nu}$-only the region on the left side of the blue lines is allowed, while for ν-only the region between the red lines is allowed.



(a) Normal Hierarchy

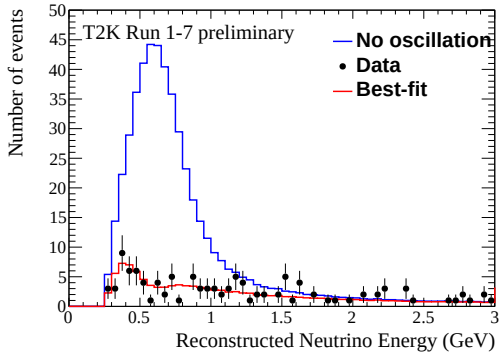


(b) Inverted Hierarchy

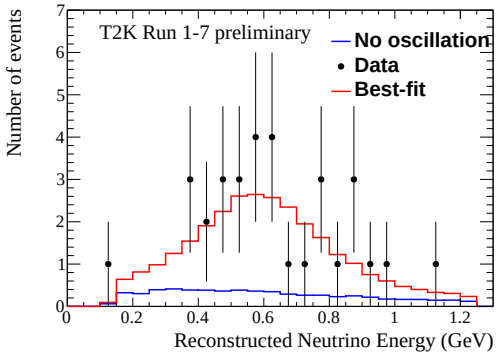
Figure 45: The measured T2K confidence intervals for $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with the full Run 1-7c data set are compared to the other experimental results. Both normal (top) and inverted (bottom) hierarchies are shown. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed independently. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 12. The measured contour of $\sin^2\theta_{13}$ vs δ_{CP} without reactor constraint, in **??** is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [76], shown by the yellow bar. For normal hierarchy the other experiment results are: Super-K (PoS ICRC2015 (2015) 1062), Minos+ (Neutrino 2014 conference), NOνA (Neutrino 2016 conference) and IceCube DeepCore (Phys.Rev. D91 (2015) 072004, arXiv 1410.7227). For inverted hierarchy the other experiment results are: Super-K (PoS ICRC2015 (2015) 1062) and Minos+ (Neutrino 2014 conference).

10.1. Best-fit spectra

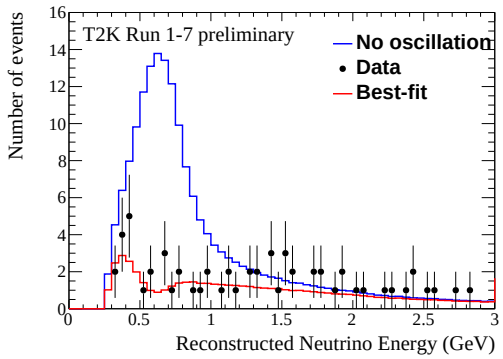
The best-fit spectra, shown in fig. 46 are computed as described in section 6.3 and are compared to the unoscillated nominal spectra. For e-like samples, where 2-dimensional templates are used, the projection onto the reconstructed neutrino energy axis is shown. All the systematic and oscillation parameters are marginalised following the procedure described in section 6.3, while δ_{CP} and the mass hierarchy were fixed to their best-fit values obtained from their simultaneous fit, i.e. fig. 37a. Indeed these oscillation parameters have the strongest effect on the measurement, since $\sin^2\theta_{13}$ is constrained with the reactor measurement.



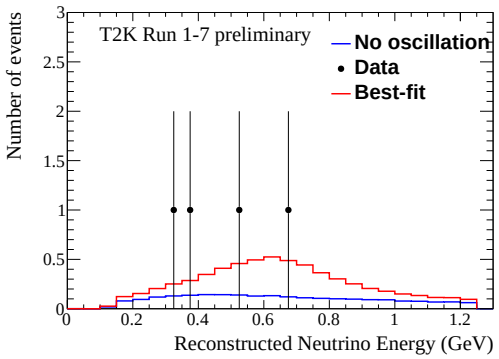
(a) Muon-like, neutrino mode



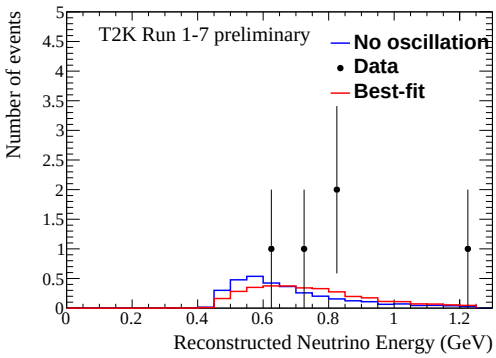
(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode



(e) ν_e CC1π^+, neutrino mode

Figure 46: The best-fit spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the Run 1-7c data set. They correspond to the fit of δ_{CP} and mass hierarchy with reactor constraint where all the nuisance parameters are marginalised. The method described in eq. (11) is used. The best-fit distributions are compared with the predicted unoscillated spectrum (all the oscillation parameters are set to 0), generated with the systematic parameters measured with the ND280 data shown in table 7, 8, 9 and the nominal Super-K systematic parameters shown in table 10.

10.2. Goodness of fit

A goodness of fit test is performed by comparing data with the best fit spectra shown in fig. 46. The goodness of fit test has been performed using a reduced binning i.e. 5 reconstructed neutrino energy bins (0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV) for neutrino mode and antineutrino mode μ-like and 1 bin for all the other Super-K samples.

The χ^2 and resulting p-value for each spectrum is shown in table 29. We see for all the spectra, the data and best fit spectra are very compatible.

Though the method and inputs are slightly different, the p-values reported here are very compatible with what was found in the previous result [11]. The number of degrees of freedom have increased by two from the previous analysis. One additional degree of freedom comes from the 1 bin ν_e CC1π^+ sample. The other comes from the reduction in free parameters in the fit from three (two) previously to two (one) in this analysis.

Mass Hierarchy d.o.f.	χ^2 / ndf	p-value
fixed	5.02391 / 12	0.96
free	5.02391 / 11	0.93

Table 29: Goodness of fit for the Run 1-7c fit of δ_{CP} and mass hierarchy where all the other parameters are marginalised. Since the mass hierarchy is a discrete parameter, the goodness of the fit test is performed assuming the mass hierarchy both as a free parameter and as a fixed parameter. $ndf = 11$ if the mass hierarchy is assumed to behave as 1 degree of freedom.

10.3. Discussion of the Run 1-7c data fit results

Using the additional ν_e CC1π^+ data set we see an excess in the number of observed appearance events which is consistent with what is seen in the neutrino mode one ring e-like sample.

The value of δ_{CP} favoured by the joint fit analysis, when the reactor constraint is applied, is -1.728 and corresponds to normal hierarchy. $\delta_{CP} = 0$ is excluded at more than 2 standard deviations and $\delta_{CP} = \pi$ more than 90% confidence which is a slight improvement on the previous analyses.

The inverted hierarchy is disfavoured at about 90%CL and is partially excluded at 3σ for values of δ_{CP} around $+\pi/2$ similarly as for 1 ring data set.

If the reactor constraint is not applied, the best-fit value of δ_{CP} is still quite close to $-\pi/2$ and $\delta_{CP} = 0$ is still excluded at about 90% CL (using the constant $\Delta\chi^2$ method).

The contour of $\sin^2\theta_{13}$ vs δ_{CP} shows a very good agreement (within 1σ), with the measurement of the reactor experiments. Although best fit value of $\sin^2\theta_{13}$ is slightly higher than the previous $1Ring$ analysis due to the observed upward fluctuation from the expectation in the ν_e CC1π^+ sample.

10.4. Comparison of δ_{CP} vs mass hierarchy Run 1-7c result with sensitivity

In order to provide a better understanding of the results of the Run 1-7c data fit, the result has been compared with the expected sensitivity obtained from a large number (10k) of toy experiments for different true values of δ_{CP} and the mass hierarchy. The nuisance parameters in the toy experiments have been randomised in accordance with the procedure used in section 8.

10.4.1. Expected sensitivity method

This section outlines the procedure to produce the expected sensitivity comparison plots detailed in this section. The example is given for true normal hierarchy and $\delta_{CP} = -\pi/2$.

- $\sim$10k toy experiments were performed for a certain true point, in this case $\delta_{CP} = -\pi/2$ and normal hierarchy, and the full χ^2_{toy} distribution was computed as a function of δ_{CP} and mass hierarchy.;
- For each bin of δ_{CP} and mass hierarchy the $\Delta\chi^2$ was obtained by calculating the mean χ^2 distribution independently for each hierarchy from all the toy experiments and the global minimum χ^2_{glob} (with respect to both hierarchies). This is what is done for each Asimov and data fit in section 9
- The $\Delta\chi^2$ distribution as a function of δ_{CP} and mass hierarchy for each toy experiment ($\Delta\chi^2_{toy}$) is taken with respect to the global minimum for that particular toy experiment ($\chi^2_{glob,toy}$) between both mass hierarchies and all the values of δ_{CP};
- From the full $\Delta\chi^2$ distribution for each toy experiment, we can draw conclusions about the expected $\Delta\chi^2$ for each value of δ_{CP} given a hypothesised true value and hierarchy.
- This distribution is similar to that labeled the Feldman-Cousins distribution, but not the same. For the construction of the Feldman-Cousins intervals the test statistic is $\Delta\chi^2_{FC} = \chi^2_{true} - \chi^2_{best-fit}$, where *true* is the value of δ_{CP} ($-\pi/2$) and mass hierarchy (normal) used to generate the toy experiments. Instead in this case the test statistic is $\Delta\chi^2_{bin} = \chi^2_{bin} - \chi^2_{best-fit}$, where χ^2_{bin} corresponds to the χ^2 value for a *bin* of δ_{CP} and mass hierarchy obtained by the fit of a toy experiment generated with $\delta_{CP} = -\pi/2$ and normal hierarchy (*true*); the distributions of $\Delta\chi^2_{bin}$ and $\Delta\chi^2_{FC}$ coincide if the *bin* is $\delta_{CP} = -\pi/2$ and normal hierarchy, i.e. the *true* point used to generate the toy experiments;
- the spread of the $\Delta\chi^2$ distribution for each point of δ_{CP} and mass hierarchy are obtained as the range of the $\Delta\chi^2_{bin}$ distribution that contains respectively the 68.27% and 95.45% of the toy experiments on its left side. The test statistic $\Delta\chi^2_{bin}$ is assumed to be 1-sided because by construction $\chi^2_{best-fit} \leq \chi^2_{bin}$.
- A plot is also produced to show the fraction of toy experiments which exclude each value of δ_{CP} to greater than 90% and 2σ significance in the Feldman Cousins corrected critical values.

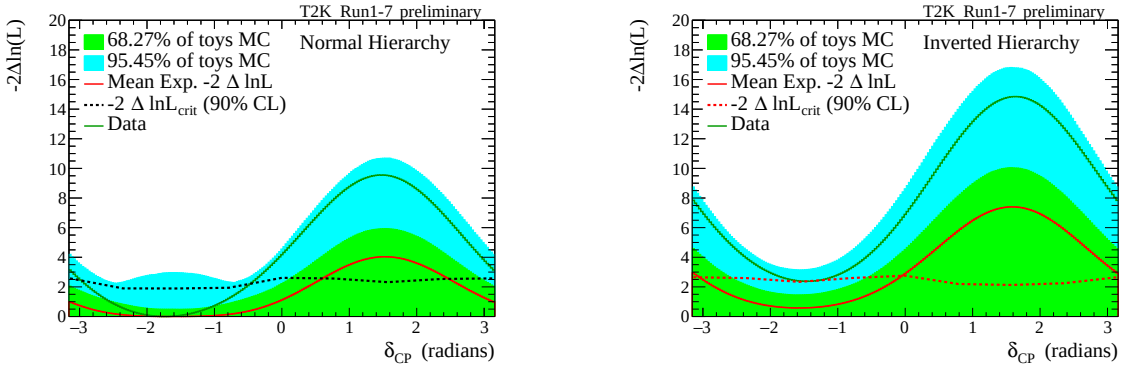
10.4.2. $\delta_{CP} = -\frac{\pi}{2}$ - normal hierarchy

In fig. 47 the Run 1-7c data result for δ_{CP} vs mass hierarchy is compared to the expected mean $\Delta\chi^2$ ($< \Delta\chi^2 >$) distribution as well as the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = -\pi/2$ and normal hierarchy.

Also in fig. 47 the distributions of $\Delta\chi^2$ Vs δ_{CP} of 10k toy experiments are shown for both normal and inverted hierarchy. These distributions are used to define the bands that contain the 68.27% and 95.45% of the toy experiments. The red curve shows the mean expected exclusion from the fake toy experiments and the data constraint is clearly stronger than the mean. However, we see that the data contour for both the normal and inverted hierarchy is within the 95% band of toy experiments. This suggests that the data constraint is consistent with a fluctuation of true $\delta_{CP} = -\pi/2$ below the 2σ level.

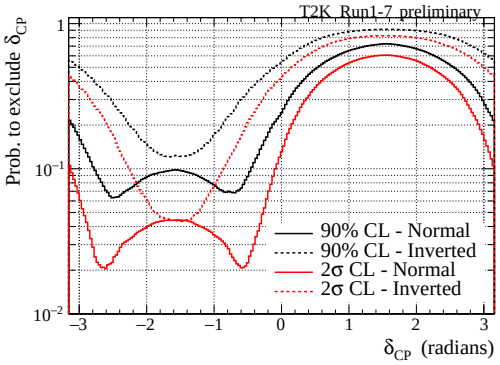
We also see an interesting feature near $\delta_{CP} = -\pi/2$ where the $\Delta\chi^2$ values have a wider distribution. This is due to the boundary at $\delta_{CP} = -\pi/2$ which causes a wider distribution when the parameter δ_{CP} would like to be pushed beyond the boundary in the fit.

Finally from fig. 47c we see that $\delta_{CP} = 0$ is excluded at 90% confidence level in about 22% of the toy experiments and the 2σ confidence level between 1% and 2% of the toy experiments. $\delta_{CP} = \pi$ is excluded in slightly fewer toy experiments in each case.



(a) Normal hierarchy

(b) Inverted hierarchy



(c) Data χ^2 probability

Figure 47: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 10k toy experiments generated with $\delta_{CP} = -\pi/2$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown. Shown are the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fraction of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman Cousins corrected critical values.

10.4.3. $\delta_{CP} = -1.728$ - *normal hierarchy*

In fig. 48 the Run 1-7c data result for δ_{CP} vs mass hierarchy is compared to the expected mean $\Delta\chi^2$ ($< \Delta\chi^2 >$) distribution as well as the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = -1.728$ and normal hierarchy which is the best fit value found in this analysis.

Also in fig. 48 the distributions of $\Delta\chi^2$ Vs δ_{CP} of 10k toy experiments are shown for both normal and inverted hierarchy. These distributions are used to define the bands that contain the 68.27% and 95.45% of the toy experiments.

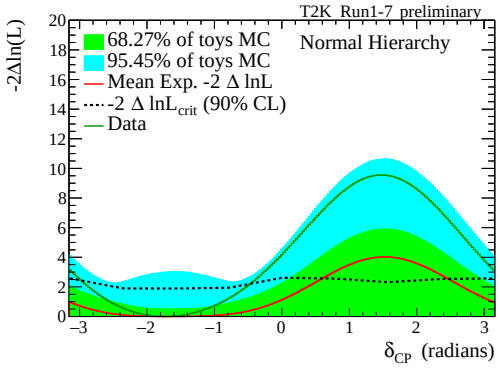
The results are very similar to those shown in section 10.4.2 as expected from the values of δ_{CP} used for the toy experiments. The exclusion found is within the 2σ regions shown in blue.

10.4.4. $\delta_{CP} = 0$ - *normal hierarchy*

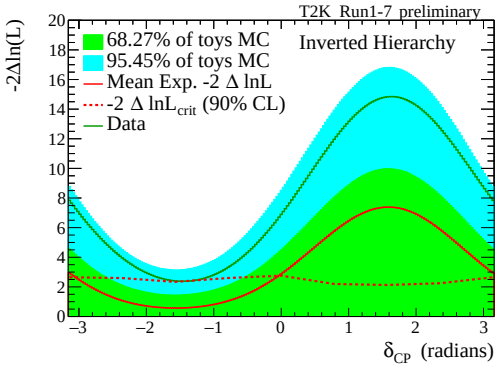
In fig. 49 the Run 1-7c data result for δ_{CP} vs mass hierarchy is compared to the expected mean $\Delta\chi^2$ ($< \Delta\chi^2 >$) distribution as well as the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = 0$ and normal hierarchy.

Also in fig. 49 the distributions of $\Delta\chi^2$ Vs δ_{CP} of 10k toy experiments are shown for both normal and inverted hierarchy. These distributions are used to define the bands that contain the 68.27% and 95.45% of the toy experiments.

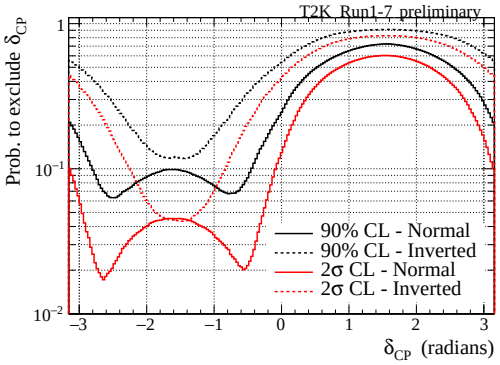
The comparison of the data and the toy experiments show that the exclusion of $\delta_{CP} = \frac{\pi}{2}$ seen in the data is unlikely to be due to a fluctuation of $\delta_{CP} = 0$. This is consistent with the exclusion of $\delta_{CP} = 0$ provided from the data fit. However the exclusion of $\delta_{CP} = \pi$ from data is within the 2σ band from the toy experiments.



(a) Normal hierarchy



(b) Inverted hierarchy



(c) Data χ^2 probability

Figure 48: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 10k toy experiments generated with $\delta_{CP} = -1.728$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown. Shown are the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fraction of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman Cousins corrected critical values.

10.4.5. $\delta_{CP} = \pi$ - normal hierarchy

In fig. 50 the Run 1-7c data result for δ_{CP} vs mass hierarchy is compared to the expected mean $\Delta\chi^2$ ($< \Delta\chi^2 >$) distribution as well as the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = \pi$ and normal hierarchy.

Also in fig. 50 the distributions of $\Delta\chi^2$ Vs δ_{CP} of 10k toy experiments are shown for both normal and inverted hierarchy. These distributions are used to define the bands that contain the 68.27% and 95.45% of the toy experiments.

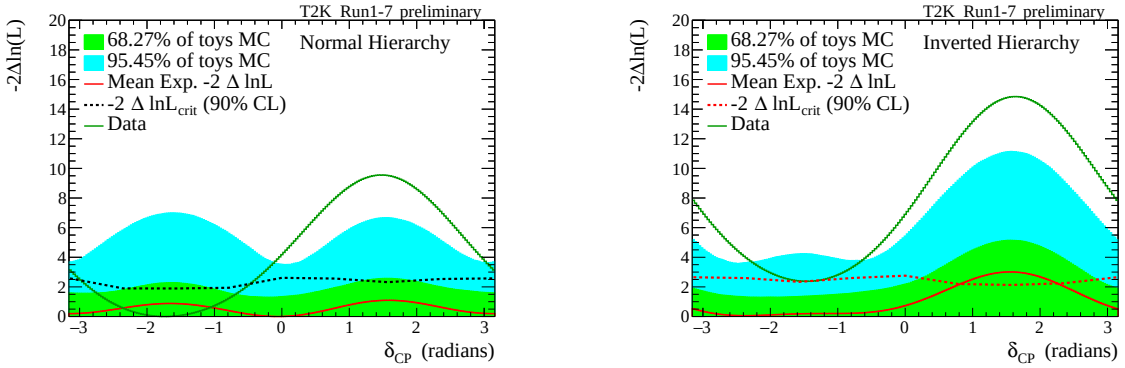
An interesting feature is the data exclusion at $\delta_{CP} = \pi$ which is within the 2σ band of exclusions from the toy experiments. Considering only that point, the exclusion found in data could be consistent with a statistical fluctuation of true $\delta_{CP} = \pi$ however the exclusion found for the inverted hierarchy and normal hierarchy $\delta_{CP} \sim \frac{\pi}{2}$ is very inconsistent.

10.4.6. $\delta_{CP} = -1.445$ - inverted hierarchy

In fig. 51 the Run 1-7c data result for δ_{CP} vs mass hierarchy is compared to the expected mean $\Delta\chi^2$ ($< \Delta\chi^2 >$) distribution as well as the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = -1.445$ and inverted hierarchy which is the best fit in the inverted hierarchy hypothesis.

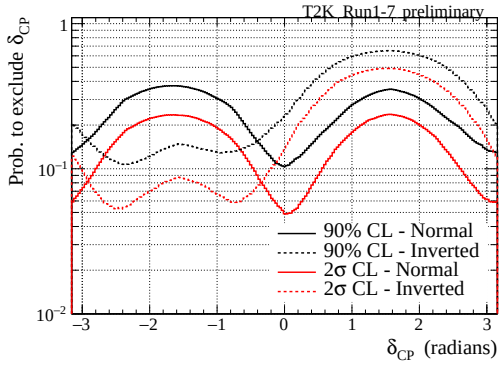
Also in fig. 51 the distributions of $\Delta\chi^2$ Vs δ_{CP} of 10k toy experiments are shown for both normal and inverted hierarchy. These distributions are used to define the bands that contain the 68.27% and 95.45% of the toy experiments.

As expected from the exclusion of the inverted hierarchy observed in the data fit, the data contours are not very consistent with the exclusion expected from inverted hierarchy toy experiments. This is the case



(a) Normal hierarchy

(b) Inverted hierarchy



(c) Data χ^2 probability

Figure 49: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 10k toy experiments generated with $\delta_{CP} = 0$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown. Shown are the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fraction of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman Cousins corrected critical values.

for both the normal and inverted hierarchy exclusions. The inverted hierarchy exclusions from the toy fits are interesting as the exclusion near $\delta_{CP} = \frac{\pi}{2}$, shown in fig. 51b, is larger than the exclusion in the normal hierarchy fits shown in fig. 51a. This is due to the mass hierarchy vs δ_{CP} degeneracy where fluctuations which favour $\delta_{CP} \sim -\frac{\pi}{2}$ also favour the normal hierarchy.

10.4.7. $\delta_{CP} = -\frac{\pi}{2}$ - inverted hierarchy

In fig. 52 the Run 1-7c data result for δ_{CP} vs mass hierarchy is compared to the expected mean $\Delta\chi^2$ ($< \Delta\chi^2 >$) distribution as well as the 1σ and 2σ uncertainties on the $\Delta\chi^2$ distributions corresponding to $\delta_{CP} = -\pi/2$ and normal hierarchy.

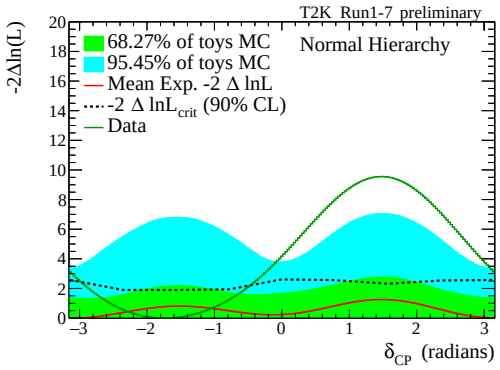
Also in fig. 52 the distributions of $\Delta\chi^2$ Vs δ_{CP} of 10k toy experiments are shown for both normal and inverted hierarchy. These distributions are used to define the bands that contain the 68.27% and 95.45% of the toy experiments.

Conclusions from this study are very similar to those in section 10.4.6.

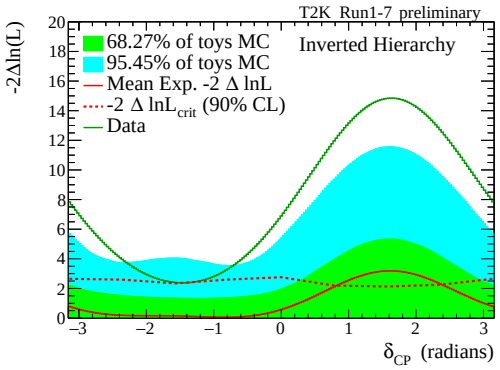
10.5. Discussion of the Run 1-7c data fit results with sensitivity

In table 30 a summary of the fraction of toys which are excluded at 90% and 2σ (defined from the Feldman Cousins critical values) for 0 and π are shown under each true δ_{CP} and mass hierarchy hypothesis studied in this section.

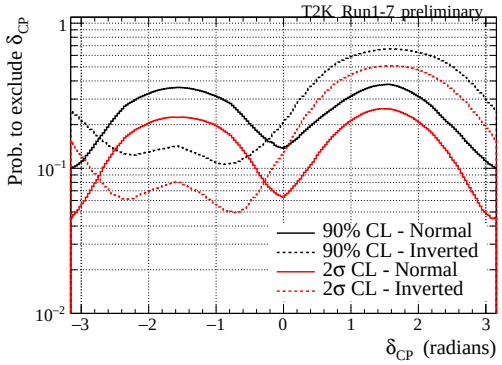
The Run 1-7c data set $\Delta\chi^2$ distribution is consistent with the hypothesis of true $\delta_{CP} = -\pi/2$ (or $\delta_{CP} = -1.728$) at about 2σ level. There is also a not negligible probability to exclude $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ with a significance of 2σ ($\sim 10\%$), if nature is $\delta_{CP} = -\pi/2$ (or -1.728) and normal hierarchy.



(a) Normal hierarchy

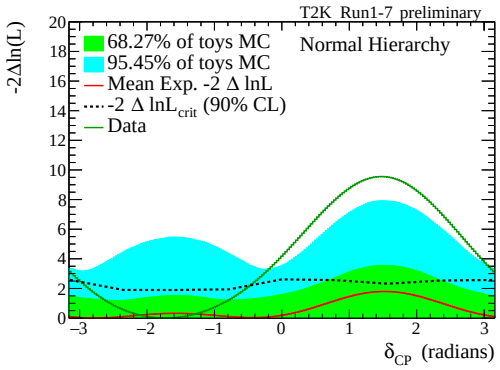


(b) Inverted hierarchy

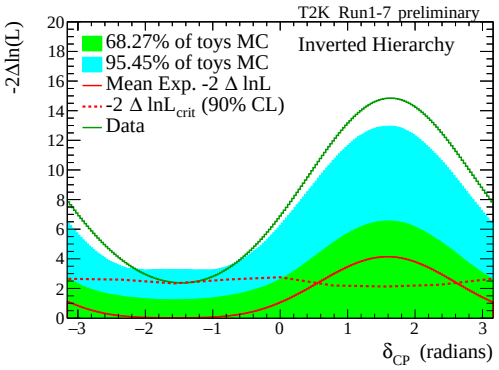


(c) Data χ^2 probability

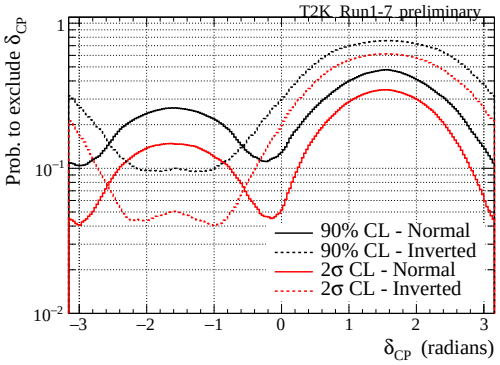
Figure 50: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 10k toy experiments generated with $\delta_{CP} = \pi$ and normal hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown. Shown are the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fraction of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman Cousins corrected critical values.



(a) Normal hierarchy

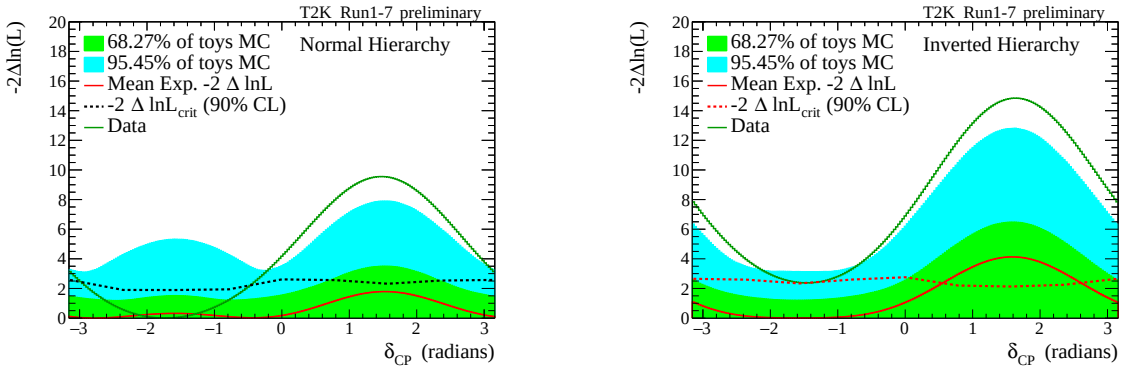


(b) Inverted hierarchy



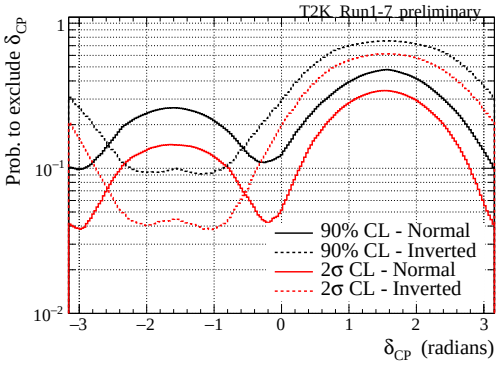
(c) Data χ^2 probability

Figure 51: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 10k toy experiments generated with $\delta_{CP} = -1.445$ and inverted hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown. Shown are the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fraction of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman Cousins corrected critical values.



(a) Normal hierarchy

(b) Inverted hierarchy



(c) Data χ^2 probability

Figure 52: The distribution of $\Delta\chi^2$ vs δ_{CP} obtained with 10k toy experiments generated with $\delta_{CP} = -\pi/2$ and inverted hierarchy is shown. The $\Delta\chi^2$ distributions obtained by fixing the mass hierarchy to normal (left) and inverted (right) are shown. Shown are the $\Delta\chi^2$ values corresponding to 68.27% and 95.45% of the toy experiments and the average $\Delta\chi^2$ values from all experiments. Also shown are the fraction of toys which exclude each value of δ_{CP} to 90% and 2σ using the Feldman Cousins corrected critical values.

True: $\delta_{CP} = -\pi/2$ - normal hierarchy			
δ_{CP}	hierarchy	90% CL	2σ CL
0	NH	0.243	0.131
π	NH	0.216	0.105
0	IH	0.542	0.425
π	IH	0.559	0.436
True: $\delta_{CP} = -1.728$ - normal hierarchy			
δ_{CP}	hierarchy	90% CL	2σ CL
0	NH	0.244	0.129
π	NH	0.209	0.100
0	IH	0.539	0.422
π	IH	0.557	0.438
True: $\delta_{CP} = 0$ - normal hierarchy			
δ_{CP}	hierarchy	90% CL	2σ CL
0	NH	0.104	0.0490
π	NH	0.130	0.0591
0	IH	0.229	0.137
π	IH	0.205	0.122
True: $\delta_{CP} = \pi$ - normal hierarchy			
δ_{CP}	hierarchy	90% CL	2σ CL
0	NH	0.138	0.0633
π	NH	0.100	0.0457
0	IH	0.206	0.130
π	IH	0.249	0.154
True: $\delta_{CP} = -\pi/2$ - inverted hierarchy			
δ_{CP}	hierarchy	90% CL	2σ CL
0	NH	0.124	0.0515
π	NH	0.102	0.0413
0	IH	0.290	0.194
π	IH	0.308	0.207
True: $\delta_{CP} = 1.445$ - inverted hierarchy			
δ_{CP}	hierarchy	90% CL	2σ CL
0	NH	0.129	0.0516
π	NH	0.109	0.0446
0	IH	0.296	0.197
π	IH	0.316	0.212

Table 30: The fraction of toy experiments for which $\delta_{CP} = 0, \pi$ and normal and inverted hierarchy are excluded at 90% and 2σ CL is shown for different true values of δ_{CP} and mass hierarchy. 10k toy experiments are used.

11. Summary

In this technical note, we have presented the results of the 3-flavour $\bar{\nu}/\nu$ joint analysis performed by the *VALOR* group, on the combined Run 1-7c data set. The goal of the analysis is to search for CP violation in the leptonic and put limits on δ_{CP} as well as measure all the other oscillation parameters to which T2K is sensitive: $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) and partially mass hierarchy, to which T2K has little sensitivity.

This analysis predicts 135.8 ± 6.9 (syst) for neutrino μ-like, 64.2 ± 3.3 (syst) for antineutrino μ-like, 28.7 ± 1.8 (syst) for neutrino e-like, 6.0 ± 0.4 (syst) for antineutrino e-like and **3.1** ± **0.47** (syst) for neutrino ν_e CC1π^+ events in Super K and observes respectively 135 for neutrino μ-like, 66 for antineutrino μ-like, 32 for neutrino e-like, 4 for antineutrino e-like and **5** for neutrino ν_e CC1π^+.

The $\bar{\nu}/\nu$ analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events and reconstructed energy versus lepton angle of all e-like ring events was used to compute the confidence intervals. All 110 systematic parameters as well as the oscillation parameters which were not parameters of interest considered in this analysis were marginalised.

From the fit of the Run 1-7c data we can see that the favoured value of δ_{CP} is -1.728, close to $-\pi/2$. The correct coverage is provided by using the "Feldman-Cousins" method with which the significance to exclude $\delta_{CP} = 0$ and $\delta_{CP} = \pi$ becomes greater than 2 standard deviations.

In case of inverted hierarchy a not negligible range of δ_{CP} around $+\pi/2$ is excluded with a significance of 3σ.

From the contour of $\sin^2\theta_{13}$ vs δ_{CP} without the reactor constraint it is clear that the T2K data are in agreement with the reactor measurements within 1σ if $\delta_{CP} \sim -\pi/2$.

References

[1] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 3.23×10^{19}-POT Muon-Neutrino Disappearance Analysis, T2K-TN-036(v4), http://www.t2k.org/docs/technotes/036

[2] C.Andreopoulos, T.Dealtry, J.Dobson, N.Grant, M.Haigh, J.Ilic, G.Pearce and A.Weber, T2K 1.431×10^{20}-POT Muon-Neutrino Disappearance Analysis T2K-TN-064(v3), http://www.t2k.org/docs/technotes/064

[3] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 1.431×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-087(v1), http://www.t2k.org/docs/technotes/087

[4] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 3.010×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-141(v10), http://www.t2k.org/docs/technotes/141

[5] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 6.3933×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-155(v2), http://www.t2k.org/docs/technotes/155

[6] C.Andreopoulos, T.Dealtry, L.Escudero, N.Grant and A.Weber, T2K 6.57×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis T2K-TN-183(v3), http://www.t2k.org/docs/technotes/183

[7] C.Andreopoulos et al., T2K 3.010×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-154(v3), http://www.t2k.org/docs/technotes/154

[8] C.Andreopoulos et al., T2K 6.57×10^{20}-POT Joint 3-Flavour Oscillation Analysis T2K-TN-175, http://www.t2k.org/docs/technotes/175

[9] C.Andreopoulos et al., T2K 2.315×10^{20}-POT 3-Flavour Muon-Antineutrino Disappearance Analysis T2K-TN-243, http://www.t2k.org/docs/technotes/243

[10] C.Andreopoulos et al., T2K 4.0108×10^{20}-POT 3-Flavour Electron-Antineutrino Appearance Analysis T2K-TN-252, http://www.t2k.org/docs/technotes/252

[11] C.Andreopoulos et al., T2K Neutrino and Anti-Neutrino 3-Flavour Joint Analysis of Run 1-6 (6.914×10^{20}-POT ν 4.011×10^{20}-POT $\bar{\nu}$), Run 1-7b (7.002×10^{20}-POT ν 7.471×10^{20}-POT $\bar{\nu}$) and Run 1-7c (7.482×10^{20}-POT ν 7.471×10^{20}-POT $\bar{\nu}$) data sets T2K-TN-266, http://www.t2k.org/docs/technotes/266

[12] C.Andreopoulos, T.Dealtry, L.Escudero and N.Grant, T2K 6.5700×10^{20}-POT 3-Flavour Muon-Neutrino Disappearance Analysis: Supplementary material on the effect of systematics, T2K-TN-183 supplement, http://www.t2k.org/docs/technotes/183

[13] T2K Collaboration (K.Abe et al.), First Muon-Neutrino Disappearance Study with an Off-Axis Beam, Phys.Rev.D85:031103, 2012; arXiv:1201.1386

[14] T2K Collaboration (K.Abe et al.), Measurement of Neutrino Oscillation Parameters from Muon Neutrino Disappearance with an Off-axis Beam, arXiv:1308.0465 [hep-ex], 2013. t2k neutrino

[15] T2K Collaboration (K.Abe et al.), Measurement of neutrino oscillation in appearance and disappearance channels by the T2K experiment with 6.6×10^{20} protons on target http://www.t2k.org/comm/pubboard/JournalPapers/oa2014prd/draft3

[16] J. Imber et al., Four Sample Joint Oscillation Analysis with T2K Run1-6 Data, T2K-TN-267, http://www.t2k.org/docs/technotes/267

[17] Konosuke Iwamoto et al., P-theta nue CC1pi+ Appearance Analysis Run1-6, Run1-7b, Run1-7c T2K-TN-293, http://www.t2k.org/docs/technotes/293

[18] Y.Hayato, A neutrino interaction simulation program library NEUT, Acta Phys.Polon.B40:2477-2489,2009.

[19] Beam WG, Flux tuning (11b v3.2), http://www.t2k.org/beam/NuFlux/FluxRelease/11brelease/11btuned-v3.2

[20] M.Otani and J.Kameda, Super-K events and Updated systematic errors for ν_μ disappearance analysis with T2K 1.431×10^{20} POT T2K-TN-065(v6), http://www.t2k.org/docs/technotes/065

[21] P. de Perio et al., NEUT nuclear effects (FSI) T2K-TN-033(v2.0), http://www.t2k.org/docs/technotes/033

[22] P. de Perio et al., Cross section parameters for the 2012a oscillation analysis T2K-TN-108(v1.3), http://www.t2k.org/docs/technotes/108

[23] P. de Perio et al., Implementation of the NIWG cross-section parametrization, T2K-TN-113(v1.2)

[24] Y.Hayato, K.McFarland, and M.Wascko (Private communication)

[25] P. de Perio et al., Pion Hadronic Secondary Interactions in Super-Kamiokande, T2K-TN-105(v1.1) http://www.t2k.org/docs/technotes/105

[26] Y. Ashie et al., Phys. Rev. Lett. 93, 101801 (2004); Y. Ashie et al., Phys. Rev. D71, 112005 (2005); J. Hosaka et al., Phys. Rev. D 74, 032002 (2006).

[27] Beam group, http://www.t2k.org/beam/NuFlux/FluxRelease/11arelease/11atuned-v2.3.

[28] M. Wilking, Status of the 2013 ND280 ν_μ analysis, http://www.t2k.org/asg/meeting/20130411/nd280numustatus/view.

[29] M. Hartz, A. Kaboth, and K. Mahn, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2013 Oscillation Analysis, T2K-TN-166(v3.0) http://www.t2k.org/docs/technotes/166.

[30] M. Hartz, A. Kaboth, K. Mahn, Jordan Myslik, Christine Nielsen and Mark Scott, Constraining the Flux and Cross Section Models with Data from the ND280 Detector for the 2014/2015 Oscillation Analysis, T2K-TN-220(v1.0) http://www.t2k.org/docs/technotes/220.

[31] M. Hartz, A. Kaboth, K. Mahn, Jordan Myslik, Christine Nielsen and Mark Scott, ND280 Flux and Cross Section Constraint (BANFF) for 2015/16, T2K-TN-230(v.1.0) http://www.t2k.org/docs/technotes/230.

[32] J. Kameda, "Updated study of the systematic error in ν_μ disappearance analysis from Super-Kamiokande", T2K-TN-159

[33] J. Albert et al., "Study on SK ν_e candidates and systematic errors with T2K 3.23×10^{19} POT data" T2K-TN-157

[34] J. Hignight et al., Super-Kamiokande events and data quality studies for T2K Run4 T2K-TN-148(v2r3), http://www.t2k.org/docs/technotes/148.

[35] A. Himmel et al., "Super-Kamiokande events and data quality studies for T2K Runs 5 and 6" T2K-TN-219

[36] J. Imber et al., "Super-Kamiokande events and data quality studies for T2K Run 7" T2K-TN-284

[37] 14a data and MC info http://www.t2k.org/t2ksk/documentation-page/MCWeighting/MCReweighting14a

[38] E. Wang, L. Alvarez-Ruso, Y. Hayato, K. Mahn, J. Nieves "Photon emission in neutral current interactions at the T2K experiment" http://arxiv.org/pdf/1507.02446v1.pdf

[39] 2015 SK splines http://www.t2k.org/asg/oagroup/inputs/201415/2015SKsplines

[40] Super-K 13a flux release http://www.t2k.org/beam/NuFlux/FluxRelease/13arelease

[41] M. Friend et al. Flux Tuning and Uncertainty Updates for the 13a Flux T2K-TN-217 http://www.t2k.org/docs/technotes/217

[42] A. Bercellie et al. Cross section parameters for 2014 oscillation analysis. T2K-TN-192 (v2.1) http://www.t2k.org/docs/technotes/192

[43] M. Dunkman et al. Updated Recommendation of the 2015 NIWG Parameters. T2K-TN-265 http://www.t2k.org/docs/technotes/265

[44] http://www.t2k.org/asg/xsec/niwgdocs/neut/NEUT5.3.2

[45] K. Iwamoto et al. Inelastic Single Pion Signal Study in ν_e Appearance using Modified Decay Electron Cut T2K-TN-233 http://www.t2k.org/docs/technotes/233

[46] T. Wongjirad, A. Minamino and J. Kameda, Muon neutrino disappearance analysis with T2K 6.57 $\times$ 10^{20} POT data, T2K-TN-164(v4), http://www.t2k.org/docs/technotes/164

[47] C. Bronner et al., Run1-4 p-theta electron neutrino appearance analysis, T2K-TN-161(v2.6), http://www.t2k.org/docs/technotes/161

[48] Y. Nishimura, J. Hignight, and S. Nakayama ν_e appearance analysis of RUN1-4 data with neutrino energy spectrum, T2K-TN-162(v2.2), http://www.t2k.org/docs/technotes/162

[49] A. Kaboth, R. Calland and D. Payne (Mach3) A Joint ND280-SK 1R$_\mu$-SK 1R$_e$ Fit using MCMC T2K-TN-171 http://www.t2k.org/docs/technotes/171

[50] K. Duffy and A. Kaboth (Mach3) A Joint ND280-SK 1R$_\mu$-SK 1R$_e$ of neutrino and antineutrino-mode data using MCMC T2K-TN-269 http://www.t2k.org/docs/technotes/269

[51] Leila Haegel et al. (MaCh3) MaCh3 run 1-7 joint fit with SK ν_e CC-1π+ sample T2K-TN-300 http://www.t2k.org/docs/technotes/300

[52] K. Duffy for the Mach3 group, http://www.t2k.org/asg/meeting/2015-06-18/mach3_nuebarapp

[53] H. Tanaka for the p-θ group, http://www.t2k.org/asg/meeting/2015-06-18/nuebar_appearance_study

[54] N Abgrall et al. Measurements of production properties of K^0_S mesons and Λ hyperons in proton-carbon interactions at 31 GeV/c Phys. Rev. C 89, 025205

[55] N Abgrall et al. Measurements of Cross Sections and Charged Pion Spectra in Proton-Carbon Interactions at 31 GeV/c. Phys.Rev., C84:034604, 2011; N. Abgrall et al. Measurement of production properties of positively charged kaons in proton-carbon interactions at 31 GeV/c. arXiv:hep-ex/1112.0150, 2011.

[56] J. Beringer et al. (Particle Data Group), Phys. Rev. D86, 010001 (2012)

[57] Lorena Escudero Sanchez for the *VALOR* group, http://www.t2k.org/asg/oagroup/meeting/20121219/VaLOR

[58] Tom Dealtry for the *VALOR* group, http://www.t2k.org/asg/meeting/2012/20121015/valornumudisapp/view

[59] Davide Sgalaberna for the *VALOR* group, http://www.t2k.org/asg/meeting/2015-05-14/view

[60] G.Cowan, K.Cranmer, E.Gross, O.Vitells, "Asymptotic formulae for likelihood-based tests of new physics" arXiv:1007.1727v2

[61] G. Cowan (1998) Statistical data analysis, Oxford University Press, ISBN 0-19-850155-2.

[62] F.James, M.Roos, Minuit: A System for Function Minimization and Analysis of the Parameter Errors and Correlations, Comput.Phys.Commun. 10:343 (1975)

[63] https://en.wikipedia.org/wiki/Gradient_descent

[64] G.J. Feldman and R.D. Cousins, A Unified approach to the classical statistical analysis of small signals, Phys.Rev.D57:3873-3889,1998; physics/9711021

[65] R.D. Cousins and V.L.Highland, Nucl.Instr. and Meth. A320:331035, 1992.

[66] J. Stuart, A. Ord and S.Arnold, Kendall's Advanced Theory of Statistics, Vol 2A (6th Ed.) (Oxford University Press, New York, 1994).

[67] N.Grant for the *VALOR* group, http://www.t2k.org/asg/meeting/2011/20111124/pvalues/view

[68] D.Karlen, p-values, distributions and low statistics, http://www.t2k.org/asg/oagroup/meeting/20120416/lowstat/view

[69] X. Qian, A. Tan, W. Wang, J. J. Ling, R. D. McKeown and C. Zhang, Statistical evaluation of experimental determinations of neutrino mass hierarchy, Phys.Rev.D86, 113011 (2012)

[70] K. McFarland for the T2K ν_μ-disappearance contour committee (C.Andreopoulos, N.Grant, J.Kameda, D.Karlen, E.Kearns, K.McFarland, M.Otani, M.Zito) Report to ASG conveners meeting, 17 May 2011, http://www.t2k.org/asg/contour/ASG-Conveners-Report/view

[71] R.Nichol, Talk at Neutrino 2012

[72] P. Adamson, et al., Measurement of Neutrino and Antineutrino Oscillations Using Beam and Atmospheric Data in MINOS, arXiv:1304.6335 [hep-ex], 2013

[73] A. Radovic, Talk at DPF 2013

[74] Y. Itow, Recent results in atmospheric neutrino oscillations in the light of large θ_{13}, Nucl. Phys. B (Proc. Suppl.) 235 79 (2013).

[75] J. Beringer et al. (Particle Data Group with 2013 update), Phys. Rev. D86, 010001 (2012)

[76] K.A. Olive et al. (Particle Data Group 2014 with 2015 update), Chin. Phys. C, 38, 090001

[77] G. Cowan, PDG statistical review (Particle Data Group with 2013 update), http://pdg.lbl.gov/2013/reviews/rpp2013-rev-statistics.pdf

[78] G.L.Fogli, E.Lisi et al., Evidence of theta(13)>0 from global neutrino data analysis, Phys.Rev.D84:053007, 2011; arXiv:1106.6028 [hep-ph]

[79] M.C.Gonzalez-Garcia, M.Maltoni, J.Salvado and T.Schwetz, Global fit to three neutrino mixing: critical look at present precision, arXiv:1209.3023 [hep-ph]

[80] Prob3++ 3-flavour oscillation probability software, http://www.phy.duke.edu/ raw22/public/Prob3++/

[81] P. Adamson *et al.* [MINOS Collaboration], Phys. Rev. Lett. **110** (2013) 25, 251801 [arXiv:1304.6335 [hep-ex]].

Appendices

A. Predicted number of events

In table 31, 32 and 33, 34 the total number of events expected for the Asimov data set B (see table 13) is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	69.747	0.035	4.335	0.002	0.151	0.001	74.270
CC1pi	17.206	0.017	1.572	0.001	0.049	0.000	18.845
CCcoherent	0.330	0.000	0.080	0.000	0.001	0.000	0.412
2p-2h	19.725	0.010	0.852	0.000	0.036	0.000	20.623
CCother	7.742	0.008	0.494	0.001	0.002	0.000	8.247
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	122.243	0.297	7.659	0.025	0.238	0.002	130.464

Table 31: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in table 13 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	7.088	0.004	13.306	0.005	0.004	0.016	20.422
CC1pi	2.718	0.003	3.066	0.002	0.002	0.003	5.794
CCcoherent	0.054	0.000	0.173	0.000	0.000	0.000	0.228
2p-2h	2.525	0.001	2.118	0.001	0.001	0.002	4.649
CCother	1.282	0.001	0.886	0.000	0.000	0.000	2.170
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.637	0.051	20.282	0.032	0.008	0.022	35.032

Table 32: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in table 13 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.062	2.129	0.001	0.088	10.811	0.121	13.211
CC1pi	0.017	0.348	0.000	0.027	1.236	0.020	1.648
CCcoherent	0.000	0.002	0.000	0.002	0.011	0.002	0.017
2p-2h	0.009	0.710	0.000	0.022	2.750	0.022	3.513
CCother	0.001	0.045	0.000	0.003	0.025	0.001	0.076
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.476	3.264	0.069	0.145	14.833	0.167	19.954

Table 33: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in table 13 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.269	1.376	2.178
CC1pi	0.002	0.040	0.001	0.060	0.045	0.163	0.312
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.022	0.029
2p-2h	0.001	0.069	0.000	0.063	0.087	0.219	0.439
CCother	0.001	0.008	0.000	0.004	0.003	0.003	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.142	0.313	0.235	0.478	0.404	1.783	3.355

Table 34: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in table 13 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	338.683	0.037	8.672	0.002	0.000	0.000	347.394
CC1pi	39.668	0.018	2.276	0.001	0.000	0.000	41.963
CCcoherent	0.678	0.000	0.136	0.000	0.000	0.000	0.815
2p-2h	74.638	0.011	1.548	0.000	0.000	0.000	76.198
CCother	8.732	0.009	0.560	0.001	0.000	0.000	9.301
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	469.893	0.301	13.518	0.025	0.000	0.000	483.737

Table 35: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	15.420	0.004	55.525	0.005	0.000	0.000	70.954
CC1pi	4.071	0.003	6.536	0.002	0.000	0.000	10.611
CCcoherent	0.074	0.000	0.612	0.000	0.000	0.000	0.687
2p-2h	4.721	0.001	7.811	0.001	0.000	0.000	12.534
CCother	1.457	0.001	1.008	0.000	0.000	0.000	2.467
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	26.713	0.052	72.225	0.032	0.000	0.000	99.021

Table 36: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

In table 35, 36 and 37, 38 the total number of unoscillated events is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.413	2.294	0.002	0.094	0.000	0.000	2.803
CC1pi	0.069	0.371	0.001	0.028	0.000	0.000	0.469
CCcoherent	0.002	0.003	0.000	0.002	0.000	0.000	0.006
2p-2h	0.095	0.759	0.000	0.023	0.000	0.000	0.878
CCother	0.004	0.046	0.000	0.003	0.000	0.000	0.054
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.970	3.504	0.071	0.154	0.000	0.000	5.698

Table 37: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.012	0.203	0.018	0.361	0.000	0.000	0.593
CC1pi	0.004	0.043	0.003	0.064	0.000	0.000	0.114
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.000	0.007
2p-2h	0.003	0.074	0.003	0.067	0.000	0.000	0.147
CCother	0.001	0.009	0.000	0.004	0.000	0.000	0.015
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.156	0.334	0.255	0.507	0.000	0.000	1.253

Table 38: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

B. Effect of systematic uncertainties

In tables 39 to 43 it is shown which systematic parameter is applied to which reaction and flavour modes for all the Super-K samples.

$f^{SK}_{E;r}$	$f^{SK+FSI}_{0;t,r}$	$f^{SK+FSI}_{1;t,r}$	$f^{SK+FSI}_{2;t,r}$	$f^{SK+FSI}_{3;t,r}$	$f^{SK+FSI}_{4;t,r}$	$f^{SK+FSI}_{5;t,r}$	$f^{banff}_{0;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{7;t,r}$	$f^{banff}_{8;t,r}$	$f^{banff}_{9;t,r}$	$f^{banff}_{10;t,r}$	$f^{banff}_{11;t,r}$	$f^{banff}_{12;t,r}$	$f^{banff}_{13;t,r}$	$f^{banff}_{14;t,r}$	$f^{banff}_{15;t,r}$	$f^{banff}_{16;t,r}$	$f^{banff}_{17;t,r}$	
✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ CCQE
✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ CC 1π
✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ CC coherent
✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ CC 2p-2h
✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ CC other
✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ/ν_τ NC $1\pi^{+/-}$
✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ/ν_τ NC $1\pi^0$
✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ/ν_τ NC coherent
✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_μ/ν_τ NC other
✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								ν_e NC 1γ
✓	✓	✓	✓															✓	✓	✓	✓	✓			$\bar{\nu}_\mu$ CCQE
✓				✓														✓	✓	✓	✓	✓			$\bar{\nu}_\mu$ CC 1π
✓				✓														✓	✓	✓	✓	✓			$\bar{\nu}_\mu$ CC coherent
✓	✓	✓	✓															✓	✓	✓	✓	✓			$\bar{\nu}_\mu$ CC 2p-2h
✓				✓														✓	✓	✓	✓	✓			$\bar{\nu}_\mu$ CC other
✓																		✓	✓	✓	✓	✓			$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$
✓																		✓	✓	✓	✓	✓			$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$
✓																		✓	✓	✓	✓	✓			$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent
✓																		✓	✓	✓	✓	✓			$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other
✓																		✓	✓	✓	✓	✓			ν_μ NC 1γ
✓					✓																		✓	✓	ν_e CCQE
✓					✓																		✓	✓	ν_e CC 1π
✓					✓																		✓	✓	ν_e CC coherent
✓					✓																		✓	✓	ν_e CC 2p-2h
✓					✓																		✓	✓	ν_e CC other
✓						✓																	✓	✓	ν_e NC $1\pi^{+/-}$
✓						✓																	✓	✓	ν_e NC $1\pi^0$
✓						✓																	✓	✓	ν_e NC coherent
✓						✓																	✓	✓	ν_e NC other
✓						✓																	✓	✓	$\bar{\nu}_\mu$ NC 1γ
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. ν_e CCQE
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. ν_e CC 1π
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. ν_e CC coherent
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. ν_e CC 2p-2h
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. ν_e CC other
																							✓	✓	$\bar{\nu}_e$ CCQE
																							✓	✓	$\bar{\nu}_e$ CC 1π
																							✓	✓	$\bar{\nu}_e$ CC coherent
																							✓	✓	$\bar{\nu}_e$ CC 2p-2h
																							✓	✓	$\bar{\nu}_e$ CC other
																								✓	$\bar{\nu}_e$ NC $1\pi^{+/-}$
																								✓	$\bar{\nu}_e$ NC $1\pi^0$
																								✓	$\bar{\nu}_e$ NC coherent
																								✓	$\bar{\nu}_e$ NC other
																								✓	$\bar{\nu}_e$ NC 1γ
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. $\bar{\nu}_e$ CCQE
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. $\bar{\nu}_e$ CC 1π
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. $\bar{\nu}_e$ CC coherent
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. $\bar{\nu}_e$ CC 2p-2h
							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓								Osc. $\bar{\nu}_e$ CC other

| | | |

	ν_μ CCQE	ν_μ CC 1π	ν_μ CC coherent	ν_μ CC 2p-2h	ν_μ CC other	ν_μ/ν_τ NC $1\pi^{+/-}$	ν_μ/ν_τ NC $1\pi^0$	ν_μ/ν_τ NC coherent	ν_μ/ν_τ NC other	ν_e NC 1γ	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC 1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC 2p-2h	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	$\bar{\nu}_\mu$ NC 1γ	ν_e CCQE	ν_e CC 1π	ν_e CC coherent	ν_e CC 2p-2h	ν_e CC other	ν_e NC $1\pi^{+/-}$	ν_e NC $1\pi^0$	ν_e NC coherent	ν_e NC other	ν_e NC 1γ	Osc. ν_e CCQE	Osc. ν_e CC 1π	Osc. ν_e CC coherent	Osc. ν_e CC 2p-2h	Osc. ν_e CC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC 1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC 2p-2h	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC $1\pi^{+/-}$	$\bar{\nu}_e$ NC $1\pi^0$	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	$\bar{\nu}_e$ NC 1γ	Osc. $\bar{\nu}_e$ CCQE	Osc. $\bar{\nu}_e$ CC 1π	Osc. $\bar{\nu}_e$ CC coherent	Osc. $\bar{\nu}_e$ CC 2p-2h	Osc. $\bar{\nu}_e$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{6;t,r}$																						✓	✓	✓	✓	✓																						
$f^{SK+FSI}_{7;t,r}$																						✓	✓	✓	✓	✓																						
$f^{SK+FSI}_{8;t,r}$																						✓	✓	✓	✓	✓																						
$f^{SK+FSI}_{9;t,r}$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓																															
$f^{SK+FSI}_{10;t,r}$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓																															
$f^{SK+FSI}_{11;t,r}$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓																															
$f^{SK+FSI}_{12;t,r}$																						✓	✓	✓	✓	✓																						
$f^{SK+FSI}_{13;t,r}$																						✓	✓	✓	✓	✓																						
$f^{SK+FSI}_{14;t,r}$																						✓	✓	✓	✓	✓																						
$f^{SK+FSI}_{15;t,r}$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓							✓	✓	✓	✓	✓																
$f^{SK+FSI}_{16;t,r}$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓							✓	✓	✓	✓	✓																
$f^{SK+FSI}_{17;t,r}$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓							✓	✓	✓	✓	✓																
$f^{banff}_{0;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{1;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{2;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{3;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{4;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{5;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{6;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{7;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{8;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{9;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{10;t,r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																						
$f^{banff}_{11;t,r}$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																						✓	✓	✓	✓	✓	✓

| | | |

	ν_μ CCQE	ν_μ CC 1π	ν_μ CC coherent	ν_μ CC 2p-2h	ν_μ CC other	ν_μ/ν_τ NC $1\pi^{+/-}$	ν_μ/ν_τ NC $1\pi^0$	ν_μ/ν_τ NC coherent	ν_μ/ν_τ NC other	ν_e NC 1γ	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC 1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC 2p-2h	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	ν_μ NC 1γ	ν_e CCQE	ν_e CC 1π	ν_e CC coherent	ν_e CC 2p-2h	ν_e CC other	ν_e NC $1\pi^{+/-}$	ν_e NC $1\pi^0$	ν_e NC coherent	ν_e NC other	$\bar{\nu}_\mu$ NC 1γ	Osc. ν_e CCQE	Osc. ν_e CC 1π	Osc. ν_e CC coherent	Osc. ν_e CC 2p-2h	Osc. ν_e CC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC 1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC 2p-2h	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC $1\pi^{+/-}$	$\bar{\nu}_e$ NC $1\pi^0$	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	$\bar{\nu}_e$ NC 1γ	Osc. $\bar{\nu}_e$ CCQE	Osc. $\bar{\nu}_e$ CC 1π	Osc. $\bar{\nu}_e$ CC coherent	Osc. $\bar{\nu}_e$ CC 2p-2h	Osc. $\bar{\nu}_e$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{0;t,r}RHC$	✓			✓								✓				✓																																	
$f^{SK+FSI}_{1;t,r}RHC$	✓			✓								✓				✓																																	
$f^{SK+FSI}_{2;t,r}RHC$	✓			✓								✓				✓																																	
$f^{SK+FSI}_{3;t,r}RHC$		✓	✓		✓								✓	✓			✓																																
$f^{SK+FSI}_{4;t,r}RHC$																					✓	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{5;t,r}RHC$						✓	✓	✓	✓	✓							✓	✓	✓	✓	✓						✓	✓	✓	✓	✓										✓	✓	✓						
$f^{banff}_{0;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{1;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{2;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{3;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{4;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{5;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{6;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{7;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{8;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{9;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{10;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	✓	✓	✓	✓	✓																	
$f^{banff}_{11;t,r}RHC$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓	✓					
$f^{banff}_{12;t,r}RHC$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓	✓					
$f^{banff}_{13;t,r}RHC$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓	✓					
$f^{banff}_{14;t,r}RHC$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓	✓					
$f^{banff}_{15;t,r}RHC$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓	✓					
$f^{banff}_{16;t,r}RHC$																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	
$f^{banff}_{17;t,r}RHC$																					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																	

| | | |

	ν_μ CCQE	ν_μ CC 1π	ν_μ CC coherent	ν_μ CC 2p-2h	ν_μ CC other	ν_μ/ν_τ NC $1\pi^{+/-}$	ν_μ/ν_τ NC $1\pi^0$	ν_μ/ν_τ NC coherent	ν_μ/ν_τ NC other	ν_e NC 1γ	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC 1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC 2p-2h	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	$\bar{\nu}_\mu$ NC 1γ	ν_e CCQE	ν_e CC 1π	ν_e CC coherent	ν_e CC 2p-2h	ν_e CC other	ν_e NC $1\pi^{+/-}$	ν_e NC $1\pi^0$	ν_e NC coherent	ν_e NC other	$\bar{\nu}_\mu$ NC 1γ	Osc. ν_e CCQE	Osc. ν_e CC 1π	Osc. ν_e CC coherent	Osc. ν_e CC 2p-2h	Osc. ν_e CC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC 1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC 2p-2h	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC $1\pi^{+/-}$	$\bar{\nu}_e$ NC $1\pi^0$	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	$\bar{\nu}_e$ NC 1γ	Osc. $\bar{\nu}_e$ CCQE	Osc. $\bar{\nu}_e$ CC 1π	Osc. $\bar{\nu}_e$ CC coherent	Osc. $\bar{\nu}_e$ CC 2p-2h	Osc. $\bar{\nu}_e$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{6;t,r}RHC$																						✓	✓	✓	✓	✓															✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{7;t,r}RHC$																						✓	✓	✓	✓	✓															✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{8;t,r}RHC$																						✓	✓	✓	✓	✓															✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{9;t,r}RHC$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓																																
$f^{SK+FSI}_{10;t,r}RHC$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓																																
$f^{SK+FSI}_{11;t,r}RHC$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓																																
$f^{SK+FSI}_{12;t,r}RHC$																					✓	✓	✓	✓	✓									✓	✓	✓	✓	✓										
$f^{SK+FSI}_{13;t,r}RHC$																					✓	✓	✓	✓	✓									✓	✓	✓	✓	✓										
$f^{SK+FSI}_{14;t,r}RHC$																					✓	✓	✓	✓	✓									✓	✓	✓	✓	✓										
$f^{SK+FSI}_{15;t,r}RHC$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓										✓	✓	✓	✓	✓					
$f^{SK+FSI}_{16;t,r}RHC$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓										✓	✓	✓	✓	✓					
$f^{SK+FSI}_{17;t,r}RHC$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓										✓	✓	✓	✓	✓					
$f^{banff}_{0;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{1;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{2;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{3;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{4;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{5;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{6;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{7;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{8;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{9;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{10;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓												✓	✓	✓	✓	✓																							
$f^{banff}_{11;t,r}RHC$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																				✓	✓	✓	✓	✓	✓

Continued on next page

Table 42 – *Continued from previous page*

| | | |

Table 42 – *Continued from previous page*

Table 42: Each row of this table represents one of the 110systematic parameters (oscillation excluded) considered in this analysis and each column represents one of the 50 MC templates used to construct the single e-like ring reconstructed energy spectrum p.d.f. in antineutrino mode. A ✓symbol denotes that the given MC template is modified when the given systematic parameter is tweaked. Only the systematic parameters that affect this sample are shown.

	ν_μ CCQE	ν_μ CC 1π	ν_μ CC coherent	ν_μ CC 2p-2h	ν_μ CC other	ν_μ/ν_τ NC $1\pi^{+/-}$	ν_μ/ν_τ NC $1\pi^0$	ν_μ/ν_τ NC coherent	ν_μ/ν_τ NC other	ν_e NC 1γ	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC 1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC 2p-2h	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	$\bar{\nu}_\mu$ NC 1γ	Osc. ν_e CCQE	Osc. ν_e CC 1π	Osc. ν_e CC coherent	Osc. ν_e CC 2p-2h	Osc. ν_e CC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC 1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC 2p-2h	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC $1\pi^{+/-}$	$\bar{\nu}_e$ NC $1\pi^0$	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	$\bar{\nu}_e$ NC 1γ	Osc. $\bar{\nu}_e$ CCQE	Osc. $\bar{\nu}_e$ CC 1π	Osc. $\bar{\nu}_e$ CC coherent	Osc. $\bar{\nu}_e$ CC 2p-2h	Osc. $\bar{\nu}_e$ CC other
$f_{E;r}^{SK}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f_{36;t,r}^{SK+FSI}$																					✓	✓	✓	✓	✓																			
$f_{37;t,r}^{SK+FSI}$																					✓	✓	✓	✓	✓																			
$f_{38;t,r}^{SK+FSI}$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓																											
$f_{39;t,r}^{SK+FSI}$	✓	✓	✓	✓	✓						✓	✓	✓	✓	✓	✓																											
$f_{40;t,r}^{SK+FSI}$																						✓	✓	✓	✓	✓	✓	✓	✓	✓														
$f_{41;t,r}^{SK+FSI}$																						✓	✓	✓	✓	✓	✓	✓	✓	✓														
$f_{42;t,r}^{SK+FSI}$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓											✓	✓	✓	✓	✓								
$f_{43;t,r}^{SK+FSI}$						✓	✓	✓	✓	✓						✓	✓	✓	✓	✓											✓	✓	✓	✓	✓								
$f_{0;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{1;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{2;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{3;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{4;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{5;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{6;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{7;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{8;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{9;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{10;t,r}^{banff}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																		
$f_{11;t,r}^{banff}$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f_{12;t,r}^{banff}$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f_{13;t,r}^{banff}$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f_{14;t,r}^{banff}$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓
$f_{15;t,r}^{banff}$											✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																					✓	✓	✓	✓	✓

| | | |

C. Method for significance calculation

The significance for a certain oscillation hypothesis given by the measured data is provided as following. First the $\Delta\chi^2$ distribution, $f(\Delta\chi^2)$, is obtained by many toy experiments (e.g. $10k$), then the $\Delta\chi^2_{data}$ value is measured with the data set. This quantities are shown in fig. 53.

First the p-value is calculated as:

$$\text{p-value} = \int_{-\infty}^{\Delta\chi^2_{data}} f(\Delta\chi^2) d(\Delta\chi^2) \tag{20}$$

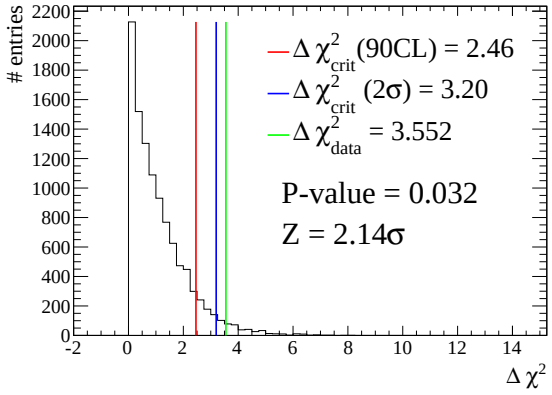


Figure 53: Distributions of $\Delta\chi^2$. $10k$ toy experiments are performed with true values of $\delta_{CP} = 0$ (left) and π (right), assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set. The $\Delta\chi^2$ value (green line) measured from the Run 1-6 data set as well as the p-value and the significance as number of standard deviations (Z), are shown as well.

Then the significance can be shown as the number of standard deviations where the measured quantity is assumed to be gaussian distributed. In fig. 54 the gaussian distribution of the measured variable μ is shown as a function of the Z-score, i.e. how many standard deviations a certain measurement x is far from the expected value:

$$Z = \frac{x - \mu}{\sigma} \tag{21}$$

If an experiment measures x, α is the area of the p.d.f. corresponding to a certain CL (90% in the example). The quantity σ is the bias of the measured value with respect to the expectation that contains $1 - \alpha = 68.27\%$ of the total area.

The significance can be reported with the Z-score in the following way:

- the p-value is measured using eq. (20)
- Assuming the p-value to be 1-sided (see fig. 53) and α 2-sided (see fig. 54), we set α = p-value
- Z, shown in eq. (21), is computed using the quantile distribution. Given the cumulative distribution function of the normal standard distribution with Mean=0 and RMS=1 ($\sigma = 1$),

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{Z}^{+\infty} e^{-t^2/2} dt = \alpha/2 \tag{22}$$

the quantile is defined as

$$Z = \Phi^{-1}(1 - \alpha/2). \tag{23}$$

The ROOT function Math::gaussian_quantile, that takes α as input, and calculate the cumulative distribution function of the lower tail of the normal (Gaussian) distribution is defined as following:

```
TF1 q("q","ROOT::Math::gaussian_quantile(1.-(x/2.)),1.)",0.,1.);
double Zscore = q.Eval(pvalue);
```

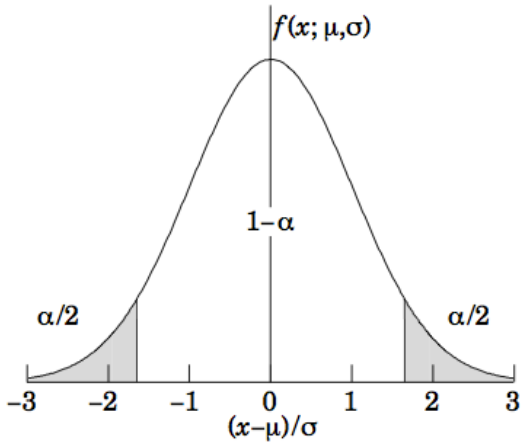


Figure 54: Illustration of a symmetric 90% confidence interval (unshaded) for a measurement of a single quantity with Gaussian errors. Integrated probabilities, defined by $\alpha = 0.1$, are shown.

D. Statistical uncertainty on weighted p-value and critical values for confidence interval

The statistical uncertainty on the p-value is calculated assuming that the number of entries in each bin of the expected distribution of the test statistic follow a poisson distribution. The p-value uncertainty is then derived from the statistical uncertainty of the population, for every given weight, in each bin. As the p-value is calculated from the fraction of trials which fall to the right of the data statistic, the p-value is expected to be binomially distributed. Given a sufficiently large number of trials (N), the p-value is expected to take a gaussian distribution with a mean of the p-value and a variance $\sqrt{p(1-p)/N}$. For a p-value of 0.01 and 10k trials $np = 100$.

The uncertainty is calculated from the first order Taylor series expansion assuming no correlation between bins and the number of events in each bin follows a poisson distribution. The calculation of the rate only uncertainty summing over all toy experiments, for the right sided p-value, is as follows:

$$p_{right} = \frac{\sum_{i=0}^{N(right-tail)} W_i \cdot n_i}{\sum_{i=0}^{N(total)} W_i \cdot n_i} = \frac{p_n}{p_d} \tag{24}$$

Where W_i is the likelihood weight, n_i is the number of times each toy appears and p_n and p_d are defined as the numerator and denominator of the p-value for convenience. The total uncertainty as a result of the statistical uncertainty on each sample is given by

$$(\sigma_p)^2 = \sum_x (\frac{\partial p}{\partial n_x}\sigma_{n_x})^2 \tag{25}$$

The uncertainty in p given uncertainty in events per bin. Considering bins to the right and left of the data statistic independently:

$$(\sigma_p)^2 = \sum_{i=0}^{N(right-tail)} (\frac{W_i p_d - W_i p_n}{p_d^2}\sqrt{n_i})^2 + \sum_{i=0}^{N(left-tail)} (\frac{W_i p_n}{p_d^2}\sqrt{n_i})^2 \tag{26}$$

Where we have used the Poisson uncertainty for the same of toys of a given weight $\sigma_{n_x} = \sqrt{n_x}$. When using the definition of p, the equation becomes:

$$(\sigma_p)^2 = \sum_{i=0}^{N(right-tail)} (\frac{W_i - W_i p}{p_d}\sqrt{n_i})^2 + \sum_{i=0}^{N(left-tail)} (\frac{W_i p}{p_d}\sqrt{n_i})^2 \tag{27}$$

Taking the limit where each toy appears only once, the uncertainty of appearance is $\sqrt{n_i} = \sqrt{1} = 1$.

$$(\sigma_p)^2 = \sum_{i=0}^{N(right-tail)} (\frac{W_i - W_i p}{p_d})^2 + \sum_{i=0}^{N(left-tail)} (\frac{W_i p}{p_d})^2 \tag{28}$$

$$(\sigma_p)^2 = (\frac{(1-p)}{p_d})^2 \sum_{i=0}^{N(right-tail)} W_i^2 + (\frac{p}{p_d})^2 \sum_{i=0}^{N(left-tail)} W_i^2 \tag{29}$$

In general the uncertainty on the p-value due to events in the left and right tails of the distribution is given by the sum of squares of the toy weights multiplied by a prefactor which depends on the p-value and total number of toys in the distribution.

Finally, when the weight of each event is 1 it simplifies to:

$$(\sigma_p)^2 = \frac{p(1-p)}{N} \tag{30}$$

Where N is the integral of the distribution. This is the expected result for a binomial process with a success rate p.

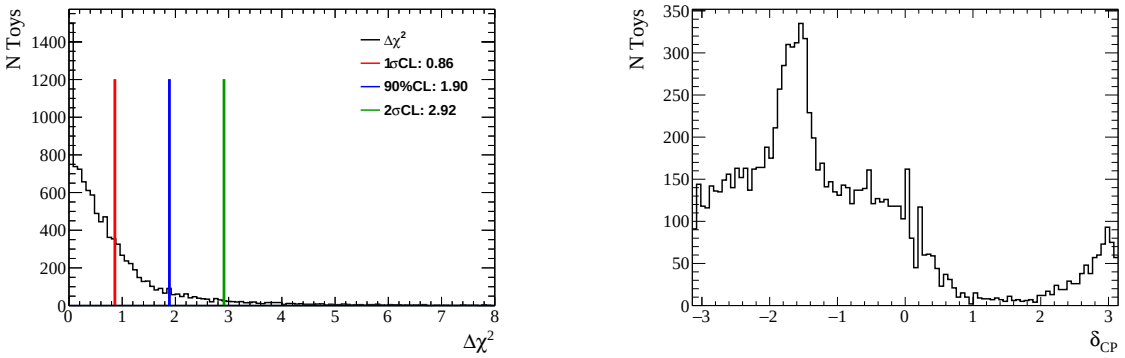
In the same way the error on the $\Delta\chi^2$ critical values can be estimated. First **??** is used to estimate the error $\sigma_{coverage}$ on the coverage, analogue to p-value = 90% for 90% CL. Then the $\pm 1\sigma$ critical values are

calculated as the $\Delta\chi^2$ values corresponding respectively to the fraction $\pm\sigma_{coverage}$ to the left and right of the central critical value.

E. FC results

This section includes supplementary plots from the 1D δ_{CP} critical value calculation. figs. 55 to 59 show the best fit values of δ_{CP} and the $\Delta\chi^2$ distributions for all fake data toy experiments with true normal hierarchy.

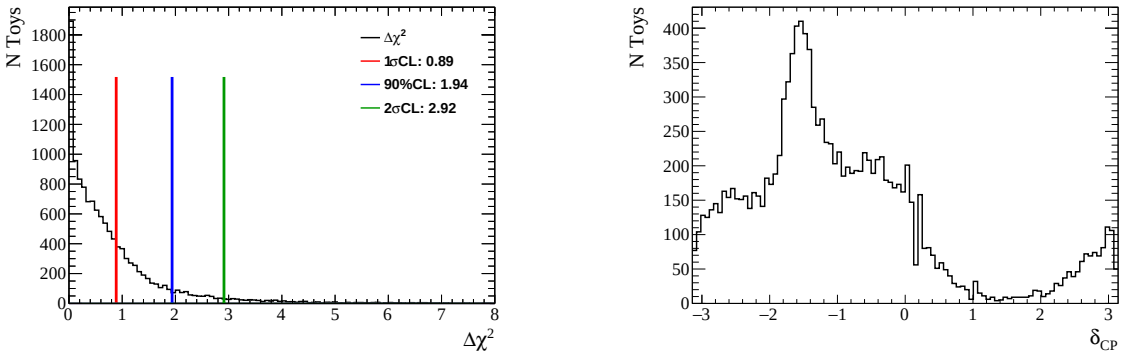
The results from the inverted hierarchy toys are found in figs. 60 to 67



(a) Best-fit distribution.

(b) $\Delta\chi^2$ distribution.

Figure 55: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = -3\pi/4$ and normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.



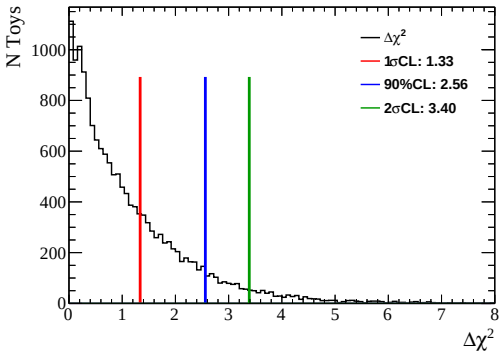
(a) Best-fit distribution

(b) $\Delta\chi^2$ distribution.

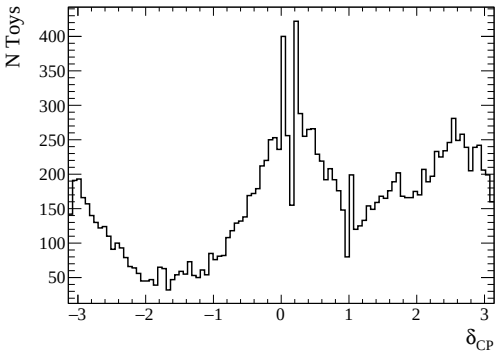
Figure 56: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = -\pi/4$ and normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

F. VALOR, MaCH3 and p-θ comparison

This section contains a comparison of the Asimov and data fits between the 3 analysis groups VALOR, MaCH3 and p-θ. The comparisons of the Asimov fits are shown infigs. 68 to 70. The comparisons of the data fits are shown in figs. 71 to 73.

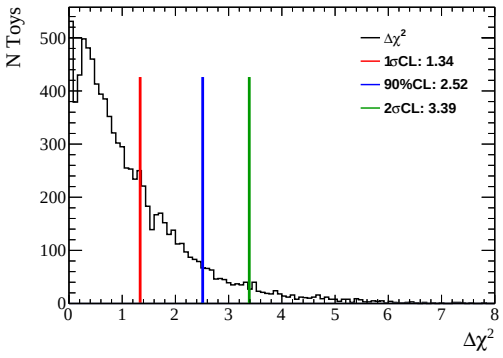


(a) Best-fit distribution

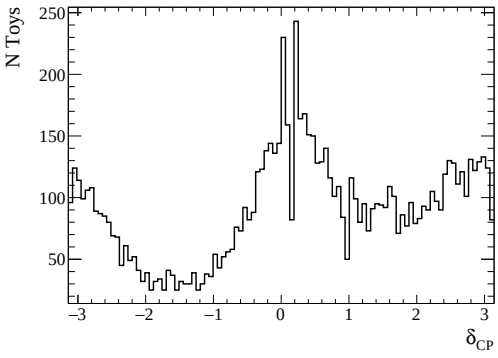


(b) $\Delta\chi^2$ distribution.

Figure 57: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = +\pi/4$ and normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

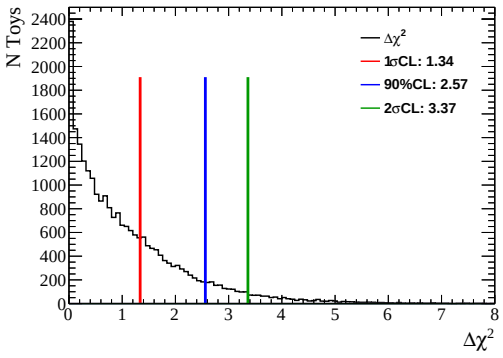


(a) Best-fit distribution.

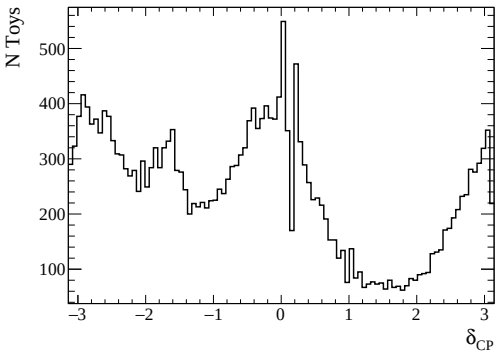


(b) $\Delta\chi^2$ distribution.

Figure 58: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = +3\pi/4$ and normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

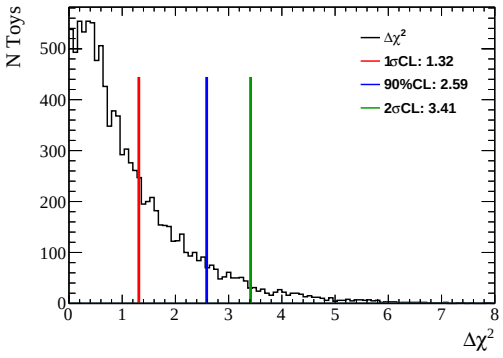


(a) Best-fit distribution.

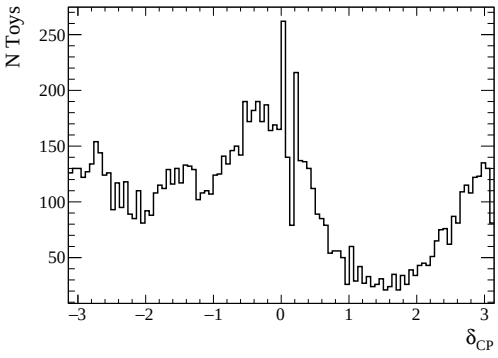


(b) $\Delta\chi^2$ distribution.

Figure 59: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = \pi$ and normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

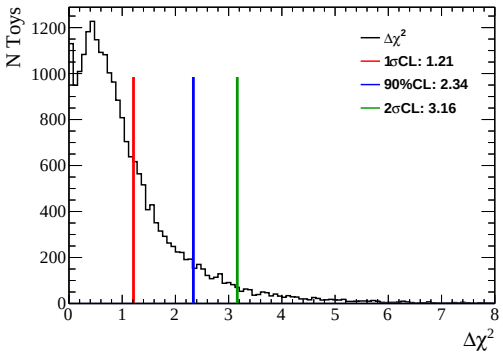


(a) Best-fit distribution.

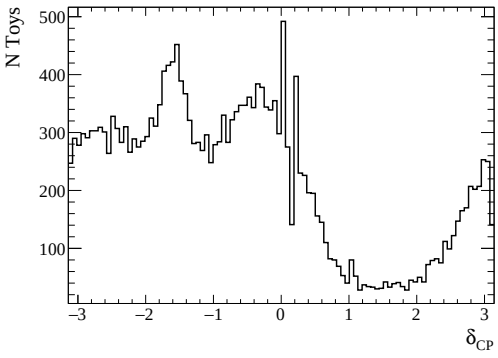


(b) $\Delta\chi^2$ distribution.

Figure 60: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = -3\pi/4$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

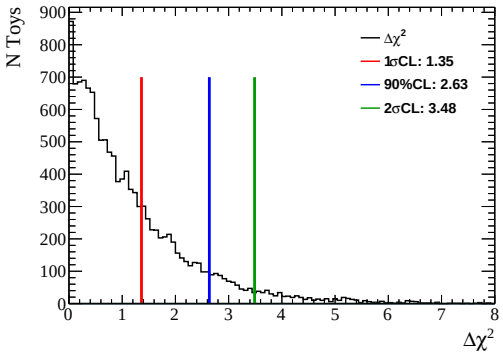


(a) Best-fit distribution.

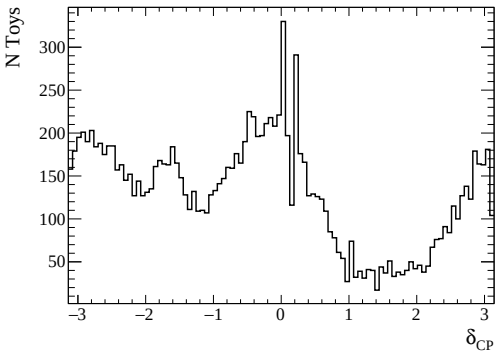


(b) $\Delta\chi^2$ distribution.

Figure 61: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = -\pi/2$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

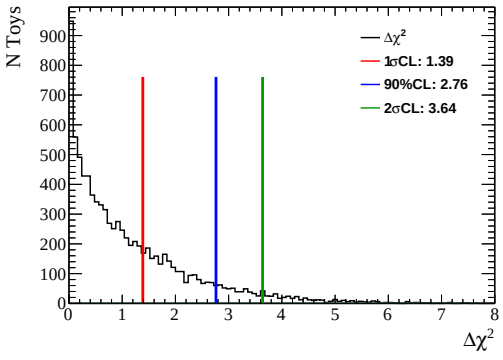


(a) Best-fit distribution

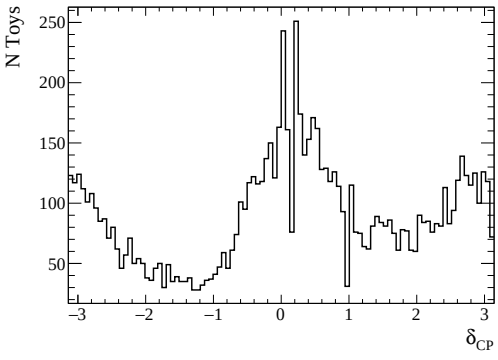


(b) $\Delta\chi^2$ distribution.

Figure 62: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = -\pi/4$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

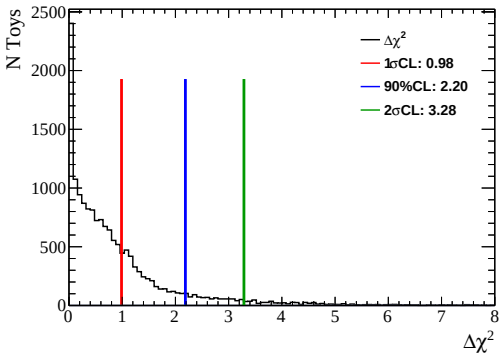


(a) Best-fit distribution

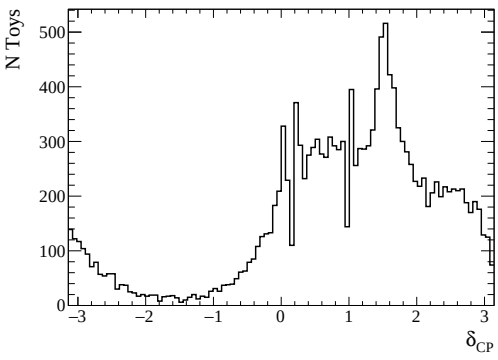


(b) $\Delta\chi^2$ distribution.

Figure 63: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = 0$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

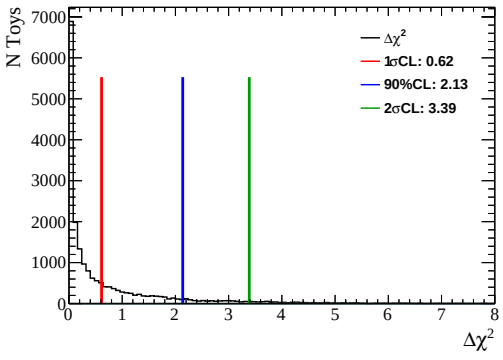


(a) Best-fit distribution

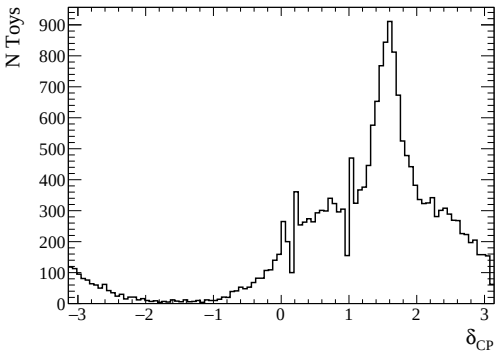


(b) $\Delta\chi^2$ distribution.

Figure 64: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = +\pi/4$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

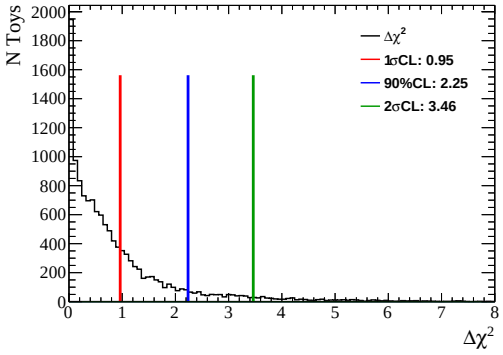


(a) Best-fit distribution.

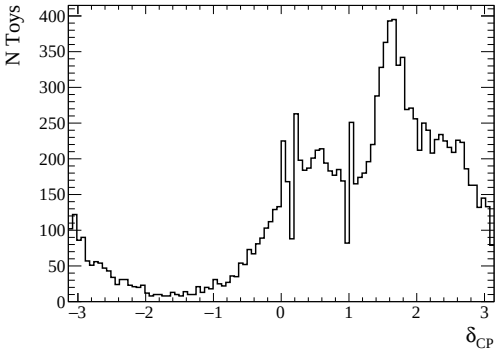


(b) $\Delta\chi^2$ distribution.

Figure 65: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿10k toy experiments are performed with a true value of $\delta_{CP}^{true} = \pi/2$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

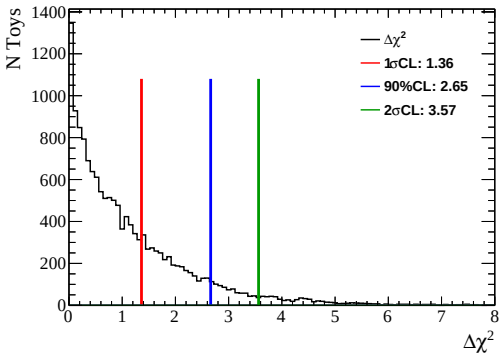


(a) Best-fit distribution.

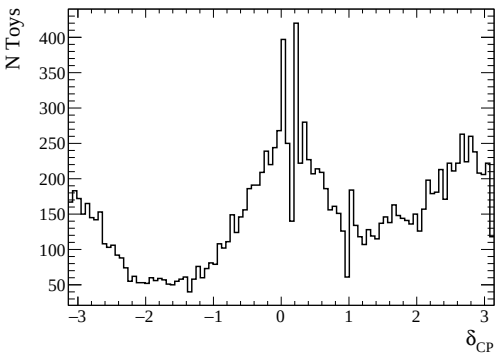


(b) $\Delta\chi^2$ distribution.

Figure 66: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿$10k$ toy experiments are performed with a true value of $\delta_{CP}^{true} = +3\pi/4$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.



(a) Best-fit distribution.

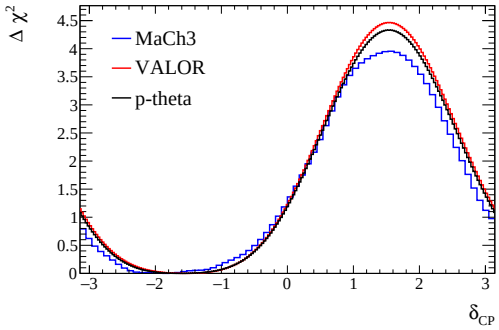


(b) $\Delta\chi^2$ distribution.

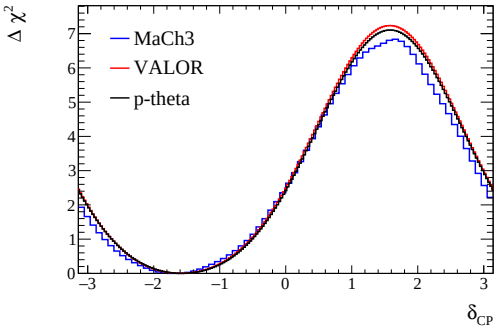
Figure 67: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). ¿$10k$ toy experiments are performed with a true value of $\delta_{CP}^{true} = \pi$ and inverted hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-7c data set.

δ_{CP}	CL	Critical Value	Coverage (Gaus)
-2.356 (normal)	1σ	0.8556	0.7416
	90%	1.899	0.9465
	2σ	2.918	0.9776
-2.356 (inverted)	1σ	1.317	0.5719
	90%	2.594	0.9108
	2σ	3.41	0.972
-1.571 (normal)	1σ	0.5011	0.8018
	90%	1.893	0.9446
	2σ	3.039	0.975
-1.571 (inverted)	1σ	1.214	0.5995
	90%	2.34	0.9297
	2σ	3.16	0.9768
-0.7854 (normal)	1σ	0.89	0.7259
	90%	1.941	0.9476
	2σ	2.918	0.9785
-0.7854 (inverted)	1σ	1.35	0.5624
	90%	2.626	0.9076
	2σ	3.476	0.9711
0 (normal)	1σ	1.333	0.5836
	90%	2.612	0.9091
	2σ	3.43	0.9731
0 (inverted)	1σ	1.391	0.5714
	90%	2.757	0.8971
	2σ	3.625	0.9681
0.7854 (normal)	1σ	1.333	0.5771
	90%	2.562	0.9126
	2σ	3.398	0.974
0.7854 (inverted)	1σ	0.9782	0.6909
	90%	2.196	0.9304
	2σ	3.277	0.9714
1.571 (normal)	1σ	1.196	0.6013
	90%	2.319	0.9295
	2σ	3.288	0.974
1.571 (inverted)	1σ	0.6245	0.7702
	90%	2.129	0.929
	2σ	3.391	0.9703
2.356 (normal)	1σ	1.342	0.5632
	90%	2.533	0.9158
	2σ	3.387	0.9737
2.356 (inverted)	1σ	0.9527	0.6985
	90%	2.247	0.9249
	2σ	3.456	0.9675
3.142 (normal)	1σ	1.343	0.5745
	90%	2.574	0.9127
	2σ	3.37	0.9742
3.142 (inverted)	1σ	1.36	0.5721
	90%	2.651	0.9054
	2σ	3.57	0.9687

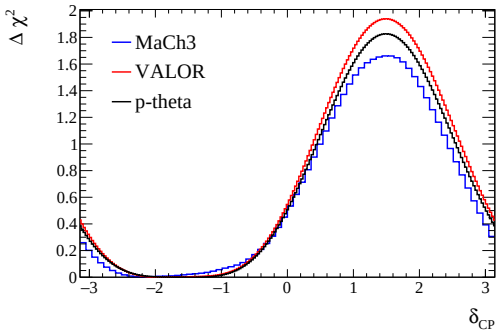
Table 44: The critical values of $\Delta\chi^2$ used to set the confidence intervals are shown for both the gaussian approximation (i.e. constant $\Delta\chi^2$) as well as the ones obtained with the Feldman-Cousins method. 10k toy experiments were used for each true point of δ_{CP}. Furthermore the coverage, defined by eq. (19), provided by the gaussian approximation for different true values δ_{CP} is shown. The true mass hierarchy is set to normal. The exposure corresponds to that of the Run 1-7c data set.



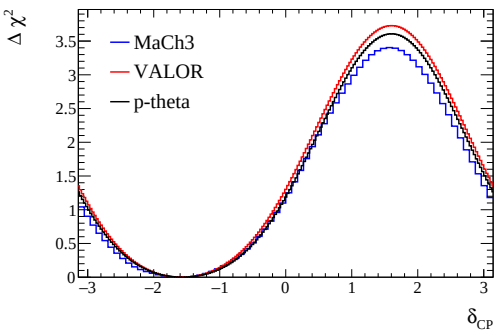
(a) Normal Hierarchy - reactor



(b) Inverted Hierarchy - reactor

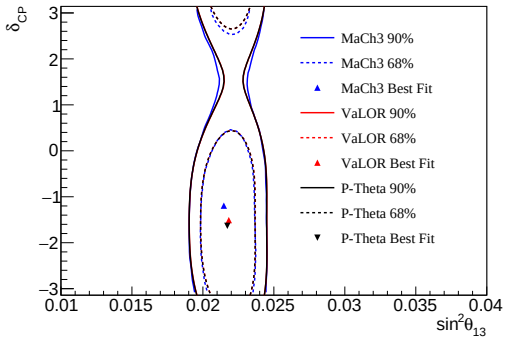


(c) Normal Hierarchy - T2K only

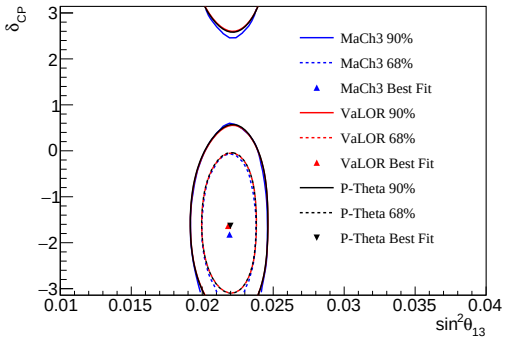


(d) Inverted Hierarchy - T2K only

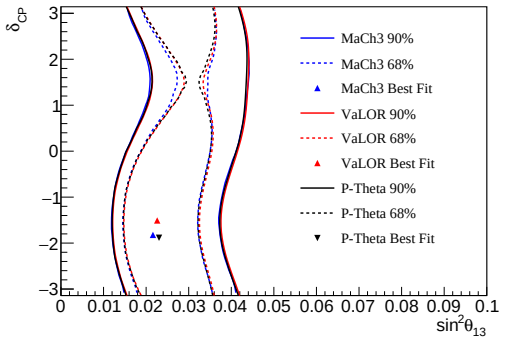
Figure 68: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} for the Asimov data set A of table 13 and a comparison between the different analysis groups is shown.



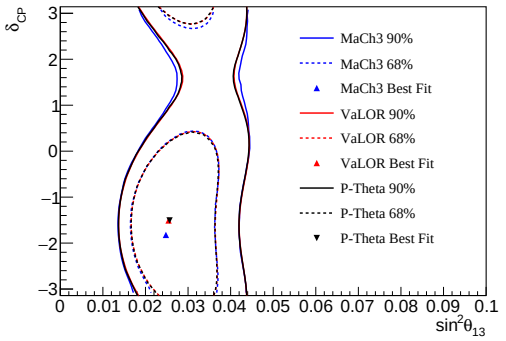
(a) Normal Hierarchy - reactor



(b) Inverted Hierarchy - reactor

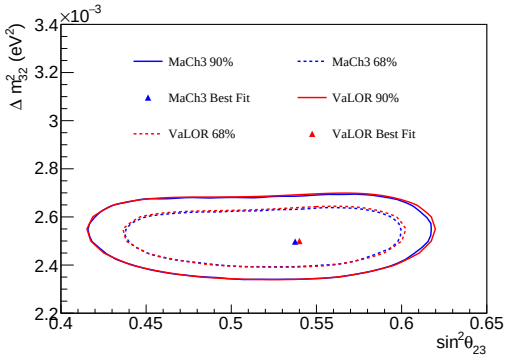


(c) Normal Hierarchy - T2K only

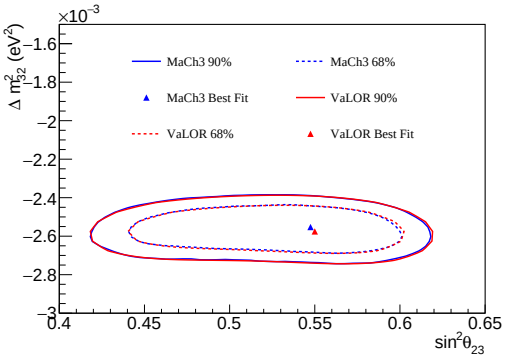


(d) Inverted Hierarchy - T2K only

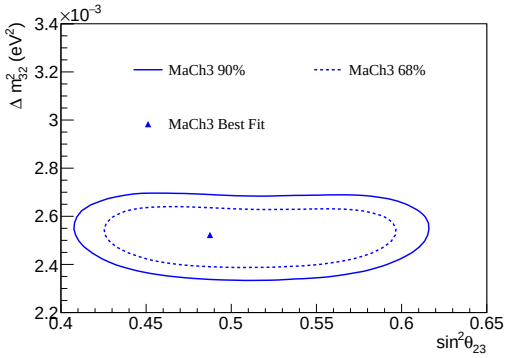
Figure 69: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} vs $\sin^2(\theta_{13})$ for the Asimov data set A of table 13 and a comparison between the different analysis groups is shown.



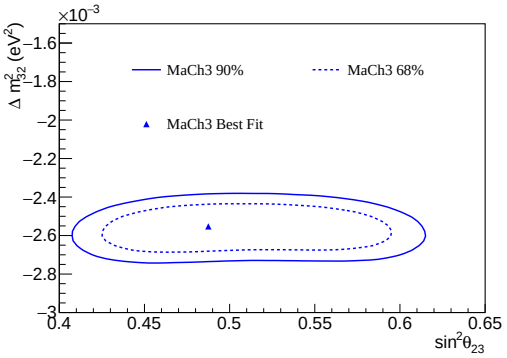
(a) Normal Hierarchy - reactor



(b) Inverted Hierarchy - reactor

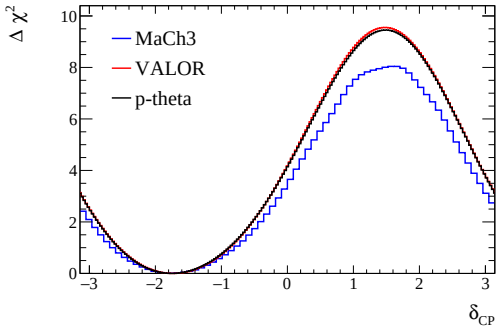


(c) Normal Hierarchy - T2K only

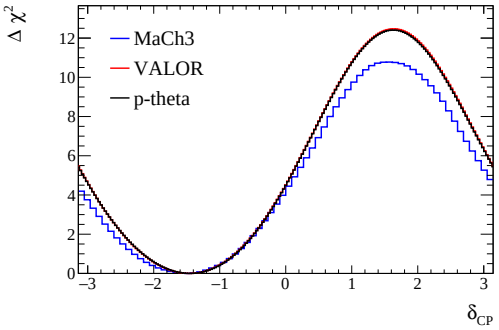


(d) Inverted Hierarchy - T2K only

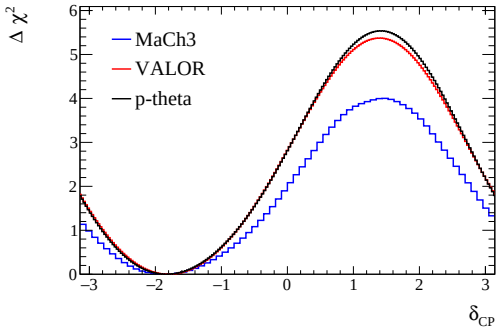
Figure 70: The expected $\Delta\chi^2$ distribution as a function of $\sin^2(\theta_{23})$ vs Δm^2_{32} for the Asimov data set A of table 13 and a comparison between the different analysis groups is shown.



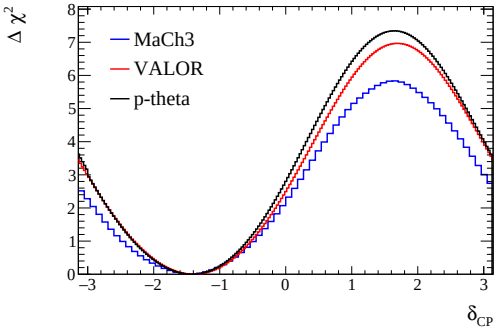
(a) Normal Hierarchy - reactor



(b) Inverted Hierarchy - reactor

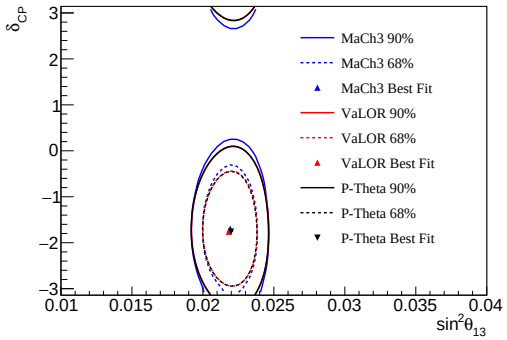


(c) Normal Hierarchy - T2K only

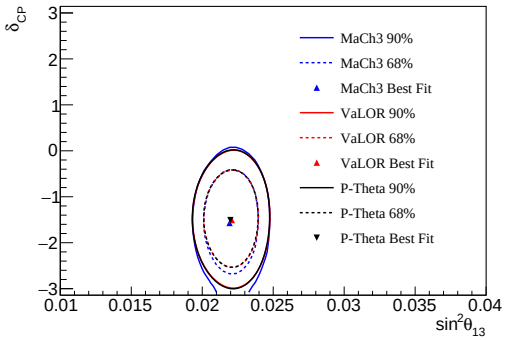


(d) Inverted Hierarchy - T2K only

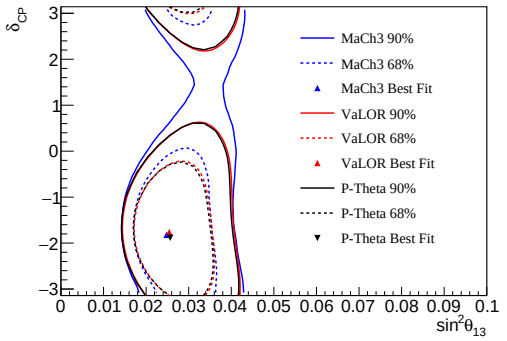
Figure 71: The $\Delta\chi^2$ distribution as a function of δ_{CP} for the data fit and a comparison between the different analysis groups is shown.



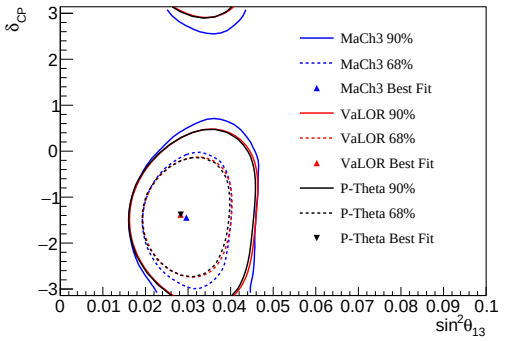
(a) Normal Hierarchy - reactor



(b) Inverted Hierarchy - reactor

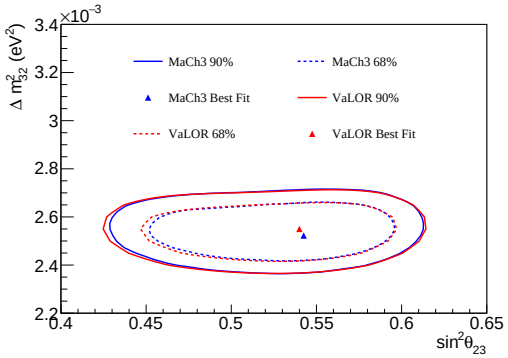


(c) Normal Hierarchy - T2K only

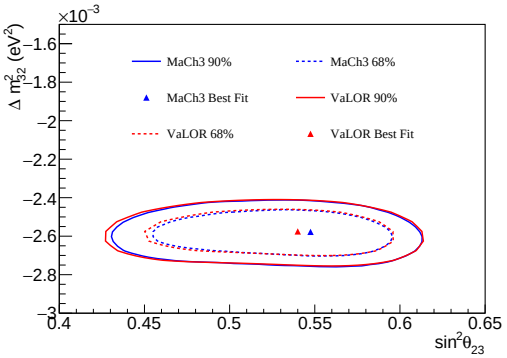


(d) Inverted Hierarchy - T2K only

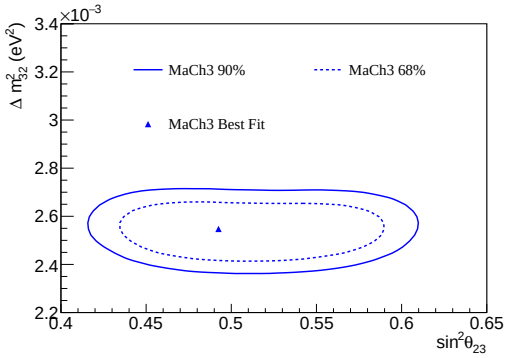
Figure 72: The $\Delta\chi^2$ distribution as a function of δ_{CP} vs $\sin^2(\theta_{13})$ for the data fit and a comparison between the different analysis groups is shown.



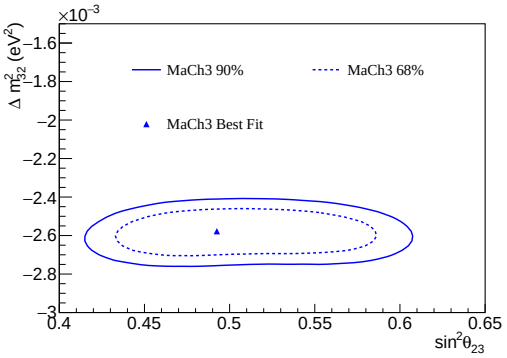
(a) Normal Hierarchy - reactor



(b) Inverted Hierarchy - reactor



(c) Normal Hierarchy - T2K only



(d) Inverted Hierarchy - T2K only

Figure 73: The $\Delta\chi^2$ distribution as a function of $\sin^2(\theta_{23})$ vs Δm^2_{32} for the data fit and a comparison between the different analysis groups is shown.

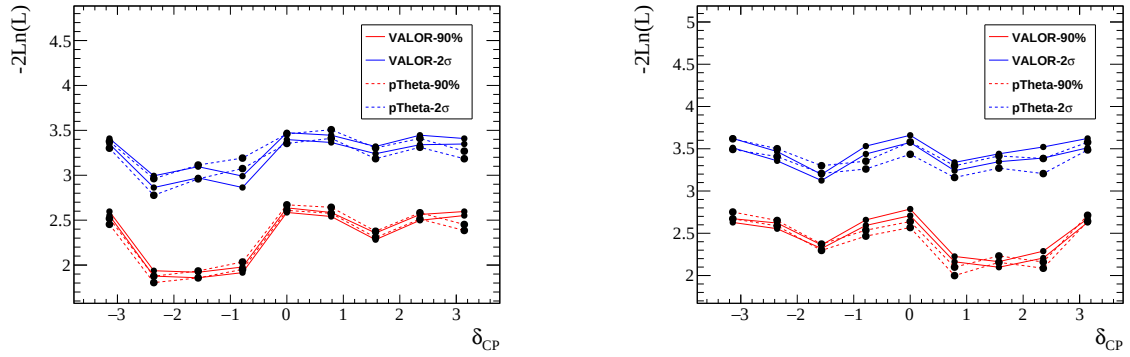
G. Further coverage studies

This section includes a summary of the validation and comparison studies which were performed to gain a better understanding of the results of the coverage studies. The VALOR and pTheta critical values and coverages were also compared.

G.1. VALOR - pTheta Comparison

This section includes a comparison of the VALOR and p-Theta critical values calculated using the MC method described in section 8. fig. 74 shows a comparison of the pTheta and VALOR critical values, the VALOR critical values are also shown considering the statistical uncertainty on the MC statistics. The normal hierarchy critical values seem to be in very good agreement between pTheta and VALOR. The small difference seen at $\delta_{CP} = \pi$ is expected as the VALOR analysis is expected to have better separation of $\delta_{CP} = \pi$ and $\delta_{CP} = 0$.

The inverted hierarchy critical values however show very small difference at the at about one to two standard deviations considering the statistical uncertainty. The difference in the critical values can be explained by the differences seen in the Asimov fits in fig. 75. The separation between the normal and inverted hierarchy $\delta\chi^2$ surfaces is slightly larger for VALOR than p-Theta so some difference in the inverted hierarchy critical values is expected.



(a) 90% and 2σ CL NH

(b) 90% and 2σ CL IH

Figure 74: A comparison of the VALOR and pTheta critical values calculated using the MC method described in section 8. For both VALOR and pTheta the lines indicate the plus and minus one sigma ranges of the critical values.

G.2. Summer vs Current analysis

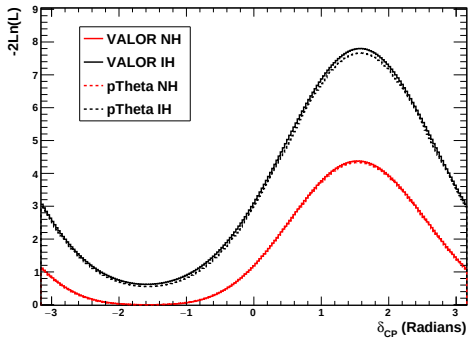
fig. 76 the comparison of the critical values in the four sample and current five sample analyses are shown. There is a difference clearly visible near $\delta_{CP} = -\pi/2$ which could arise from two sources; the addition of the new sample or the method to estimate critical values. Both of which were changes made from the four sample to the five sample analysis. The effects were investigated in the following studies.

G.2.1. Study of the new sample

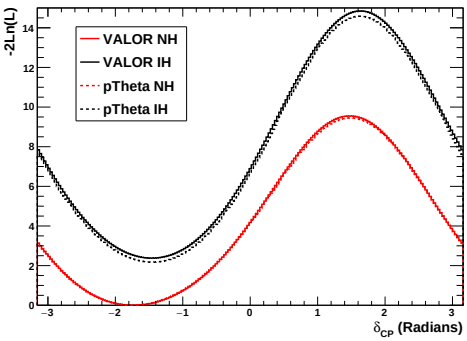
In this study, the critical values for normal hierarchy and $\delta_{CP} = -\pi/2$ were found for two different cases.

- Four 1-Ring samples with the Run 1-7c statistics. The neutrino mode statistics are then scaled up in accordance with the additional appearance events expected, assuming Asimov data set A, in the additional ν_e CC1π^+sample.
- Four 1-Ring samples and the additional ν_e CC1π^+sample with the Run 1-7c statistics.

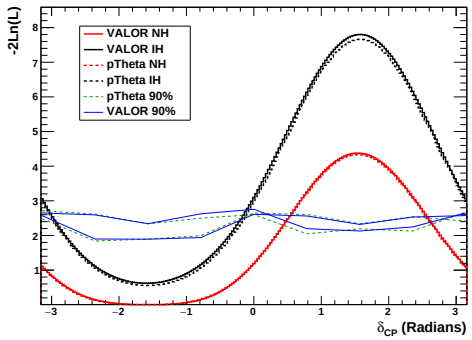
In each case 10k toy experiments were performed as with the coverage studies in section 8. A comparison of the critical values are shown in table 45. We find that the differences between the critical values found in both methods are well within the statistical uncertainty due to the MC statistics. This indicates that the difference in the critical values we seen between the previous and new analysis is not an artefact of the new ν_e CC1π^+sample.



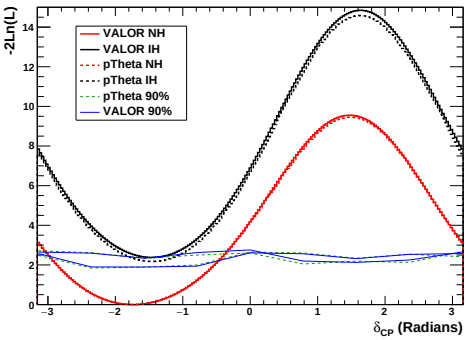
(a) Asimov A



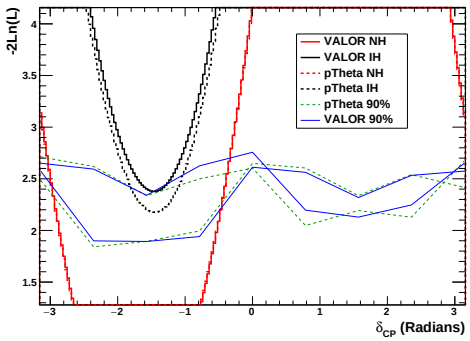
(b) Data



(c) Asimov A, with FC



(d) Data, with FC



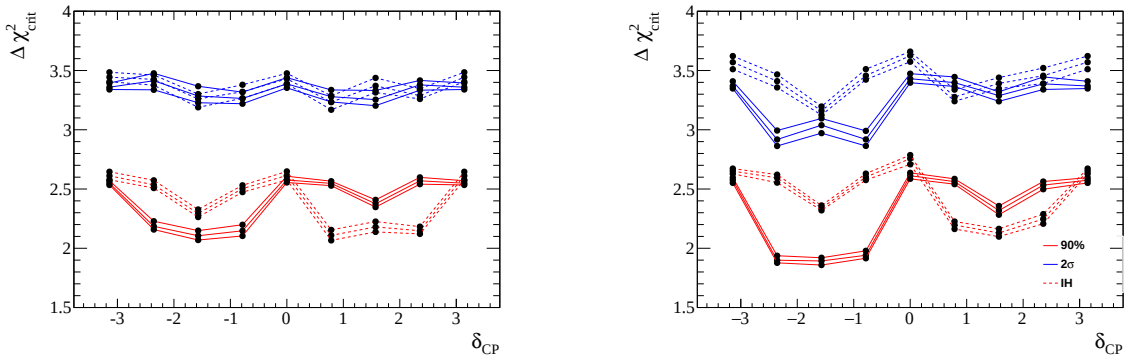
(e) Data, with FC zoomed

Figure 75: A comparison of the VALOR and pTheta $\delta\chi^2$ surfaces for the Asimov data set A and the Run 1-7c data set. The Feldman-Cousins critical values are also shown for both pTheta and VALOR for reference.

G.2.2. MC toy extraction method

A major change in the coverage studies between the previous and current analysis was that in the current analysis fake MC toy experiments were chosen based on the constraint in the current analysis. Specifically, the values of $\sin^2\theta_{23}$ and Δm^2_{32} for the toy experiments were selected based on the 2 dimensional likelihood from the current data set. More details are given in section 8. In the previous analysis, this was not done and the values of $\sin^2\theta_{23}$ and Δm^2_{32} were selected randomly from the prior distributions detailed in table 12. The old method is labeled prior sampling and the new data fit sampling.

This study is a comparison of the critical values found for normal hierarchy and $\delta_{CP} = -\pi/2$ with the new and old method for the 1 ring samples only. A study was performed by the pTheta group comparing the case when all 5 samples are used and can be found in [17]. A similar study was also performed in the



(a) Run 1-7c statistic, four 1 Ring samples

(b) Run 1-7c statistic, 5 samples

Figure 76: Critical values obtained with the Feldman-Cousins method for Run 1-7c for the 4 sample joint fit and the new 5 sample for several true values of δ_{CP} and both normal (solid line) and inverted (dashed line) hierarchy. Both 90% (red) and 2σ (blue) CL critical values are shown. Greater than $10k$ toy experiments are performed for each point. All the other oscillation parameters are marginalised. The $\pm 1\sigma$ error on the critical values is shown as well and is represented by the lowest and highest set of critical values, with the same colour of the central ones.

ăCL	4 Sample + stats	5 Sample
ă1 Sigma	$0.4990^{+0.0200}_{-0.0149}$	$0.5011^{+0.0103}_{-0.0122}$
ă90% CL	$1.8438^{+0.0451}_{-0.0482}$	$1.893^{+0.0267}_{-0.0346}$
ă95% CL	$2.9197^{+0.0871}_{-0.0558}$	$2.8523^{+0.0777}_{-0.0522}$
ă2 Sigma	$3.1002^{+0.0678}_{-0.1333}$	$3.0386^{+0.0567}_{-0.0671}$

Table 45: A comparison of the critical values found when considering the four 1-ring samples with increased statistics and the full 5 sample fit. The uncertainties given are calculated due to the statistical uncertainty in the number of toy experiments used in each study.

summer but not for $\delta_{CP} = -\pi/2$. From table 46 we see that there is a significant difference between the critical values found in the two methods which can not be explained by the statistical uncertainty.

The MC fake data toys used in the data fit sampling method are different to those in the prior sampling method in two ways. The muon like samples in the fake data set are closer to maximal mixing in the data fit sampling method and there are on average more events in the electron like sample. These combined effects lead to the difference in the critical values. The coverage study is done to adjust the critical values because of the deviation from gaussian coverage as a result of physical boundaries and the systematic uncertainties. In the data fit sampling method, more toys are generated at maximal disappearance therefore we expect this to have some impact on the critical values.

The critical values also become more non gaussian as we increase the data in our samples and the result becomes more affected by the systematic uncertainties. This is the case of the neutrino mode electron like sample. This can be seen by again considering fig. 76 where we see that only the critical values near $\delta_{CP} = -\pi/2$ are significantly different from those found in the summer. This is because only near $\delta_{CP} = -\pi/2$ are there significant statistics in the neutrino mode electron like sample to see this effect.

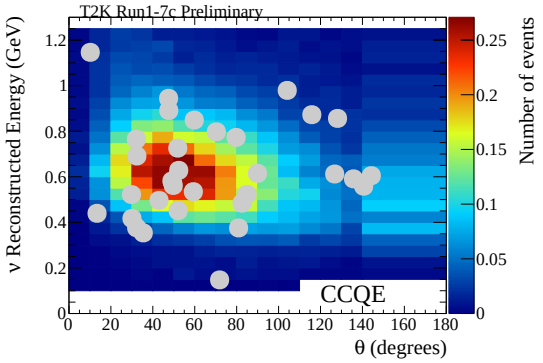
The inverted hierarchy critical values are also evidence for this effect as they are consistent between the summer and current analysis. Swapping from normal to inverted hierarchy has the effect to reduce the number of events in the neutrino mode electron like sample therefore statistical uncertainty is more significant than systematic in the inverted hierarchy case. We therefore expect the inverted hierarchy critical values to decrease around $\delta_{CP} = -\pi/2$ as we increase the exposure.

ăCL	Prior sampling	Data fit sampling
ă1 Sigma	$0.5881^{+0.01626}_{-0.0222}$	$0.51233^{+0.01213}_{-0.01412}$
ă90% CL	$2.1243^{+0.07458}_{-0.0531}$	$1.9322^{+0.04027}_{-0.04286}$
ă95% CL	$3.1222^{+0.0558}_{-0.0524}$	$3.0114^{+0.06206}_{-0.07667}$
ă2 Sigma	$3.2321^{+0.0842}_{-0.0663}$	$3.1701^{+0.06086}_{-0.07468}$

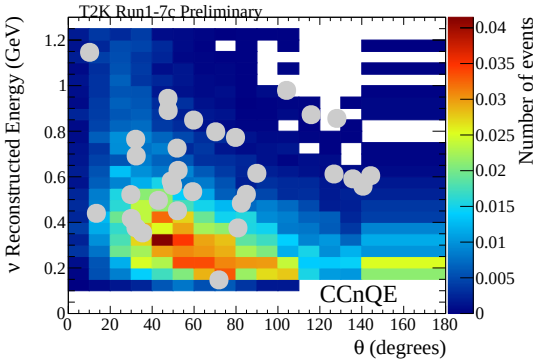
Table 46: A comparison of the critical values found when considering the four 1-ring samples with increased statistics and the full 5 sample fit. The uncertainties given are calculated due to the statistical uncertainty in the number of toy experiments used in each study.

H. Signal vs background

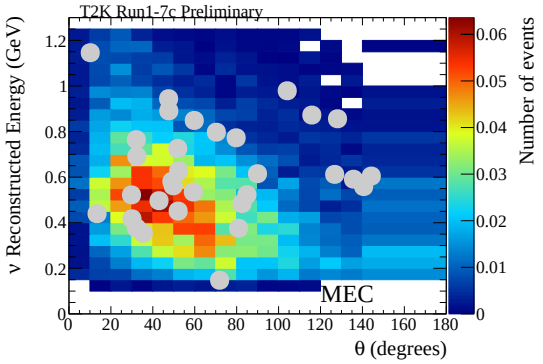
This section includes a comparison of the observed events to the signal and background templates for the electron like samples. The comparisons are shown in figs. 77 to 79.



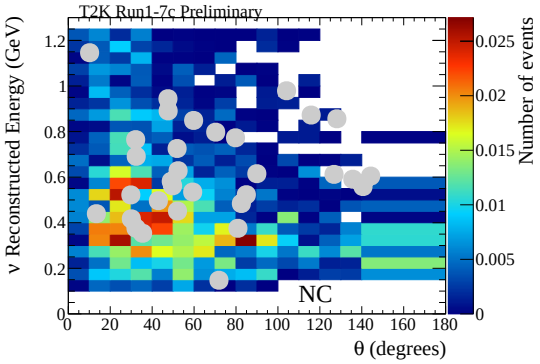
(a) CCQE



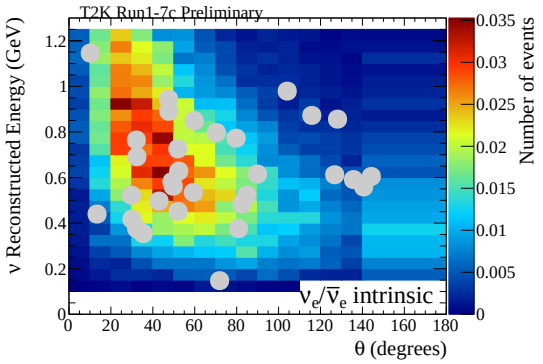
(b) CC non QE



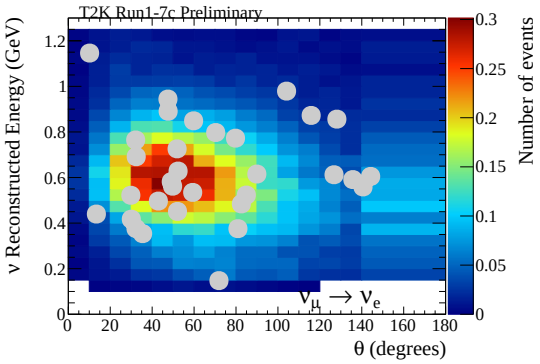
(c) 2P-2H



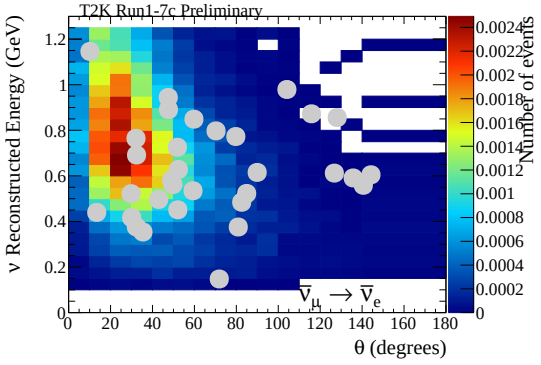
(d) NC



(e) Beam ν_e

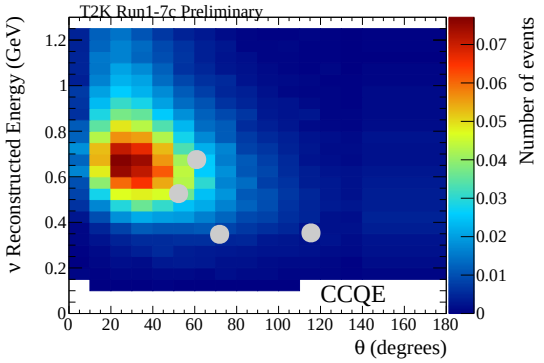


(f) $\nu_\mu \rightarrow \nu_e$

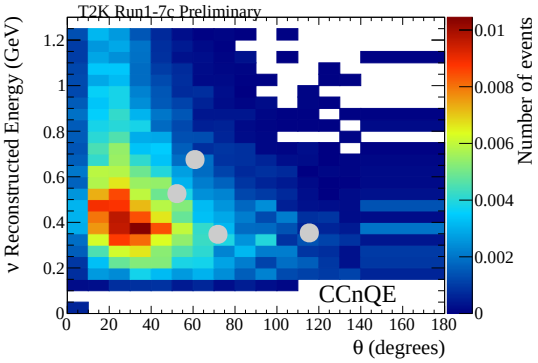


(g) $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$

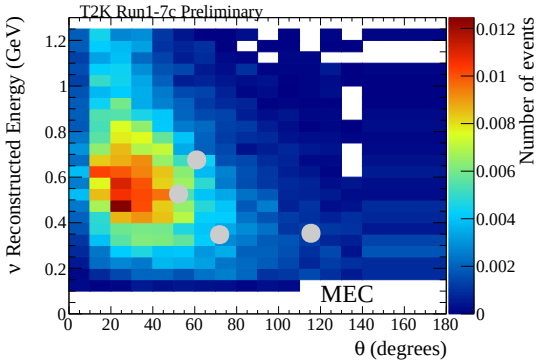
Figure 77: A comparison of the signal and a breakdown of the Asimov data set A prediction by neutrino type and interaction mode for the neutrino mode electron like sample.



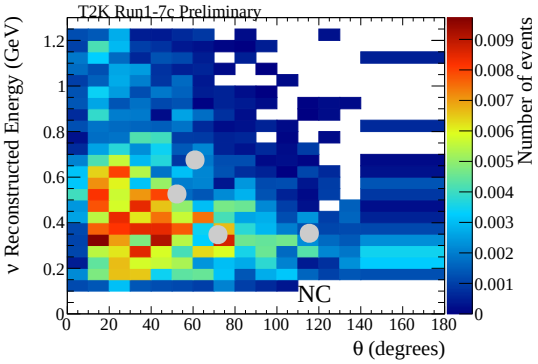
(a) CCQE



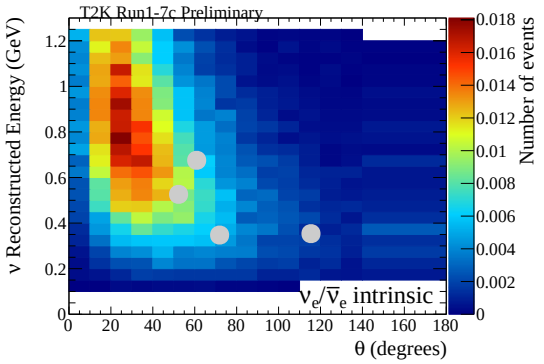
(b) CC non QE



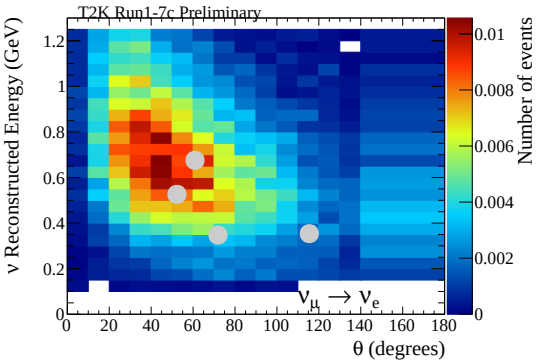
(c) 2P-2H



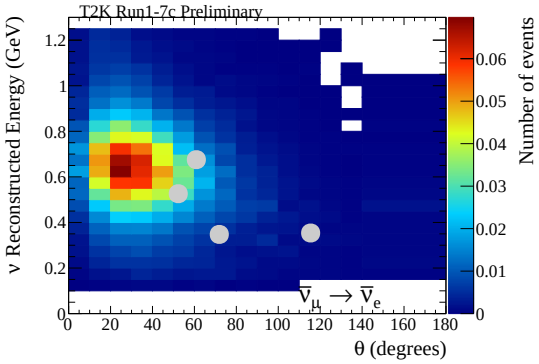
(d) NC



(e) Beam ν_e

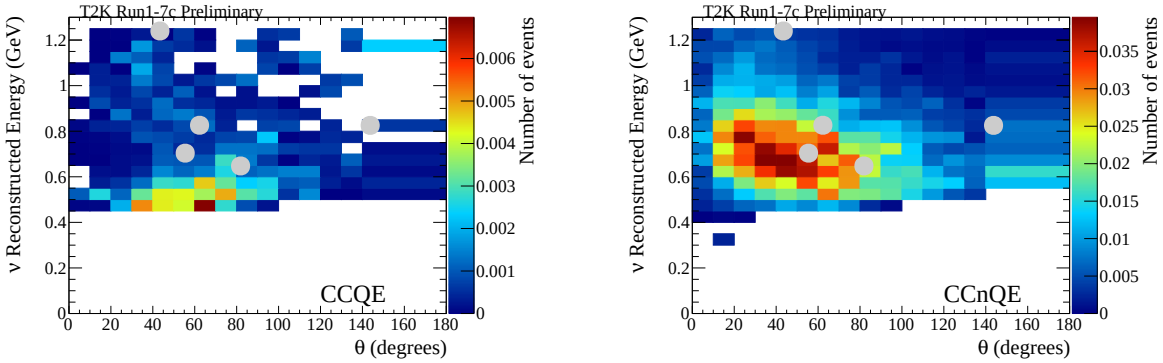


(f) $\nu_\mu \rightarrow \nu_e$



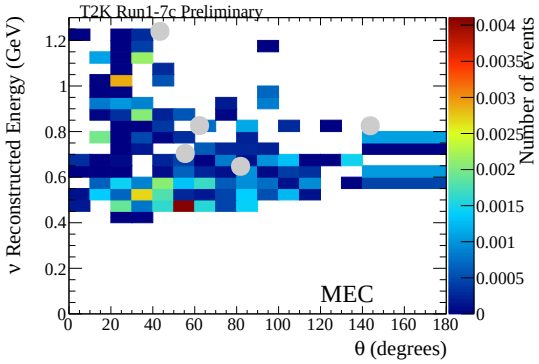
(g) $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$

Figure 78: A comparison of the signal and a breakdown of the Asimov data set A prediction by neutrino type and interaction mode for the antineutrino mode electron like sample.

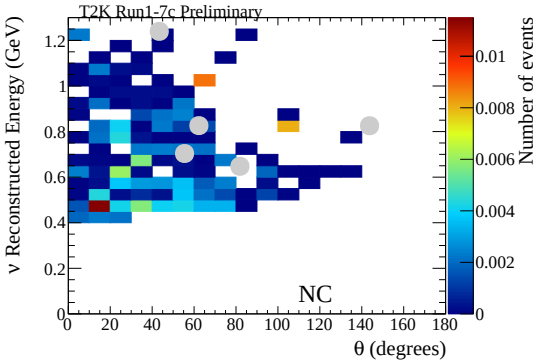


(a) CCQE

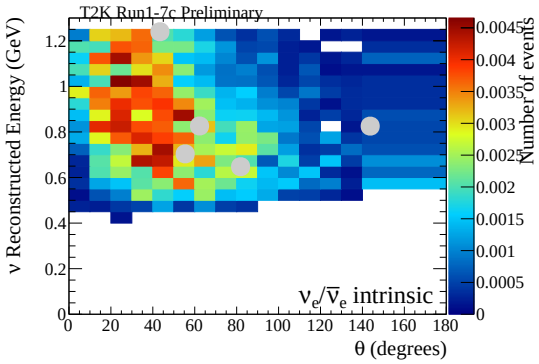
(b) CC non QE



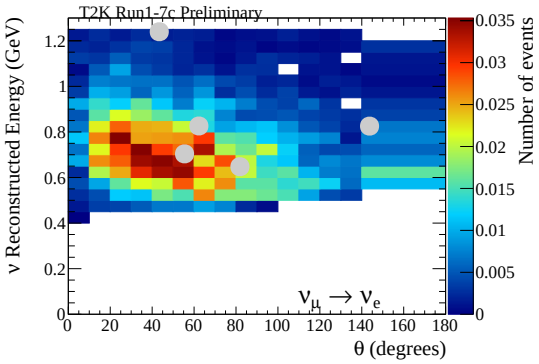
(c) 2P-2H



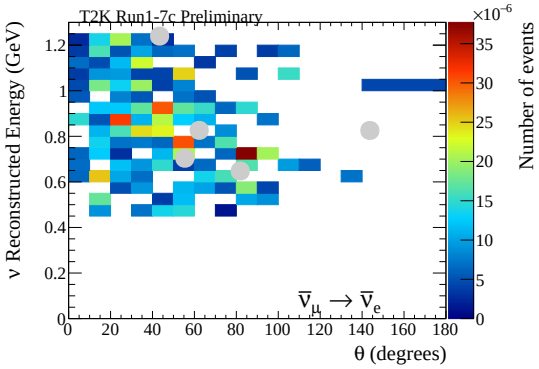
(d) NC



(e) Beam ν_e



(f) $\nu_\mu \rightarrow \nu_e$



(g) $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$

Figure 79: A comparison of the signal and a breakdown of the Asimov data set A prediction by neutrino type and interaction mode for the neutrino mode ν_e CC1π^+ sample.